United Nations Development Programme

Country: Indonesia

PROJECT DOCUMENT

Project Title: Advancing Indonesia's Lighting Market to High Efficient Technologies (ADLIGHT)		
Country: Indonesia	**Implementing Partner:** Ministry of Energy and Mineral Resources (MEMR)	**Management Arrangements:** National Implementation Modality (NIM)
UNPDF/Country Programme Outcome: UNPDF/CPD Outcome 3 -- By 2020, Indonesia is sustainably managing its natural resources, on land and at sea, with an increased resilience to the effects of climate change, disaster and other shocks.		
UNDP Strategic Plan Output: CPD Indicative Outputs -- 2.2.1. National energy policies and guideline developed and integrated into sub-national development plan.2.2.2. Sub-national authorities and key partners are able to implement programmes, mobilize resources and develop public-private partnership for RE/EE, which will contribute to the reduction of national greenhouse gases emission.		
UNDP Social and Environmental Screening Category: Moderate	**UNDP Gender Marker: 2**	
Atlas Project ID/Award ID number: 00098585	**Atlas Output ID/Project ID number:** 00101814	
UNDP-GEF PIMS ID number: 5721	**GEF ID number: 9493**	
Planned start date: 1 June 2020	**Planned end date: 1 June 2023**	
LPAC date: 5 February 2020		

Brief project description:

ADLIGHT aims to promote the increased deployment of high efficiency lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions. The project is expected to lead to increased market penetration of high efficiency lighting through capacity building for the country's lighting industry, thereby enhancing the local manufacturing capability to locally-produce highly quality lighting systems in line with MEPS, through introducing quality and minimum energy performance regulation, procurement policies and improved customs procedures and through introduction of new business models and awareness raising. This will bring about cumulative electricity savings of 310 GWh and cumulative GHG emissions reduction of 250 kt CO_2 by year 3 of the project and increase the impact as the selected demonstrations to enhance market penetration will catalyze more replication of these EE lighting technologies for the coming years and beyond to have a total potential of 1,646 kt CO_2 emission reduction. This will be achieved through implementation of three Components, namely, (1) Support to local lighting industry to transform the market for high quality, high efficient lighting systems, (2) Regulatory mechanisms and market monitoring, verification, and enforcement, and (3) New business models and awareness raising for high efficiency lighting technology penetration. During the implementation, in addition to the GEF fund of USD 3,895,872 that will be registered at the Ministry of Finance as grant to the government of Indonesia, co-financing for parallel activities will be provided by UN Environment (UNEP) to an amount of USD 60,000, UNDP USD 80,000, from the government (MEMR, BSN and City Government) to an amount of USD 27,889,337 and from private lighting

manufacturers to an amount of USD 9,065,000. The total co-financing of USD 37,094,337 will not be registered as part of grant to the government of Indonesia as it will be used directly by each of co-financier to conduct parallel activities complementary to ADLIGHT.		
FINANCING PLAN		
TOTAL BUDGET ADMINISTERED BY UNDP		
(1) **GEF Trust Fund**	USD 2,633,372	
Total Budget administered by UNDP	USD 2,633,372	
Total GEF Trust Fund for ADLIGHT project	**USD 2,633,372**	
PARALLEL CO-FINANCING		
GEF Trust Fund UNEP Outcome 2 (2)	USD 1,262,500	
UNDP (3a)	USD 80,000	
UNEP (3b)	USD 60,000	
Government of Indonesia (3c)	USD 27,889,337	
Industry (3d)	USD 9,065,000	
(2) Total co-financing (3a- 3d)	USD 37,094,337	
(3) Grand-Total Project Financing (1)+(2)+(3)	USD 40,990,209	
SIGNATURES		
Signature: **Ir.F.X. Sutijastoto, M.A.** **Director General NREEC**	**Agreed by:** **Ministry of Energy & Mineral Resources**	**Date/Month/Year:**
Signature: **Christophe Bahuet** **Resident Representative**	**Agreed by:** **UNDP**	**Date/Month/Year:**

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List of Abbreviations & Acronyms

Abbreviation	*Meaning*
ADLIGHT	Advancing Indonesia's Lighting Market to High Efficient Technologies
AILKI	*Asosiasi Industri Luminer & Kelistrikan Indonesia (*Association of Indonesia Luminaires and Electricity Industry*)*
ALINDO	*Asosiasi Luminer Indonesia (*Indonesia Luminaires Association*)*
ALINDO	*Asosiasi Luminer Indonesia* (Indonesian Luminaire Association)
APAEC	ASEAN Plan of Action Energy Cooperation (2016-2025)
APERLINDO	*Asosiasi Industri Perlampuan Listrik Indonesia* or Indonesian Electrical Lighting Industry Association
ASEAN-SHINE	Association of Southeast Asian Nations - Standards Harmonization Initiative for Energy Efficiency
ASKRINDO	*PT Asuransi Kredit Indonesia or* Credit Insurance Company
BAPPENAS	*Badan Perencanaan Pembangunan Nasional* (National Development Planning Agency)
BMP	*Bobot Manfaat Perusahaan* (Weight-based company benefit)
BPD	*Bank Pembangunan Daerah* or Regional Development Banks
BPPT	*Badan Pengkajian dan Penerapan Teknologi* (Research Agency for Technology Development and Implementation)
BRESL	Barrier Removal to the Cost Effective Development and Implementation of Energy Efficiency Standards and Labeling
BSN	*Badan Standardisasi National* (National Standardization Agency)
CFL	Compact Florescent Lamp
DANIDA	Danish International Development Agency
DGEED	Directorate General of Electricity and Energy Development
DGNREEC	Directorate General of New-Renewable Energy and Energy Conservation
EE	Energy efficiency
EECCHI	Energy Efficiency and Conservation Clearing House Indonesia
EEL	Energy Efficient Lighting
ES&L	Energy standards and labeling
ESCO	Energy Service Company
FGDs	Focused group discussions
FSP	Full Sized Project
GAMATRINDO	*Gabungan Industri Manufaktur Lampu Terpadu Indonesia* (Indonesian Integrated-Lamp Manufacturing Industry Association)
GEF	Global Environment Facility
GEFSEC	Global Environment Facility Secretariat
GHG	Greenhouse gas
GIZ	Die Deutsche Gesellschaft für Internationale Zusammenarbeit (German Agency for International Development)
GLP	Global Leapfrogging Program
HS Code	Harmonized Commodity Description and Coding System
IEC	International Electrotechnical Commission

Abbreviation	*Meaning*
IIGF	Indonesia Infrastructure Guarantee Fund
JICA	Japan International Cooperation Agency
KAN	*Komite Akreditasi Nasional* (National Accreditation Body of Indonesia)
LED	Light-emitting diode
LFA	Logical Framework Analysis
LKPP	*Lembaga Kebijakan Pengadaan Barang/Jasa Pemerintah* or the National Public Procurement Agency
MASKEEI	*Masyarakat Konservasi dan Efisiensi Energi Indonesia* or IECES (*Indonesia Energy Conservation* and Efficiency *Society*)
MEMR	Ministry of Energy and Mineral Resources
MEPS	Minimum Energy Performance Standard
MSP	Medium Sized Project
MVE	Monitoring, verification and enforcement
NAMA	Nationally Appropriate Mitigation Action
OJK	*Otoritas Jasa Keuangan* or Financial Services Authority
P3TK EBTKE	*MEMR* (Research and Development of DGNREEC)
PFAN	Private Financing Advisory Network
PIF	Project Identification Form
PIR	GEF Project Implementation Report
PLN	*Perusahaan Listrik Negara* (State Electricity Company)
POPP	Programme and Operations Policies and Procedures
PPG	Project Preparation Grant
PPP	Public-Private Partnership
PT SMI	*PT Sarana Multi Infrastruktur* - Persero (PT SMI)
RAD-GRK	Local Action Plan for Greenhouse Gas Emission Reduction as a supplement to *RAN-GRK*
RAN-GRK	*Rencana Aksi Nasional Penurunan Emisi Gas Rumah Kaca* (National Action Plan to Reduce Greenhouse Gases Emission)
SDGs	Sustainable Development Goals
SE4ALL	Sustainable Energy for All
SNI	*Standar Nasional Indonesia* (Indonesia National Standard)
STAP	GEF Scientific Technical Advisory Panel
TKDN	*Tingkat Kandungan Dalam Negeri* (Local Content level)
TWh	Terawatt-hour
U4E	United for Efficiency
UNDP-GEF	UNDP Global Environmental Finance Unit
UNEP	United Nations Environment Program
USAID	United States Agency for International Development

**Italicized* abbreviations/acronyms and titles are from Indonesian titles with their corresponding English title equivalent

II. DEVELOPMENT CHALLENGE

Context and Global Significance

1. With a population of over 250 million and a growth rate of over 5.8% in 2013, Indonesia is facing several interconnected problems. The country is witnessing a high growth in its energy demand (7% per year) and the national electrical generation capacity is failing to keep pace, resulting in underserved areas and frequent power cuts and brownouts. In 2014, 20% of the population did not have access to electricity, and many others only had restricted supplies. This has a negative impact on entrepreneurship, education, health and safety. In addition, Indonesia's over-reliance on fossil fuels, which accounts for 71% of the country's energy mix, results in rising greenhouse gas emissions.

2. The proliferation of energy efficient and affordable lighting is important to human development because of the significance of lighting in the life of women and men, whether indoor or outdoor in all sectors of the economy. Access to energy efficient lighting (EEL) products and systems is very relevant to national development priorities, global environment and adaptation issues. The proposed project responds to sustainable development goals (SDGs): #7- affordable and clean energy; #9- Industry, innovation and infrastructure; #11- sustainable cities and communities and #12- Responsible consumption and production.

3. *Government Regulation of the Republic of Indonesia No. 70/2009 on Energy Conservation* sets the legal basis for introduction of energy efficiency standards and labeling for electronic appliances. The Ministerial Regulation of the Minister of Energy No.18/2014 came into force in June 2014 and mandates that all compact fluorescent lamps (CFLs) must meet energy efficiency performance standards and must be labeled. The key energy efficiency program of the Ministry includes issuance of energy efficiency regulations, awareness building, energy audits and energy manager program, energy efficiency standards and labeling for appliances and implementation of smart street lighting in cities. Indonesia is also active in the Association of South-East Asian Nations - Standards Harmonization Initiative for Energy Efficiency (ASEAN-SHINE) project regarding performance and testing method for air conditioners, which soon will cover high-efficient lighting as well. The Ministry has set an energy efficiency (EE) target to achieve 1% reduction of energy intensity annually and reduction of GHG emission of 14.71 million tCO_2 annually by 2019.

4. The Directorate General of New-Renewable Energy and Energy Conservation (DGNREEC) of Indonesia was established in August 2010 under the Ministry of Energy and Mineral Resources (MEMR). From March 2011, it began enforcing provisions under Government Regulation 70/2009, which requires that all entities with energy consumption of more than 6,000 tons of oil equivalent per year shall establish an energy efficiency program, appoint an energy manager within the organization, conduct regular energy audits, and implement and report the audit recommendations. DGNREEC established the Energy Efficiency and Conservation Clearing House Indonesia (EECCHI), a facility to promote and share best practices knowledge in energy efficiency and conservation.

5. The ratification of the Paris Agreement through Law No.16/2016 marks a critical turning point for global climate action in the country. Indonesia's NDC submitted to UNFCCC in November 2016 outlined its transition to a low-carbon future, committing to an unconditional emissions reduction of 29 percent by 2030 compared to the business-as-usual (BAU) scenario, which is projected

approximately 2,869 $GtCO_2e$ in 2030; and up to 41% emission reduction subject to availability of international support for finance, technology transfer and capacity building. Under unconditional commitment, energy sector is targeted to reduce 314 $MtonCO_2e$ or 11% of total BAU emission in 2030. Within this target, the energy efficiency programme will contribute through achieving 17% energy efficiency target by 2025 according to the National Energy Plan. Among other efficiency programme, the Ministry of Energy targets to regulate energy consumption level of 12 home appliances considering energy consumption in residential is one of the highest.

6. Considering the above, the Government of Indonesia is proposing a project entitled "Advancing Indonesia's Lighting Market to High Efficient Technologies (ADLIGHT)" for funding by the Global Environmental Facility (GEF) through the United Nations Development Programme (UNDP) and the United Nations Environment Programme (UNEP) which is in line with the above-mentioned energy efficiency national policies and plans. The ADLIGHT project aims to promote the increased deployment of high efficiency lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions. The project is expected to lead to increased market penetration of high efficiency lighting through capacity building for the country's lighting industry, thereby enhancing the local manufacturing capability to locally-produce highly quality lighting systems in line with MEPS, through introducing quality and minimum energy performance regulation, procurement policies and improved customs procedures and through introduction of new business models and awareness raising.

7. The ADLIGHT Project of Indonesia is among ten (10) national child projects that will receive support for the development program using an integrated policy approach through the multi-country arrangement under the "Leapfrogging Markets to High Efficiency Products Program (for appliances, including lighting, and electrical equipment)", also a GEF-supported program, also called as the "Global Leapfrogging Program (GLP)". The GLP contributes to the UN Secretary General's Sustainable Energy for All (SE4ALL) initiative's Lighting and Appliance & Equipment Accelerators. The SE4ALL Global Project has formed a global partnership, named United for Efficiency (U4E), which is a global effort supporting developing countries and emerging economies to move their markets to energy-efficient appliances and equipment. Further, by the end of the SE4ALL project, it will have the commitment from at least thirty developing countries and emerging economies, including Indonesia, to transform their markets to energy efficient lighting, appliances, and equipment.

8. The Global Leapfrogging Program will utilize the tools already developed under SE4ALL Global Project, such as country assessments and best practice policy guides in order to increase the number of countries committing to advance energy efficient products considering their respective national context and needs. Further, as a child project under the GLP, the ADLIGHT Project will benefit from the support to be provided by the GLP e.g. by following the consensus recommendations on the policy framework when developing technical guides and training, such as in MEPS and energy labeling. **Fig. 1** illustrates the relationship among the SE4ALL Global Project, the Global Leapfrogging Program and the ADLIGHT Project.



Fig. 1: Relationship among the SE4ALL Global Project, Global Leapfrogging Program and the ADLIGHT Project of Indonesia among other Child Projects

Thus, for the participation in the SE4ALL Global Project, the Project Concept Note (PCN) for ADLIGHT was submitted by the Government of Indonesia in March 2016 and includes: (a) capacity building on MEPS for local lighting industry, (b) regulatory mechanisms and MVE and (c) high efficiency lighting technology penetration. The PCN is herein being elaborated and developed as the ADLIGHT Project Document for a Full Size Project (FSP) to be proposed with GEF 6 funding.

Barriers to Implementation of Programs in the Local Manufacturing and Application of Energy Efficient Lighting Systems

9. The archipelagic nature of Indonesia, consisting of 18,307 islands (Indonesia National Institute of Aeronautics and Space – LAPAN 2002), is prone to porous borders and allows the local market to be confronted by imported low quality Energy Efficient Lighting (EEL) products. This situation hampers consumer's access to good quality of LED products due to relatively easy access to substandard EELs (oftentimes unknowingly) entering the market in Indonesia. Without Minimum Energy Performance Standard (MEPS) being enforced, people are buying substandard EEL products and miss the benefits of EEL products. Without MEPS, the penetration of good quality of EEL products in the market will face difficulty to compete with the low-priced substandard lighting products. The process to issue MEPS requires national standardization procedures set by the National Standardization Agency (BSN). According to the Government Regulation (PP) number 102/2000 regarding National Standardization, standard development can be approached through 2 different methods; 1) consensus, and 2) scientific evidence. For standards with a development dimension value, such as the MEPS, consensus is likely to be the method used in the process. In creating consensus, approval from all relevant stakeholders is crucial and local manufactures will have a bigger percentage in the overall portion in consensus. Currently, the comparison percentage between local and international EEL brands in 3 associations (GAMATRINDO, APERLINDO and ALINDO) is 76% (local manufacturers) and 24% (imported assemblers). Support to local manufacturers on the principles of MEPS and capacity building for local manufacturers will be crucial in order to realize a comprehensive draft of MEPS.

10. The e-catalogue, an electronic platform that provides information access to goods and services for public procurement in Indonesia through e-tendering and e-purchasing, was created by the Indonesia Public Procurement Agency (LKPP) to increase transparency and accountability of public procurement

in Indonesia. The e-catalogue covers approximately 25% of total public procurement in 2016 (LKPP 2016). The use of the e-catalogue in public procurement is growing steadily since its first launch in 2010, and it was re-enforced under Presidential Regulation number 70/2012 – an upgrade to the previous Presidential Regulation number 54/2010 regarding e-procurement. Despite the practical use of the e-catalogue, which cut significantly the lengthy tendering process in public procurement, the platform still requires improvements. E.g. for public procurement in EEL equipment, the determined specifications are disfavoring local EEL manufacturers. The reason behind the challenge to enter e-tendering is not a matter of quality or technical standards, but more of administrative barriers. The introduction of MEPS should create a level playing field with equal participation of international and local EEL manufacturers.

11. The Government of Indonesia is committed to tap the vast economic and environmental benefits of the transformation to energy efficient appliances and lighting products. Three related projects and initiatives in removing barriers regarding introduction of energy efficient lighting systems consist of the UNDP-GEF BRESL Project, the UNEP's ASEAN-SHINE Project and the UNEP En.lighten initiative (details will be discussed further below in the Baseline Scenario section). These projects and other similar projects and their outcomes altogether have been shaping the country's lighting industry and its markets through the progress of their barrier removal efforts. The lighting industry in the region is evolving rapidly and is moving to a need for further development of EE lighting towards high energy efficient lighting by means of LED products.

    For the BRESL project, the following were some of the significant barriers that were addressed and can be complemented by the ADLIGHT Project, including: (a) low public awareness on energy efficiency and absent consumer knowledge on life time costs (consumers choose the cheapest product which is usually not efficient); (b) low energy price (electricity tariff) due to government subsidy; (c) lack of standards and regulations that incentivize energy efficient lighting production, commercial demand and public procurement; (d) lack of local manufacture and capability to implement EE&C technology; and (e) lack of testing laboratory for energy performance. The achievements of the BRESL project laid the foundation of the standards development, capacity building, laboratory and testing procedures, policies, road map and action plan for efficient lighting system as it started to focus on the CFL transition.

    ASEAN SHINE (2016-2020) aims to support the establishment of the ASEAN Economic Community by removing non-tariff barriers to trade through harmonization of product standards. ASEAN SHINE currently works on standards harmonization across ASEAN for air conditioners and lighting products. In the area of lighting, the achievements include: the establishment of ASEAN SHINE Lighting Technical and Policy workgroup and the agreement of ASEAN Member States to establish harmonized standards for lighting products in the region.

    The UNEP En.lighten initiative focused on the barriers related to the classification and quality status of the numerous EEL products and applications; the monitoring, verification and enforcement (MVE) issues and the tapping of the energy saving potential in the lighting sector. It examined the status of the efficient lighting MVE activities and programs in six ASEAN countries.

    As initially identified by the abovementioned related projects, the barriers to energy efficient lighting deployment have been or are still being addressed and can be complemented by the proposed ADLIGHT Project.

12. A variety of barriers to advancing Indonesia's lighting market to high efficient technologies were identified during the numerous focus group discussions and bilateral meetings during the development of the ADLIGHT Child project under the UNEP-GEF Global Leapfrogging Program:

- *Limited capacity of lighting manufacturers*: Although Indonesia is home to an important number of lighting manufacturers; most local manufacturers have limited capacity to meet standards for energy efficiency and quality, and do not have the technical and financial capacity to innovate, thus reducing their capability to produce affordable, high quality LEDs and meet future MEPS standards. Local businessmen do not seem to be capable of developing business transformation plans for shifting from conventional lighting manufacture to high efficiency lighting systems such as LEDs. Without such plans in place, financial institutions develop a risk-averse attitude and therefore, are not able to fund local lighting manufacturers for upgrading production lines.
- *Absence of MEPS for LEDs*: at the moment, Indonesia does not have any Minimum Energy Performance Standards (MEPS) in place for lighting products other than CFLs. Indonesian consumers will buy any lamp available in the market which usually leads to purchasing low cost, albeit inefficient lamps. At the same time, there is still limited capacity with Customs officers and Ministry of Trade staff to curtail the import of inefficient lamps into the country even as the adoption and implementation of the MEPS for CFLs has already started. As long as low priced - although of low quality - conventional lighting products continue to be distributed, Indonesian consumers still prefer them over LED lighting products. This discourages the manufacturers to produce high energy efficient lighting products.
- *Absence of guidelines for public procurement*: because no guidelines exist for public procurement, street lighting is often quite inefficient in Indonesia. By having a standard minimum quality requirement available in the guidelines for public procurement, consumers, manufacturers and importers of LED lights in Indonesia will have reference in complying with standards requirement, and therefore, the quality of LED lighting to be made available in Indonesia can be rightly monitored, verified and enforced (in an MVE system). With the standard minimum quality that is aligned with international standards, local LED manufactures have the chance for their products to be included for public procurement as listed in the e-Catalogue. According to statistics data of the state-owned electricity company or *Perusahaan Listrik Negara* (*PLN*), the street lighting in Indonesia consumed 3.2 TWh in 2013, which is equal to the emission of 2.6 m tCO_2 eq. In 2014, MEMR piloted smart street lighting in two cities and developed a Nationally Appropriate Mitigation Actions (NAMAs) proposal ("Smart Street Lighting Initiative (SSLI)") that aimed at reducing GHG emissions by increasing the energy efficiency of street lighting systems in Indonesian cities. Application of high efficient street lighting has been one of the priority programs of the Ministry and is in

line with another program on Green City Development in cooperation with the Ministry of Public Works.

- *Absence of regulatory mechanisms for MVE and lack of capacity of testing laboratories*: there are currently no regulatory mechanisms for efficient lighting monitoring, verification and enforcement (MVE) in Indonesia including for mitigation measurement. To date, the Secretariat of *RAN-GRK* (or the National Action Plan to Reduce Greenhouse Gases Emission) with support from GIZ (or the German International Development Agency) and Japan International Cooperation Agency (JICA) has issued the *Guideline for Monitoring, Evaluation and Reporting* under the National and Local Action Plans to Reduce Greenhouse Gases Emission *(RAN-GRK* and *RAD-GRK)* at a broader MEMR ministerial level. This is certainly an initial step towards a measurement, reporting and verification system, but there is currently no institutional capacity for its implementation, let alone the enforcement. On the other hand, the capacity of existing testing laboratories needs to be improved with updated testing standards on MEPS for CFL to check non-compliant products as well as prepare to cater to the desired demand growth of high efficiency lighting systems including the LEDs.

13. Further, in the Logical Framework Analysis (LFA) Workshop conducted in May 22-23, 2017, in Bali, Indonesia during the PPG stage of the ADLIGHT project development, the following concrete problems and challenges were confirmed as the main issues that affect the local manufacture and application of energy efficient lighting systems:

a. LED lamp products in the local market have uncertain quality:
    - High number of complaints on supposed to be high efficient lighting products
    - Lack of standards on EE lighting systems
    - Lack of accredited testing laboratories
    - Accredited laboratory infrastructure and facilities are not established yet
    - No MEPS on LED systems
    - Lack of quality control on EE lighting
    - Regulation and supervision throughout Indonesia is not optimal
    - Low quality and quantity of monitoring of LED products
    - Lack of sharing of international knowledge on EE lighting manufacturing
    - Low budget allocation for monitoring and MEPS development.

Sugandi Pernama, Aziz & Siong (2014). Is mom energy efficient? A study of gender, household energy consumption and family decision making in Indonesia. Energy Research & Social Science 6 (2015) 78-86
http://www.academia.edu/20862146/
Is_mom_energy_efficient_A_study_of_gender_household_energy_consumption_and_family_decision_making_in_Indonesia

http://web.unep.org/sites/default/files/ggeo/ggeo_summary_report_final.pdf

http://iesr.or.id/2017/01/fgd-gender-mainstreaming-in-the-energy-sector/?lang=en

https://www.adb.org/sites/default/files/publication/182935/ino-paper-16-2016.pdf

https://www.energytoday.net/economics-policy/policies/women-energy-economic-empowerment-applying-gender-lens-amplify-impact-energy-access/

Standard annex to project documents for use in countries which are not parties to the Standard Basic Assistance Agreement (SBAA).

May also be designated Project Co-ordinator or Chief Technical Adviser, as appropriate.

b. Lack of clear policy and institutional support to promote manufacture and application of high efficient lighting technology:
    - Lack of market data on production and application of lighting in Indonesia, to inform policy development
    - Lack of support in manufacturing from government
    - Lack of government investments on EE lighting projects
    - Lack of coordination among related ministries and local regional/provincial government
    - Import policy does not support local LED industry
    - Lack of infrastructure to support LED industry growth (logistics, etc.)
    - Lack of studies on cost-benefit analysis on LED to support policy making
    - Limited national and local budget allocation for EE lighting investment
    - Lack of EE lighting implementation incentives from government
    - Lack of harmony among related ministries and agencies regarding LED manufacture and application
    - Lack of legal basis for establishing a coordination mechanisms in LED
    - Government procurement policy is still on least cost approach and the projects are meant for public service and not for making profits (it is not necessarily oriented for life-cycle costing that can justify higher first cost but could be more profitable in the long run)
    - EE lighting cannot be a priority in budget allocation in government procurement as long as procurement regulations are based on least up-front costs instead of life-cycle costs
    - Lack of tax policy support from government for EE lighting investment and manufacture

c. Small market share for locally produced EE lighting products
    - Government market for EE lighting is limited by availability of budget
    - Low absorption of EE lighting systems by government and other users
    - Low preference to select local EE lighting products
    - EE lighting budget line is not prioritized compared to other infrastructure
    - Low uptake from ESCO on EE lighting
    - Benefit from EE lights deployment are not recognized
    - Market confidence is not yet there to use local EE lighting products
    - Absence of updated procurement guidelines containing standard minimum quality. Government procurement uses a reference listing of registered products and suppliers as contained in the e-Catalogue but this reference coding guide consists of imported products mostly since local producers do not easily become a registered supplier in E-catalogue for government projects.
    - Lack of capacity to prepare EE lighting investment projects for public private partnership
    - Not many financing institutions or local banks would like to give loans for energy efficient projects
    - Lack of kWh meters to measure saving in energy and monetary benefits

The Problem Tree resulting from the LFA Workshop is seen in **Annex Q**:

14. These barriers and possible solutions were further elaborated in the ADLIGHT Validation Workshop conducted with stakeholders and relevant agencies and ministries on July 19-20, 2017. Follow-up

focus group discussions (FGDs) on one-on-one basis were also conducted to supplement and update the information and the basis for the detailed project design. These barriers will be addressed by the project towards further resolving them to achieve the outcomes and overall objective of the project.

Baseline Scenario

15. In 2016, Indonesia's on grid electricity consumption was 217.83 TWh. Lighting is estimated to account for 18.8% (ca. 41 TWh) of this electricity consumption in 2016 as shown in **Fig. 2**. Details can be found in **Annex L**.

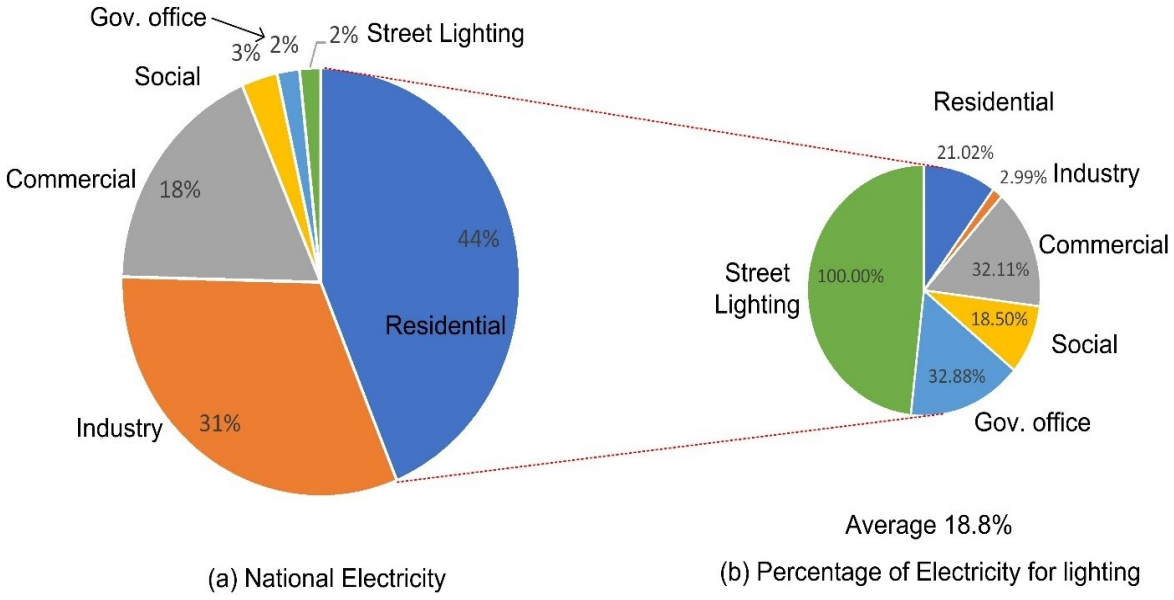


Fig. 2: Estimated Share of Electricity for Lighting in Indonesia for each Sector in Year 2016

16. In terms of EE implementation, the MEMR has allocated budget for programs that involve the implementation of energy efficiency activities such as energy audits, fiscal policies, and regulations focusing ES&L. The MEMR, in close coordination with the Ministry of Finance, has implemented import tax reduction for companies that import high-energy-efficiency lights to the country, in order to provide an incentive for lighting industries in the country to manufacture or to assemble energy-efficient products. Specific activities of these programs will be subsumed into ADLIGHT as baseline activities.

17. Indonesia implemented several projects related to energy efficient lighting during the period 2009 to 2016. The UNDP-GEF BRESL project - Barrier Removal to the Cost Effective Development and Implementation of Energy Efficiency Standards and Labeling - laid the foundation of the standards development, capacity building, laboratory and testing procedures, policies, road map and action

plan for efficient lighting system as it started to focus on the CFL transition. Indonesia was among six countries participating in this project including Bangladesh, China, Pakistan, Thailand and Vietnam. The project has assisted in developing energy efficiency standards and labeling for 6 home appliances including CFL; improved the capacity of testing laboratories so there are at least 4 appointed testing labs for CFL; and built capacity of 5 CFL manufacturers/assemblers to meet the mandatory performance standard. Other results of the BRESL project in Indonesia are the establishment of a database of energy efficient products and implementation of a consumer education scheme. Through the influence of the BRESL project, the government has started to come up with a procurement scheme to go for energy efficient products. The BRESL project also offered training to selected personnel from appliance manufacturing companies regarding ES&L as well as MEPS and enables them to apply the concepts learned in workshops. Such trainings are expected to reduce the barrier related to the inadequate technical know-how in implementing energy efficiency measures when going for ES&L.

18. The Switch-Asia project ASEAN SHINE is implemented under the steering of the ASEAN Member States and has been recognized as a key Dialogue Partner by the ASEAN Ministers of Energy Meeting (AMEM). ASEAN SHINE adopts a holistic approach to market transformation, by using mechanisms including policy, regulations, capacity building along the supply chain, and awareness raising among end-users. The ASEAN SHINE lighting-regional market assessment[2] has been done by International Institute for Energy Conservation (IIEC) in year 2015 in all ASEAN member states. The assessment of Indonesia shows that the residential sector was the largest end-use sector (accounting for 41% of total national annual electricity consumption) for 2013. The electrification ratio in Indonesia in year 2013 was 74% (65 million households). Lighting points per household varies from 8 to 12 lamps. Common lamp types are (a) T8 fluorescent lamps (tube and circular), (b) screw-base omnidirectional: CFLs (5 – 8w), LED (5-9W), incandescent, and (c) pin-base directional: Halogen. The estimated lamp stock is 394 million. Based on the survey, it was shown that the lamps in the residential sector were dominated by CFL (71%), followed by LED (20%) and Incandescent (0.5%) and Halogen (0.5%) lighting.

    In a related development, LED lamps were collected from nine ASEAN countries in December 2016 (including Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand and Vietnam) for an LED benchmarking exercise[3], funded by the Australian Government, as part of the ASEAN SHINE initiative on Lighting. The test covered non-dimmable integral LED general service (non-directional) lamps, 3 -15 watts with normal cap size and color temperature according to local preference. The testing of these products using CIE 5025 test method demonstrated that while good quality LED products are available, significant numbers of poor quality products remain available in the market. The comparative results for Indonesia can be used to benchmark the LED production improvements to suit local and international market demand.

19. In 2014, the UNEP En.lighten initiative performed a survey on lamp sampling in Cambodia, Indonesia, Lao DPR, the Philippines, Thailand and Vietnam, which showed that in Indonesia the most popular wattage for CFL and LED lamps are 8 watts (W) and 7W, respectively. It was observed that there are quite a number of local brands available in Jakarta's lighting market. However, based on on-shelf frequency, international brands still dominate the lighting market in the capital city of Indonesia. For CFL lamps, it is evident that 5W dimmable light (DL) is the most popular model in Indonesia, followed by 8W DL, 14W DL and 18W DL. Philips dominates the Indonesia lighting market, especially in the

[2] ASEAN SHINE lighting-regional market assessment. IIEC.February 2016

[3] ASEAN LED Lamp Benchmarking: Test Results Analysis. August 2017.

capital city and Osram is the second brand. The preferred wattages for majority Indonesia are in the low-wattage range (5-10W), followed by mid-wattage range (11-18W) and high wattages. There are more than 18 different wattages of lamps available in the Indonesian market. The wide diversity of lamps in Indonesia clearly shows the size of Indonesian lighting market. Indonesian people are more likely to purchase lower wattage than higher wattage lamps. This fact is different from other ASEAN countries. On the other hand, for LED lamps, 7W DL dominates majority shelf in Jakarta, followed by 9W DL and 11W DL. Here, Philips dominates the efficient lighting market of Indonesia and followed by Osram and Panasonic. Daylight is the popular color for the Indonesian market.

The UNEP En.lighten initiative also examined the status of the efficient lighting MVE activities and programs in the above-mentioned six ASEAN countries. Through its country lighting assessments in 2010, UNEP En.lighten estimated that Indonesia has a lighting saving potential of 6.32% of total on-grid electricity use. This is equivalent to a saving of 9.35 TWh or 7.17 Mt of CO_2 emissions.

20. The shift to CFL with its MEPS adoption and implementation as initiated by the BRESL Project is a big step in the right direction and is moving to a need for further development of EE lighting towards high energy efficient lighting by means of LED products. The ADLIGHT will provide the critical bridging of gaps for the energy efficient lighting technology development program to materialize.

21. UNDP in collaboration with MEMR of Indonesia is currently implementing a five-year (2017-2021) project called the “Market Transformation for Renewable Energy and Energy Efficiency” (MTRE3) or also referred to as the Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector”. The project’s main objective is to support the design and implementation of appropriate climate change mitigation action in the energy generation and energy end use sectors. In the energy efficiency sector, it is expected that a 20% reduction on Energy Consumption Intensity (IKE) in the commercial building sector, particularly in office buildings, would be achieved and maintained. The project targets a total of 50,000 m^2 of commercial floor space, and could expand to other building floor space up to 500,000 m^2, in addition to the baseline target. In this commercial building sector, lighting systems consume significant shares of total energy consumption. Thus, retrofitting the conventional lamps into energy efficient lamps is one of the solutions to meet the target. In addition, one of the activities to meet the project objectives it to establish a Business Center or ‘meeting point’ for ESCOs and the private sector. The activities in the MTRE3 related to energy efficient lighting and ESCO models are subsumed into, and considered as, among the baseline activities in the ADLIGHT project, as discussed below.

22. In order to provide more efficient street lighting in Indonesia, ADB, the state-owned utility *PT PLN* and MEMR collaborated on implementation of a pilot LED Municipal Streetlight and PLN Substation Retrofit Project (Pilot LED Project). The objective of the Pilot LED Project was to demonstrate reduction in the peak electricity demand in the selected municipalities and pilot PLN substations by focusing on demand side management through street lamp retrofit using relatively new energy-efficient LED technology. The pilot consisted of two municipalities in Central Java (Batang and Semarang) and PLN substation and power generation facilities in West Java province. A total of 1,439 LEDs have been retrofitted. The post-retrofit meter reading shows that the energy savings from two municipalities, Batang, Semarang, and PLN substation are up to 59%, 44%, and 44%, respectively. The investment payback period, where cost saving for lower maintenance of LEDs is taken into account, are 3.6 years for Batang and 7.0 years for Semarang, respectively. The significant longer payback period for Semarang was due to the increased cost of LEDs lamps needed to meet the city’s unusually high light level requirements and the addition of control technology in the lamps. This

project has addressed four significant challenges and recommendations explained as follows. First, municipalities are often not willing or able to implement LED retrofits due to lacking benefit from the energy savings. This is because, in general, PLN does not bill the municipalities for street lighting based on actual electricity consumption but on a flat fee (lump sum) basis which is not related to actual electricity consumed, even where *PLN* meters are installed. In fact, street lighting energy consumption is usually not metered. The second challenge is limited knowledge of LEDs and capacity to retrofit, operate and to maintain LEDs. The recommendation for this is to develop a national standard on street lighting that addresses LED technology and incorporates the best practices and lessons learned in the project. The Ministry of Transportation of Indonesia is currently drafting a regulation on street lighting with ADB support. The implementation of multiple demonstration LED projects with different retrofit and financing structures and nationwide capacity-building are also recommended to increase understanding and to build capacity within municipalities. The third challenge is the absence of sufficient municipality budgets and access to financing for citywide LED retrofit programs. It is recommended to develop a new financial product that allow financial institutions and third-party entities like ESCOs to provide the needed long-term financing such as national credit guarantee to back up loans, capital cost and insurance of the street lighting retrofit program. Finally, the fourth challenge is difficulty of municipalities and government agencies to be able to use ESCO services. This is due to absent formalized regulation to support multi-year contracts with ESCOs, government's procurement procedures based on least cost selection instead of life cycle cost, and budget disincentives.

23. The ADLIGHT Project will build upon relevant ongoing EE lighting projects/programs in the country and draw lessons including from those already completed and ongoing. The following were identified during the PPG phase that could be relevant to ADLIGHT, which are mostly in the application of local efficient lighting products:

Project Name - Sponsor	Areas of Interest/Status	ADLIGHT Additional Outputs*
Market Transformation through Design and Implementation of Appropriate Mitigation Actions in the Energy Sector (MTRE3)-UNDP	The project will put in place enabling environment and removing barriers to sustainable market of renewable energy and energy efficiency. To achieve this purpose, the project will focus its interventions in renewable-based electricity generation and energy efficiency in commercial building interventions. The project will consist of 3 main components: (1) Climate change mitigation options for the RE-based energy generation and energy efficiency; (2) Market transformation through implementation of appropriate mitigation actions; (3) Measurement, Reporting and Verification (MRV) system and national registry for mitigation actions - Ongoing 2017-2021	• Market transformation to energy efficient lighting with enhancements on local EEL production capability • Robust policy and guidelines on EEL production and utilization in buildings and street lighting application • Strengthening in MRV and MVE systems with adoption of SNI/MEPS and testing laboratories to support them
ASEAN SHINE	Standards harmonization across ASEAN for air conditioners and lighting products in support of the ASEAN Plan of Action for Energy Cooperation (2016-2025)	• MEPS and energy labels in place for high efficient energy lighting products

Project Name - Sponsor	Areas of Interest/Status	ADLIGHT Additional Outputs*
UNEP en.lighten Initiative	Status of the efficient lighting MVE activities, programs in the selected six ASEAN countries, including Indonesia, the classification and quality status of the numerous EEL products and application, benchmarking and tapping of the energy saving potential in the lighting sector	• Regulatory mechanisms for efficient lighting monitoring, verification and enforcement (MVE) *
Super-Efficient Equipment and Appliance Deployment (SEAD) Initiative - Clean Energy Ministerial and the International Partnership for Energy Efficiency Cooperation	Aiming to make it easier for governments and the private sector to capitalize on opportunities regarding appliance and equipment (including lighting) efficiency to reduce energy demand and carbon emissions while lowering energy costs for consumers, businesses, and institutions - Ongoing	• Policy and guidelines for public procurement of LED lighting products (residential, commercial and outdoor) developed and adopted
Indonesia Clean Energy Development (ICED) Phase II - USAID	Establish an effective policy, regulatory and incentive environment for low-emission growth in the energy sector, while simultaneously attracting public and private sector investment in clean energy development. Contributing to the Government of Indonesia's (GOI) targets for increasing access to energy, while concurrently supporting national efforts to curb GHG emissions. Focus on Energy Sustainable Finance, Previous project on EE Street Lighting Pilot in Makasar, ICED in collaboration with GIZ SSLI – Ongoing 2015 -2020	• Capacity development for policy makers, enforcement &custom officials and other relevant government agencies on market control procedures
AFD EE street Lighting - Agence Française de Développement (AFD)	Contribute to Indonesia's efforts in the fight against climate change (appropriate infrastructure development, establishment of dedicated financial instruments). Relevant project in EE Street Lighting in Bogor and Mataram – Ongoing at a preliminary stage	• Regulatory mechanisms for efficient lighting monitoring, verification and enforcement (MVE) including testing standards defined and implemented
LED Street Lighting Project in Indonesia – Asian Development Bank	Implement LED Retrofit Project in Batang and Semarang (Pilot Cities) and several power stations of State Owned Utility (PLN) - Completed 2016.	• Pilot demonstrations for accelerated LED lamp deployment in buildings and for street lighting in the context of sustainable cities
Smart Street Lighting Initiative (SSLI) NAMA Indonesia - Die Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)	Program to increase energy efficiency of SL through implementing efficient technology (lamp, armature, control system, etc) focusing on study in 22 cities and pilot in several cities – piloting completed in 2014; development phase (2015-2017); transformation phase (2017-2019)	• Pilot demonstrations for accelerated LED lamp deployment in buildings and for street lighting in the context of sustainable cities
Environmental	Developing inclusive and sustainable growth	• innovative financial mechanisms

Project Name - Sponsor	Areas of Interest/Status	ADLIGHT Additional Outputs*
Support Program Phase 3 (ESP3) - Danish International Development Agency (DANIDA)	through improved environmental management and climate change mitigation and adaptation. Supports in implementation and monitoring energy efficiency, energy conservation and renewable energy including pilot projects that will provide demonstration and experience of energy efficiency, energy conservation and renewable energy in Indonesia - Ongoing	enabling the accelerated penetration of advanced lighting systems, including ESCO models
Green Banking - *PT Sarana Multi Infrastruktur* - Persero (PT SMI) in partnership with MTRE3 Project	Green infrastructure bank transforming into an infrastructure bank to help fund roads, railways and seaports with private backing and interested in street lighting financing - Ongoing	• Business transformation plans of selected local lighting manufacturers to convert their production lines to the production of more energy efficient LED lamps
Synergy Efficiency Solutions (SES) – Private Company	Assists companies in the implementation of savings solutions by minimizing investment costs through its ESCO financing options focusing on energy efficiency, including EE Lighting – On going	• technical assistance program for selected local EEL product manufacturers to upgrade manufacturing facilities
Promoting Industrial Energy Efficiency through System Optimization and Energy Management Standards in Indonesia – UNIDO/GEF	Promoting industrial energy efficiency through system optimization approach and introduction of ISO energy management standards – For completion in December 2017	• business transformation plans of selected local lighting manufacturers to convert their production lines to the production of more energy efficient LED lamps
Private Financing Advisory Network (PFAN) – USAID Climate Technology Initiative (CTI) and UNFCCC Expert Group on Technology Transfer	Bridge the gap between investments and clean energy businesses. Identify promising clean energy projects at an early stage and provide mentoring for developing a business plan, investment pitch, and growth strategy, significantly enhancing the possibility of financial closure for the projects – Completed 2012-2014	• Implemented awareness and promotion program and information system explaining the benefits of high energy efficient lighting technologies

*Almost all ADLIGHT outputs are commonly applicable to and supplementing the results of the baseline projects listed above through the barrier removal activities of ADLIGHT in pursuit of its goals and objectives.

24. According to the definition of the Ministry of Trade, a local manufacturer is an entity producing products that contain at least 40% locally sourced components. The local LED lighting industry in Indonesia is represented by three major industry associations: GAMATRINDO, APERLINDO, and ALINDO. The following table describes number of EE lighting local manufacturers under the three LED industry associations (GAMATRINDO, APERLINDO and ALINDO) with the 2016 production of LED lamps and average efficiency (lumens per watt) both indoor and outdoor types.

	Total LED Industry under the Three Associations in Indonesia
Total no. of members	68

	Total LED Industry under the Three Associations in Indonesia
No. of members in • Indoor light • Outdoor (Street Light)	 68 6
2016 Production • Indoor light • Outdoor (Street Light)	 98.2 million LED bulbs 51.8 thousands LED-SL
2016 Average Efficiency, lumens per watt • Indoor light • Outdoor (Street Light)	 70 - 90 100 – 110

Notes:

1. *GAMATRINDO* has ten member manufacturers, all of them produce indoor light lamps and only 1 produces outdoor (streetlight)
2. *APERLINDO* has 48 member manufacturers, all of them produce indoor light lamps and only 1 manufacturer produces outdoor (streetlight)
3. *ALINDO* has ten member manufacturers, all of them produce indoor light lamps and only 1 manufacturer produces outdoor (streetlight).

Based on data collected from three associations of local EE lighting manufacturers, in the last 8 years from 2007 to 2015, the production of in-door lamps increased by 4% per year for production without LED and by 7% per year for production with LED. By 2015, total in-door lamp production without LED lamps from domestic lighting companies reached 360 million lamps which was still dominated by CFL with a share of 63.6% followed by LED (18.2%), fluorescent lamps (17.1%) and incandescent bulbs (1.1%). The production of CFL lamps, however, increased more than 3 times from 100 million lamps in 2007 up to 340 million lamps in 2014 although in the following year (2015) there was a decrease of 50 million lamps. This seems to be in line with the development of LED lighting technology.

In general, the following table shows significant changes in production patterns as a reflection of market changes or consumer changes in lamp utilization. In addition, two developments are particularly noteworthy: the drastic decline in production of incandescent and fluorescent lamps and the production increases for CFL and LED light bulbs.

Production capacity of locally produced electric lamps in Indonesia[4]

Year	Indoor (type of lamps), number of lamp ('000)				Total indoor ('000)		Out door (Street lighting), number of street lighting (PJU)	
	Incandescent	Fluorescent	CFL	LED	Total without LED	Total with LED	Non-LED Street lighting	LED Street lighting
2007	90,000	65,000	100,000	-	255,000	255,000	NA	NA

[4] Data is aggregated from three Association of EE local manufacturers namely *APERLINDO*, *ALINDO* and *GAMATRINDO*. Unfortunately this data is still temporary because not all data from the three associations are derived from all its members, e.g. data from Alindo is only obtained from one of its 10 member companies (*PT Solaren Ledindo*). Therefore, in the implementation of ADLIGHT, more detailed and accurate data are needed in Component 1 activities, especially with regards to market share periodic activity.

2008	70,000	75,000	120,000	-	265,000	265,000	NA	NA
2009	60,000	75,000	160,000	-	295,000	295,000	NA	NA
2010	50,000	75,000	200,000	-	325,000	325,000	NA	NA
2011	40,000	75,000	260,000	-	375,000	375,000	NA	NA
2012	40,000	75,000	320,000	-	435,000	435,000	NA	NA
2013	25,000	75,000	330,000	15,000	430,000	445,000	NA	NA
2014	10,000	80,000	340,000	40,000	430,000	470,000	NA	5,938
2015	5,000	75,000	280,000	80,002	360,000	440,002	NA	4,321

Source: Aggregated from three Associations of EEL local manufacturers

III. STRATEGY

Project Rationale and Policy Conformity

25. The ADLIGHT project aims to promote the increased deployment of high efficiency lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions. The project is expected to lead to increased market penetration of high efficiency lighting through capacity building for the country's lighting industry, thereby enhancing the local manufacturing capability to locally-produce highly quality lighting systems in line with MEPS, through introducing quality and minimum energy performance regulation, procurement policies and improved customs procedures and through introduction of new business models and awareness raising. The project proposes to have an innovative approach that covers assistance to the upstream portion of the LED value chain, across the midstream of delivery and marketing in to the downstream application of the LED products. Thus, the theory of change includes consideration of internal factors (relating to project design and implementation) and external factors (relating to other partners, stakeholders and the entire value chain participants) that will be critical for achieving the expected change in increasing the national market penetration of affordable, high quality LEDs. Current achievements in the state-of-the–art of the LED technology will support the overall development strategy, including inputs and findings from evaluation and other credible research. With ADLIGHT being aligned with the GEF-supported programmatic global projects, i.e. the Global U4E Project and the Global Leapfrogging Program being implemented by UNEP, the proposed project will gain the necessary working linkages as well as knowledge, good practices and lessons learned to be shared from said multi-country projects. Likewise, previous work and ongoing projects by UNDP, UNEP and others, in this country and in other relevant regional contexts, e.g. ASEAN will add to the available resources for ADLIGHT. The ADLIGHT project should carefully examine the dynamics of the entire LED industry from production to application of the LED lamps.

26. The proposed project focusing on lighting is in line with the Indonesia's NDC submitted to UNFCCC in November 2016. Under unconditional commitment, energy sector is targeted to reduce 314 MtonCO2e or 11% of total BAU emission in 2030. Within this target, the energy efficiency programme will contribute through achieving 17% energy efficiency target by 2025 according to the National Energy Plan. Among other efficiency programme, the Ministry of Energy targets to regulate energy consumption level of 12 home appliances considering energy consumption in residential is one of the highest.

    In the energy sector, the Indonesian Energy Law 30/2007 and Presidential Regulation 22/2017 on National Energy Plan renewable energy is targeted to contribute 23% of the primary energy mix in 2025 and energy efficiency in industries, buildings and households is expected to save 17% of the total primary energy needed in 2025. In addition, the Government of Indonesia is targeting 100% electrification by 2020, with special annual budget allocation for Rural Electrification Program. Efficient lighting will play a key role in meeting these targets.

Alternative Scenario

27. The overall goal of the ADLIGHT project is to reduce the growth rate of annual GHG emissions from lighting energy use by means of reducing electricity consumption in the energy end-use sectors. This is done with a gender-responsive approach, responding to the needs and capacity of women and men.

    The project objective of ADLIGHT is to increase the penetration of high quality energy-efficient lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions. Although LED penetration is gradually increasing in Indonesia, these are often low quality products thereby harming the consumer confidence in LED. The specific situation of Indonesia being an archipelago leads to the situation where it is very easy for low quality products to enter the market. The situation in Indonesia is therefore still challenging in terms of creating a market for high quality LED lighting that is at the same time affordable to consumers and progresses gender equality.

    From the barrier analysis described above, the key barriers that need to be addressed are the following:

    - Limited capacity of local lighting manufacturers to produce affordable, high quality energy efficient lighting which meet future MEPS and can lead to higher penetration of these products
    - Lack of clear policy, regulatory mechanisms and institutional support to promote import, manufacturing and application of high efficient lighting technology
    - Limited demand and available business models for energy efficient lighting products resulting in a small market share for locally produced EE lighting products
    - Lack of gender-analysis on the LED lighting market in Indonesia, and therefore lack of gender-sensitive approaches within the market.

    The project builds on the NDC commitment of Indonesia and proposes an alternative scenario by supporting Indonesia to advance its market to energy efficient lighting technologies. It also builds upon the proven integrated approach that has been successfully developed and introduced during the implementation of the GEF-financed UNEP implemented En.lighten project and together with results of other earlier initiatives will guide the country to successfully transition to a greatly expanded EE lighting market. The project design is closely following the Concept Note which was earlier agreed on by the MEMR as the Implementing Partner and UNEP, brought about by the SE4ALL Global Project and the Global Leapfrogging Program as described above. Correspondingly, the ADLIGHT approach comprises the following strategies, each addressing the above-mentioned key barriers to this transition in attaining the objective, consistent with the theory of change discussed above:

    Support to local lighting industry to improve the quality and efficiency of lamps and ballasts: The project will collaborate with the Indonesian lamp manufacturing associations to provide support to Indonesian lighting manufacturers to increase their capacity to produce high quality energy efficient lighting that can meet MEPS designed and implemented as part of the project. The strategy focuses mainly on increasing the capability of local manufacturers to expand their production lines e.g. by making them capable of developing good business transformation

plans and submitting good bankable proposals to local commercial banks. Since the local manufacturers need financial resources to upgrade their production lines, the proposed project will help them to develop high quality business transformation plans. The project will further establish partnerships with the local banks to foster investment in high efficiency lighting production in Indonesia.

Regulatory mechanisms and market monitoring, verification, and enforcement (MVE): The project will capitalize on the initiatives in place and support MEMR to develop and implement minimum energy performance standards (MEPS) in coordination with the Ministry of Industry (MoI) and the national standards organization regulatory mechanisms. The ambition is to have these standards harmonized within a regional approach. The project will also support Indonesia in developing a well-functioning system of monitoring, surveillance, control, and testing facilities for effective implementation of the MEPs and improved public procurement procedures to higher uptake of LED lighting. Activities under this component will closely coordinate with the ongoing efforts of as far as monitoring and verification are concerned. When it comes to the enforcement of testing standards for high efficiency lighting systems, the project will work closely with the lamp manufacturing associations and respective local industries to educate them on the applicable national and regional standards to be followed in terms of quality and efficiency and enforced by the relevant government agencies and customs authorities in the numerous ports that is characteristic of the unique porous archipelagic situation of the country. This will ensure products comply with MEPS and reduce the number of non-compliant products entering the market to an absolute minimum in collaboration with Ministry of Industry and Ministry of Trade.

High efficiency lighting technology penetration: The proposed project will study available financial mechanisms and explore for their suitability in the project. The pilot projects will concentrate on the ESCO business models (both guaranteed and shared savings model) as well as develop guidelines for public and private procurement procedures as a gateway to higher uptake of efficient lighting products (residential, commercial and outdoor) in support of the most beneficial business approaches. Since there is a large scope for reducing the energy demand for public lighting, the project will closely work with selected cities and municipalities to scale up the smart street lighting initiative. The technical and financial assistance that the project will provide to building owners and city governments in the demos will aid in their conversion to efficient street lighting. These will involve such upfront costs in formulating and initiating the EE program that usually become barriers affecting possible EE investment initiatives in view of the potential benefits that have been identified. The project will remove these barriers and pave the way to follow this developmental route for more efficient and sustainable operations. Once these developmental concerns are addressed, the EE lighting program will be put in place for MEMR to follow through and sustain through the effective implementation of the replication plan that will be initiated within the project period. The project will take a gender-responsive approach to consultations with relevant stakeholders and the public in order to consider the needs, particularly of women, related to the usage of energy efficient lighting technologies and systems. Sex-disaggregated data will be gathered whenever possible. Further, the project will develop communication campaigns, distribution campaigns, and demonstration projects to raise awareness on the benefits of high efficiency lighting considering gender specific aspects to increase the effectiveness of communication. The project will further implement pilot projects with the Green Cities project and the Local Government

Association to introduce high efficient street lighting. Innovative financial mechanisms will be built around approaches such as ESCO models to stimulate the deployment of efficient street lighting in wider Indonesian cities. While the project will focus on commercial buildings and street lighting in most of the demonstration and replication activities, it also recognizes the big potential of the high efficiency lighting systems in the residential applications which will need different approaches for the necessary promotion and market penetration. The project will focus on developing and applying business/financing models through ESCOs that can be relevant in commercial buildings, street lighting, whereas the residential market will be addressed in the project's demonstration, communication and marketing campaigns to bolster overall EEL market penetration.

The ADLIGHT project design strategy is based on the results of the Logical Framework Analysis (LFA) Workshop conducted by UNDP for the stakeholders from the lighting industry of Indonesia and involved key agencies in the form of stakeholder consultation using the cause-and-effect "problem tree" analysis methodology and then transformed to an "objective tree" that became the basis of the ADLIGHT Logical Framework (or Log Frame). The LFA workshop which was conducted on May 22-23, 2017 was supplemented by one-on-one focus group discussions (FGDs) conducted by the Team with the stakeholders to continue discussions and update the information to complete the ADLIGHT Log Frame and list down the project activities under each Output and Outcome.

Fig. 3 shows the project theory of change illustrating how the three strategies (three project Outcomes) lead to addressing the problems and realizing the three main Outcomes towards the achievement of the Project Objective and Goal. The Project Results Framework in **Section VI** presents a more complete, integrated and purpose-driven theory of change in tabular form with the success indicators, baselines, targets, means of gauging project success and assumptions. UNDP's responsibility extends only to Component 1 and Component 3 of the Theory of Change while UNEP is responsible for Component 2. The Components described in this ProDoc therefore only reflect the activities under Component 1 and Component 3. Component 2 is under UNEP as GEF Agency and captured in their Project Collaboration Agreement signed with the government of Indonesia.

28. In line with the above interrelated strategies that will result in the desired outcomes, the various project outputs that will bring about the ADLIGHT Alternative Scenario are composed of the following:

<u>Outcome 1: Improved quality, energy efficient and affordable locally-produced EEL products and systems.</u>

The focus is on increasing the capability of local manufacturers to expand their production lines, with gender considerations, e.g. by making them capable of developing good business transformation plans and submitting good bankable proposals.

- Output 1.1: Establishment of knowledge center and systems that helps manufacturers in their production planning and policy makers in reviewing enabling environment

- Output 1.2: Adopted and implemented business transformation plans of selected local lighting manufacturers to produce high quality energy efficient lighting which meet future MEPS

- Output 1.3: Completed capacity development program for banking/financial institutions on the evaluation and financing of lighting industry modernization projects

Outcome 2: Improved conditions for fair market competition of EE lighting products, informed by robust policy and institutional framework - *Component 2 is under UNEP as GEF Agency and captured in their Project Collaboration Agreement signed with the government of Indonesia*

Outcome 3: Increased penetration of high quality and efficient lighting

This component will focus on determining the most appropriate among the many available financial mechanisms referring to the various experiences abroad and existing instruments in the country. Based on the initial studies, there is a critical need to have pilot demonstrations that would concentrate on the ESCO business model and the big potential for building and street lighting application of LEDs to realize increased EEL market share.

- Output 3.1: Development of an innovative financial model enabling accelerated penetration of advanced lighting systems, focusing on the development of ESCO business models

- Output 3.2: Pilot demonstrations for accelerated LED lamp deployment in buildings and for street lighting in the context of sustainable cities as well as in residential sector

- Output 3.3: Implemented awareness and promotion program and information system explaining the benefits of high energy efficient lighting technologies, taking into account gender specific aspects in developing and implementing the programmes.

The barrier removal activities in each of the ADLIGHT project components comprise the incremental activities of the proposed project that will either be fully or partly funded by GEF resources. All these GEF-funded incremental activities will supplement the baseline activities under the EEL program of the country particularly focusing on the relevant LED manufacturing and application projects by the partners that will be subsumed under ADLIGHT. Altogether, the combined incremental and baseline activities are expected to result in the realization of the expected project outcomes.

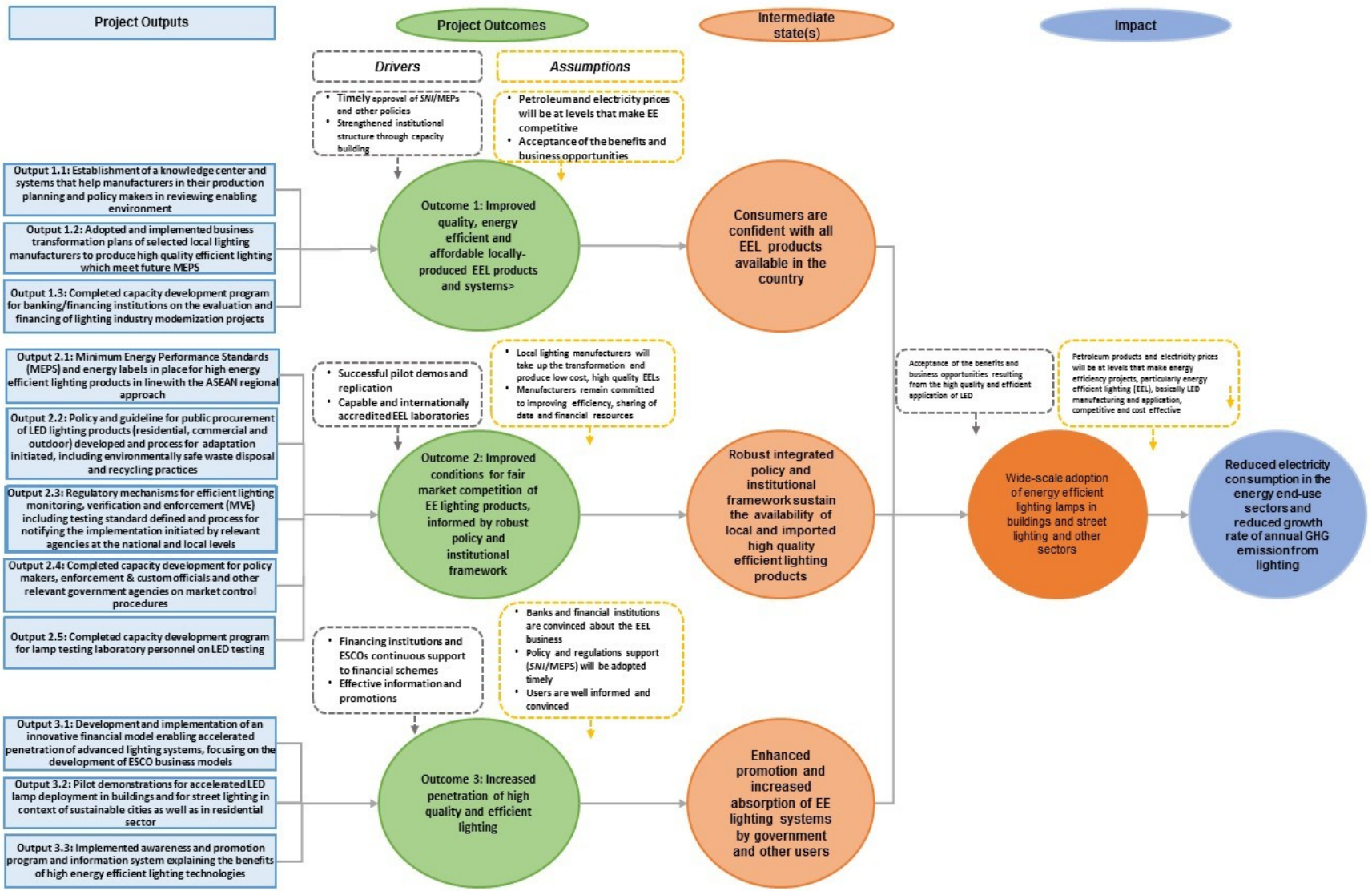


Fig. 3: Theory of Change: Advancing Indonesia's Lighting Market to High Efficient Technologies (ADLIGHT) – - *Component 2 is under UNEP as GEF Agency and captured in their Project Collaboration Agreement signed with the government of Indonesia*

OUTCOME 1: Improved quality, energy efficient and affordable locally-produced EEL products and systems.

Table 1 presents major Activities for delivering the planned Outputs that will contribute to the achievement of Outcome 1.

Table 1: Activities that will deliver the Outputs resulting to Outcome 1

OUTPUTS	ACTIVITIES
1.1: Establishment of knowledge center and systems that helps manufacturers in their production planning and policy makers in reviewing enabling environment.	1.1.1: Identification, design and establishment of an Institute to support energy efficient lighting transformation in Indonesia 1.1.2: Periodic (inception, mid-term and terminal) gender sensitive market surveys of EEL production and application in Indonesia 1.1.3: Development of LED lighting development roadmap for Indonesia, with gender equality goals included 1.1.4: Review and development of recommendations for government policies to provide level playing field to manufacturing of high quality LEDs
1.2: Adopted and implemented business transformation plans of selected local lighting manufacturers to produce high quality energy efficient lighting which meet future MEPS	1.2.1: Gender-responsive assessment of needs and development of capacity building to prepare local EEL manufacturers to upgrade production facilities to produce high quality, efficient lighting that can meet future MEPS 1.2.2: Development of gender-responsive business transformation plans of local EEL manufacturers to produce high quality LED lamps that can meet future MEPS requirements. 1.2.3: Organization and conduct of workshops and business forums for local manufacturers for the adoption and implementation of the business transformation plans, with gender-sensitive capacity building components.
1.3: Completed capacity development program for banking/financing institutions on the evaluation and financing of lighting industry modernization projects	1.3.1 Design, planning and conduct of training for financing institution officials on assessment of financial proposals on energy efficiency, focusing on production upgrades of lighting manufacturers, while specifically addressing gender-gaps in the market.

The following paragraphs describe the Activities that will be conducted to deliver the expected Outputs to contribute to realization of Outcome 1.

- **Output 1.1: Establishment of knowledge center and systems that help manufacturers in their production planning and policy makers in reviewing enabling environment.**

Activity 1.1.1: Identification, design and establishment of an institute to support energy efficient lighting transformation in Indonesia. This will involve the establishment of an Efficient Lighting Knowledge Center for collecting market information on lighting products, quality of products, consumer perception, including gender-disaggregated consumer data where available and other related information regarding efficient lighting technology and technology providers. Existing agency with relevant mandate and organization, such as *BSN*, will be assisted in preparing a strategy and design to meet the requirements and purpose of the Knowledge Center and in establishing it in the selected agency that will continue to be responsible in carrying out a sustainable program ushered by

the project in providing not only knowledge but also a categorized quality product listing (QPL) and database in cooperation with a network of qualified laboratories. Collaboration with LINTAS EBTKE (Information and Investment Service Center) at the DG-NREEC will also be pursued under this Activity to facilitate wider dissemination of information and investment potential in energy efficient lighting sector. The Knowledge Center to be established within the selected existing agency in Indonesia will complement to the virtual Centre of Excellence and other relevant mechanisms under the Parent Program, with the title: "Leapfrogging Markets to High Efficiency Products" (GEF ID: 9436) as implemented by UNEP in coordination with the partnerships established in the Parent Program involving international organizations and private sector companies.

GEF support will be provided for the development and implementation of the Knowledge Center.

Activity 1.1.2: Periodic (inception, mid-term and terminal) gender sensitive market surveys of EEL production and application in Indonesia. This activity will involve the conduct of market surveys and analysis to ensure availability of critical market information to support the local LED lighting manufacturers' information needs in production and application of LEDs as well as to inform the development of a LED lighting roadmap for Indonesia. The periodic market assessment and consumer surveys of EEL production and application in Indonesia includes the gathering and analysis of market data, such as gender-sensitive consumer preferences, trends in market prices and the availability of EEL products and applications including market share of both local and imported EEL lamps manufacturers. The market survey will gather information directly from consumers about their preferences, habits, their purchasing power and other important experiences especially with locally produced EEL, factoring in gender considerations, that will provide local EEL manufacturers with insight about their target customers, such as how much money they spend on certain types of products, product preferences and the interest level for new products. On the other hand, data about EEL products and market share, production facilities and desire to upgrade, or perception of future demand, or other related information will be collected through surveys among manufacturers. In the design of the surveys, the needs of the government for such information shall also be incorporated so that the results of the surveys in terms of generic data gathered will be useful for government policy formulation and decision making.

During the implementation of ADLIGHT, the market assessment and consumer surveys will periodically be undertaken on 3 stages: the first being a baseline survey and the other two are aimed at monitoring project outputs and outcomes in line with mid-term and terminal project evaluations. For the first survey, at the beginning of first year (2018), the baseline market survey aims to get the existing data of EEL production and application in Indonesia. In addition, baseline market survey consists of two activities (a) industry scanning of local lighting industries and market survey of efficient lamps in Indonesia and (b) assessment of physical production facilities and needs in producing efficient and cost-effective LED products and systems per market demand. For the second survey, the market survey will be conducted in mid-term (2020) in order to assess the achievement of the project targets after two years of implementation. For the third survey, an end-of-project market survey (2021) is aimed to evaluate overall ADLIGHT project implementation through which all the outputs and outcomes are consistently achieved towards the overall project goal and objective. The range of the surveys, the list of respondents, the data sets and other requirements to be included in the survey designs can be determined in more detail at the start of the project implementation based on the needs analysis that will be conducted at the outset.

The periodic surveys will be undertaken by ADLIGHT to fully assess the market situation of the local energy efficient lighting in selected cities in general and specifically in city governments, state-owned bank and government buildings and selected residential areas who participated as the hosts in pilot demonstrations. The surveys will involve traditional markets, outlets, minimarkets, malls, residences and the three local EEL manufacturer associations, namely *GAMATRINDO*, *ALINDO* and *APERLINDO*. This activity will also integrate with Activity 1.2.2 on dissemination of the benefits to shift from conventional lighting, including compact fluorescent lamps (CFLs), to LED lamps through local smart lighting corners in hyper markets. The survey results and synthesis of market situation will be provided to the EEL manufacturers to assist them in their own market assessment and production planning.

GEF support will be provided for these three market surveys and synthesis of results on high efficient lamps market in selected cities and the pilot hosts.

Activity 1.1.3: Development of LED lighting development roadmap for Indonesia, including gender equality goals. On the basis of market data collected in Activity 1.1.1, complemented with national statistical data, a modeling exercise and consultation with relevant stakeholders, an LED lighting development roadmap for Indonesia will be developed as agreed among relevant ministries and agencies. In the development of the roadmap, potential gender and environmental impacts from manufacturing, installation, maintenance, operations and disposal of LED lighting technologies and systems will be considered. The development of MEPS and energy labeling in activities under Component 2 (UNEP) will be aligned with the LED lighting roadmap, which will be updated along with the three phases of data collection under Activity 1.1.2.

GEF support will be provided for the LED roadmap development and consultation meetings during the development of the roadmap.

Activity 1.1.4: Review and development of recommendations for government policies to provide level playing field to manufacturing of high quality LEDs. This activity will involve policy studies and recommendations on appropriate import duties on assembled and components of LEDs, CFLs, etc., taxation of lighting manufacturing, and other related policy matters that impact the competitiveness of manufactures vis-à-vis imports.

GEF support will be provided for the conduct of policy studies and consultation meetings leading to the recommendation of policy measures to relevant government bodies.

- **Output 1.2: Adopted and implemented business transformation plans of selected local lighting manufacturers to produce high quality energy efficient LED lamps that can meet future MEPS**

Activity 1.2.1: Gender-responsive assessment of needs and development and implementation of capacity building to prepare local EEL manufacturers to upgrade production facilities to produce high quality, efficient lighting that can meet future MEPS. This activity will focus on increasing the capability of local manufacturers to expand their production lines by making them capable of developing good business transformation plans and submitting good bankable proposals. This activity will involve evaluation of the existing EEL production facilities of local manufacturers in order to upgrade their manufacturing facilities, production techniques and the capacity of human resources to handle the high technology equipment to produce affordable, high quality EEL products. This

assessment consists of a series of activities such as (a) assessment of human resources capability to handle high technology equipment, including assessment of women's and men's needs and capabilities for safe and gender equal workplaces; (b) organization and conduct of a workshop for a broad range of local manufacturers on the potential of EEL and upgrading production facilities, ensuring that women staff active participation is encouraged and promoted; (c) assessment of production techniques of the selected local EEL manufacturers to design and produce high EEL systems including LED and (d) assessment of quality assurance needs and how to provide needed technical assistance in local EEL production.

This activity will involve in the needs assessment process, the participation of selected EEL local manufacturers (under three above-mentioned LED industry associations), DGNREEC – MEMR, National Standardization Agency of Indonesia (*BSN*), Ministry of Trade Ministry of Industry and other relevant agencies, including gender equality-focused institutions. Based on the needs analysis conducted, manufacturers will (a) undergo a series of three progressive trainings on the potential and benefits of EEL and possibilities for upgrading the production facilities in their factories, (b) provide and share utilization of common service utilities, such as die cast moulding for LED street light housing, and (c) receive technical assistance on LED global standard adoption in line with ASEAN approach to local EEL industry.

GEF supports the travel expenses for the field trips, selected local EEL manufacturers, workshops/FGDs and needs assessment, capacity building to handle the high technology equipment, series of three trainings to disseminate the potential and benefits of energy efficient lighting production to all local manufacturers and other related activities.

Activity 1.2.2: Development of gender-responsive business transformation plans of local EEL manufacturers to convert their production lines to the production of high quality LED lamps that can meet future MEPS requirements. This activity is designed to develop business transformation plans of local EEL manufacturers to convert their production lines to the production of high quality, energy efficient LED lamps. This includes the preparation of feasibility studies (FS) on LED business transformation plans at the individual company level as well as the industry integration at the program level. These preparatory works for the gender-responsive transformation plans will be done by the experts to be hired by the project together with an active participation of and consultation with both female and male staff of the local EEL manufactories. As part of the expert study, a selective technical field visit can be organized (benchmarking) in EEL industry abroad, particularly on how the manufacturers have coped with global standards (MEPS) adoption. At the individual company level, the project will provide technical assistance by hand holding the local manufacturers in developing business transformation plans that will be of sufficient high quality so that financial institutions will be able to assess related loan requests in accordance with agreed bank standards and development needs. A training/workshop is to be developed and conducted by the project experts with the required modules on how to develop business plans and what (financial) information to include in a loan request and other pertinent information. Local manufacturers female staff participation in the training/workshop will be particularly encouraged. As the local manufacturers learn individually from this activity and establish their confidence on the transformation program, the confidentiality of business plans will be followed as this issue surfaced as a major hesitancy by the manufacturers. At the industry level, the overall program sustainability will be tackled. At the aggregated level, the consultant will provide guidance and technical assistance in developing an EEL business

transformation program and sustainability plan for use by the MEMR in cooperation with other relevant ministries at the national level.

This activity involves selected EEL local manufacturers (under the three above-mentioned LED industry associations), DGNREEC – MEMR, state-owned banks, *OJK*, *BSN* with the assistance of a national consultant hired by the ADLIGHT project. This activity will integrate activities in Output 2.1 on MEPS test methods and energy labels that will be adhered to in the transformation of the facilities.

GEF supports the preparation of feasibility studies and a training/ workshop, follow-up FGDs in the development and related technical studies to support EEL transformation plans.

Activity 1.2.3: Organization and conduct of workshops and business forums for local manufacturers for the development of bankable investment proposals of the business transformation plans, with gender-sensitive capacity building components. This activity, as a follow-up of the approved business transformation plan, is intended to organize and conduct a series of three progressive workshops targeted for local manufacturers to prepare bankable investment proposals (including gender responsive budgeting) within the framework developed in Activity 1.2.2. This is in line with the Indonesian context of selected local EEL manufacturers to convert production lines to the production of more LED lamps.

This activity involves selected EEL local manufacturers (under the three above-mentioned LED industry associations), the DGNREEC – MEMR, state-owned banks, *OJK*, *BSN* with assistance of a national consultant hired by ADLIGHT project. This activity will integrate activities in Output 2.1 regarding the necessary support for the adoption of MEPS, test methods and energy labels for EELs in line with the ASEAN regional approach.

GEF supports a series of three progressive workshops for local manufacturers across the country to prepare efficient and cost-effective EEL designs, business transformation plan and bankable investment proposals.

- **Output 1.3: Completed capacity development program for banking/financing institutions on the evaluation and financing of lighting industry modernization projects.**

Activity 1.3.1 Design, planning and conduct of training for financing institution officials on assessment of financial proposals on energy efficiency, focusing on production upgrades of lighting manufacturers, while specifically addressing gender-gaps in the market. This activity will design, plan and conduct training for women and men financing institution officials on assessment of financial proposals on energy efficiency, focusing on production upgrades of lighting manufacturers. The activity will involve the provision of a feasibility template that will be developed by the project on the evaluation and financing of selected local EEL manufacturers' modernization projects, while addressing barriers to gender equality. In addition, this activity will conduct dissemination trainings for financial institutions (credit officers and risk managers) on assessment and evaluation of business transformation plans and bankable investment proposals focusing on production upgrades of EEL manufacturers /modernization projects. This will also involve dissemination trainings for the eight participant banks of Indonesia First Movers on Sustainable Banking and related Regional Development Banks or *Bank Pembangunan Daerah* (*BPD*) of Smart Cities Street Lighting pilots. Women staff participation in the trainings will be particularly encouraged. The Indonesia Finance Service

Authority (*OJK)* will be in the best position to understand and lead the affiliated banks into fully supporting the realization of the LED industry business transformation plan and necessary elements for it to be successfully and timely implemented. The project will in turn provide the necessary technical assistance to develop its capacity, guidance and tools to equip the banks to effectively receive, evaluate and approve loan applications from interested LED manufacturers.

This activity involves, DGNREEC – MEMR, the eight participant banks of Indonesia First Movers on Sustainable Banking and related Regional Development Banks (*BPD*) of Smart Cities Street Lighting pilots, *OJK* (Financial Services Authority), National Standardization Agency of Indonesia (*BSN*) and a national consultant hired by ADLIGHT project. This activity will integrate with activities in Output 1.2, and Output 3.1.

GEF supports Design, planning and conduct of training for financing institution officials and related activities.

OUTCOME 2: Improved conditions for fair market competition of EE lighting products, informed by robust policy and institutional framework - *Component 2 is under UNEP as GEF Agency and captured in their Project Collaboration Agreement signed with the government of Indonesia*

OUTCOME 3: Increased penetration of high quality and efficient lighting

Table 3 presents major Activities for delivering the planned Outputs that will contribute to the achievement of Outcome 3.

Table 3: Activities that will deliver the Outputs resulting to Outcome 3

OUTPUTS	ACTIVITIES
3.1: Development of an innovative financial model enabling accelerated penetration of advanced lighting systems, focusing on the development of ESCO business models	3.1.1: Comparative evaluation of international best practices of financial support schemes for EEL financing
	3.1.2: Development of ESCO business models to be demonstrated in the pilot projects
	3.1.3: Mainstreaming of the existing national policies and regulations for the banking sector to accelerate penetration of EEL
3.2: Pilot demonstrations for accelerated LED lamp deployment in buildings and for street lighting in context of sustainable cities as well as in residential sector	3.2.1: Design, approval, capacity development and implementation of pilot demonstrations using ESCO business model
	3.2.2: Design, planning and conduct of training for local government municipalities and cities and pilot demonstration hosts
3.3: Implemented awareness and promotion program and information system explaining the benefits of high energy efficient lighting technologies, taking into account gender specific aspects in developing and implementing the	3.3.1: Development of a gender-responsive EEL awareness and promotional program on the benefits of good quality LED lighting
	3.3.2: Development and implementation of information support network system, database and website on EEL product distribution, sales and performance in the market for various consumer groups

programmes	

The following describes the Activities that will be conducted to deliver the expected Outputs to contribute to realization of Outcome 3.

- **Output 3.1: Development of an innovative financial model enabling accelerated penetration of advanced lighting systems, focusing on the development of ESCO business models**

***Activity 3.1.1: Comparative evaluation of international best practices of financial support schemes for EEL financing*.** This activity will involve a study on the local and international experiences and best practices on finance for EE lighting application in various sectors. The study will focus on evaluating various available schemes being practiced in many countries. In Indonesia, there are existing instruments or schemes that could be applicable to LED lighting projects as they are already being applied to Indonesia's other infrastructure projects such as: (a) Public Private Partnership (PPP) mechanism for EE lighting, (b) Bankable Investment Proposal for ESCO financing, (c) Credit Guarantee Facility for EEL/LED investment projects, (d) means of bundling LED lighting products/ small transactions to enable them as accepted collaterals for banks.

This activity will involve the participation of DGNREEC-MEMR, *PT SMI*, PPP-task force (*KPBU Terpadu BAPPENAS*), Indonesia Financial Service Authority (*OJK*), *PT Asuransi Kredit Indonesia* (*ASKRINDO*, an insurance credit company) and *Perum Jaminan Kredit Indonesia* (*JAMKRINDO*, a state-owned credit guarantee company); selected government smart city pilots, PT Sarana Multi Infrastructure (PT SMI, an infrastructure financing company and an accredited entity for the Green Climate Fund), selected state-owned banks, and PT Nico Securities Indonesia or other similar local investment bank in Indonesia. This activity will further deepen the understanding and appreciation by the stakeholders on actual global best practices and lessons learned regarding the proposed innovative financing support schemes which will listed in Activity 3.1.2. This activity will involve the assistance of two national consultants (a technical expert and a financial expert) and an international consultant.

GEF will provide support the intensive studies including travel expenses, hiring of experts for development of innovative financial support schemes, synthesis of report and recommendations, conduct of FGDs and other related expenses.

Activity 3.1.2: Development of ESCO business models to be demonstrated in the pilot projects. This activity will identify financial support instruments or schemes as evaluated in Activity 3.1.1 that are most relevant and could be practical in expanding the market for EEL in Indonesia. Based on the initial PPG studies, ESCO models are recommended and can be implemented by the project in pilot projects under Activity 3.1.3. In order to strengthen the viability of the ESCO arrangements for various LED applications in buildings and street lighting, the existing instruments in the country as mentioned under Activity 3.1.1 will be explored as to how they could be integrated with the ESCO models. The identified financial support scheme should be able to stimulate and incentivize the implementation of full scale ESCO business model (and/or in combination with other business models) and should be in line with economic and environmental objectives of the project. To be fully supported by the government in the ESCO's adoption and qualification for a loan, the project will provide support for the development, drafting and issuance of a financial and bank policy through a decree by the Indonesia Finance Service Authority (*OJK*). This activity is also linked to Output 3.2.

Schemes to be explored in configuring the financial support instruments or schemes to strengthen ESCO models in various LED lighting applications are:

a. PPP framework for smart cities LED street lighting projects
b. Tools to prepare a Bankable Investment Proposal including Feasibility Study (FS) and Financial Modeling for Financing ESCO business model
c. Credit Guarantee Facility for ESCOs implementing shared savings EEL/LED investment projects
d. Other financial support schemes such as an alternative/ possibility to bundle multiple EE lighting projects as acceptable collateral for EEL Financing, as maybe deemed necessary for the project's objectives.

Activity 3.1.3: Mainstreaming of the existing national policies and regulations for the banking sector to accelerate penetration of EEL. This activity will involve up to three nationwide banking forums for Bank credit officers, credit analysts and risk managers of banks and key stakeholders including dissemination of project results in line with the overall market transformation for EE lighting. This will be done in cooperation with *OJK*, BKF of Ministry of Finance of Indonesia (the Centre for Climate Change Financing and Multilateral Policy), Ministry of Finance and DGNREEC MEMR. Dissemination training and forums on "EE financing and LED lighting investment" will be designed to demonstrate exemplary good practices of EEL financing to equip and enable bankers to evaluate and support lighting industry modernization project. The targeted training participants are Credit Officers and Risk Managers (both divisions construct the "Four Eyes Principles" in credit approval process in a bank) from eight bank participants of the First Movers of Sustainable Banking (an *OJK* program) for sustainable finance. Particularly, this will be in view of the EE lighting projects of commercial office buildings, plus Regional Development Banks (*BPD*) of the selected pilot demonstration of street lighting smart city pilot hosts, PT Nico Securities Indonesia, PT SMI, and Indonesian Banking Associations (*HIMBARA* and *PERBANAS*). This activity is also linked to the achievement of several other relevant outputs namely Output 1.1, Output 1.3, Output 2.1, Output 2.2 and Output 2.3.

GEF support is for the incremental requirements that may be required in the following activities: drafting of the procedures, the conduct of three nationwide banking forums and expert consultations.

- **Output 3.2: Pilot demonstrations for accelerated LED lamp deployment in buildings and for street lighting in context of sustainable cities as well as in residential sector**

Activity 3.2.1: Design, approval, capacity development and implementation of pilot demonstrations using ESCO business models in street lighting, buildings and residences. This activity will encourage end users to shift from conventional lamps to LED in selected applications in commercial buildings and street lighting through the following pilot demonstrations as described below. The pilot demonstrations will address technical, investment and financial support requirements which constitute the up-front costs in formulating and initiating the EE program that usually become hurdles for coming to EE investment initiatives even while the potential benefits may have been identified. The project will remove these barriers and pave the way to attract numerous building owners and city governments to follow this developmental route for more efficient and sustainable operations. Once these developmental concerns are addressed, the EE lighting program for government buildings will be properly launched and the sustaining mechanisms will be put in place for MEMR to follow through and sustain through the effective implementation of the replication plan that will be initiated within the project period. The project will provide assistance in terms of investment support and technical

assistance as cost-sharing with the energy user and investor. The project will also draw from the demo experiences valuable inputs in developing the procurement policy and guidelines and engagement of ESCOs for EE lighting projects which are expected to be sustainable owing to the indicated energy savings and GHG reduction. The involvement of *LKPP* as the government procurement agency is strategic in the development and adoption of the procurement policy and guidelines applicable to EELs. After addressing the program-related up-front costs that will be supported by GEF project, the EE lighting program will continue to be supported by the MEMR as part of its regular administrative and monitoring function while the targeted energy users can already budget for its own EE lighting transformation requirement that are sufficiently justified and implemented under the approved procurement plans and guidelines resulting from these project barrier removal activities. The Demo Project Concepts are described in **Annex K** in terms of project development, demo implementation, performance contracting and replication preparation.

a. ESCO business models for public and commercial buildings

This activity will involve the implementation of an ESCO business model at a government agency office (initially identified as the Government Procurement Agency or *LKPP* building) and commercial office buildings of the eight participant banks of the 'First Movers of Sustainable Bank' (a program initiated by the Indonesia Financial Service Authority) to give a worthwhile 'learning by doing' experience for those institutions and to demonstrate the workability of the ESCO business model. Demo Concept Notes can be found in **Annex K.1** and **K.2** for public building and commercial building, respectively. This implementation consists of two pilot demonstrations: a) a pilot to prove the workability of the ESCO business model for "guaranteed savings" of EE lighting in *LKPP* Building in cooperation with the Property Management/ General Affairs, Accounting Division and Internal Audit Division; and b) pilot demonstrations to demonstrate the workability of the "shared savings" ESCO business model in commercial office buildings in cooperation with Fixed Asset Management Division as well as an Energy Sector Specialist, and Risk Management Division which are the entities in-charge in creating a loan product specialized for ESCO financing.

In the case of the public building demo, the challenges for public building administrators to enter into contracts with ESCOs through the usual procurement procedures pose certain barriers since that practice is not yet in place for EELs and the confidence in ESCOs to guarantee the savings still need to be established. This describes a situation where some entities would be hesitant to convert to EELs for several reasons including unbudgeted financial requirement for the upfront cost of lamps and lack of technical knowhow to implement and measure the impact of the conversion. In this demo, the experiences gathered will lead to two important barrier removal steps: the determination of the necessary elements for developing and approving a government procurement scheme for the government to engage ESCOs and the identification of the needs of the ESCO to strengthen itself to be able to qualify in the enhanced EEL procurement procedures and be able to guarantee the savings. But because the ESCO financial capability is still to be developed and the government does not have the budget to pay as restricted by the current procurement regulations, the project will shoulder the initial up-front cost of the lighting equipment to trigger this scheme. *LKPP*, being in a strategic position as an energy user and policy maker on government procurement, will pave the way through this barrier removal process. In the future, the project envisions ESCOs are strengthened technically and financially and the government sector could budget for engagement with ESCOs through a government procurement

policy involving standard Energy Performance Contracting (EPC) that will be developed through the project.

The second EEL demo scheme involves ESCO Shared Savings model in commercial buildings. The ESCO assumes the investment for the initial EEL equipment outlay and enters into an EPC arrangement with the building owner. Depending on the agreed investment scheme, the proportion of the sharing in the savings after cost recovery between the commercial building owner and the ESCO will be determined at the outset and reflected in the EPC. The EPC will stipulate the pay back arrangement as monthly installments to cover for cost recovery and the ESCO service fees as well as the sharing formula for the resultant savings.

This activity will document, monitor and evaluate the technical and financial performance of the two models, prepare reports on advantages and disadvantages, lessons learned and best practices and develop manuals for the effective implementation of each of the models. The development strategy also looks at the possibility of the public sector being able to budget and pay for its EEL conversion and also enter into contracts with ESCOs with the option of using Shared Savings scheme as already being practiced in many places in Europe.

This activity will involve participation of *LKPP*, ESCOs, three local lighting manufacturers associations (*ALINDO*, *APERLINDO*, *GAMATRINDO*), DG NREEC – MEMR, Laboratory testing of *P3TEK* of DGNREEC – MEMR, eight participant banks of the First Movers of Sustainable Banking (to be initiated by *OJK*) for the EE lighting projects of Commercial Buildings. This activity will be undertaken by ADLIGHT project under the supervision of DGNREEC-MEMR, Ministry of Information and Communication and *OJK*.

GEF supports the implementation of this pilot project to shift from conventional lighting to LED lamps in the selected banks office buildings.

b. ESCO Guaranteed Savings model and revolving fund for street lighting

This activity will involve the implementation phase of the selected partner City Governments to conduct a pilot using the ESCO Guaranteed Savings model for EE Street Lighting and setting up a revolving fund to finance future EEL projects in municipalities. The project will provide technical advisory and assistance in design and planning and the financial support for the acquisition and installation of LED lamps replacing less efficient lamps, incorporating women's and men's different needs and perceptions of street lighting. The project will also support the initial investments in the municipal EEL transformation and ESCO development. The energy savings resulting from the conversion will be deposited in a revolving fund that will accommodate the financing of future EEL projects. Each municipality can set up its own revolving fund, where initial project financial assistance allow for using the savings to capitalize the revolving fund and use it for their future projects.

This implementation consists of two pilot demonstration projects. A first pilot project will be in Java Island to prove the workability of the ESCO model for EE Street Lighting. This will include the development of an energy metering and reporting system to measure Energy Savings and Budget Savings (of Local Government Annual Income and Budget/*APBD*) for EE Street Lighting and to submit the evaluation and recommendation report to the City Government and House of Regional Representatives/ DPRD. A second pilot demonstration of street lighting will be outside Java island

under a different setting but typical of other areas using different procurement procedures. The Demo Project Concept is described in **Annex K.3**. The Government of Indonesia is currently assessing 10 up to 25 candidates for piloting of Smart Cities including those outside Java Island. This activity will involve participation of City Governments, ESCOs, DGNREEC – MEMR, three local lighting manufacturers associations, Lab test P3TEK of DGNREEC – MEMR , Regional Development Banks (*BPD*) of the selected pilot of Smart Cities. This activity will be undertaken by the project under supervision from DGNREEC-MEMR, Ministry of Information and Communication and Financial Service Authority (*OJK*).

GEF supports the implementation of this pilot project to shift from conventional lighting to LED lamps in the selected Smart City Street Lighting and other municipalities and technical support in setting up revolving funds.

c. *On-bill financing model for selected government-owned residential buildings*

This will involve the implementation in selected government low-income residential housing which are owned and managed by the government. The pilot demonstration will consist of replacing 250,000 existing lighting with 3-watt LED lighting. The cost of the LED replacement will be included by the housing management in the tenants' monthly maintenance fees. Lamps will be replaced and costs for the lamps will be recovered from maintenance fee through the regular management and administration of the residential buildings. The activity will also involve training of the residential building administrator on the relevant financing and billing tasks regarding the effective conversion to EEL so as to orient them on the new responsibility to manage and sustain the EEL conversion. Lessons learned from implementation can be used to guide the possible improvement of financial support schemes as necessary for the deployment of more replication projects. In its important role for introducing EEL, government can encourage its housing projects to participate in proliferating EELs. The transition towards efficient lighting in residential sector will be ushered by the government to expedite the uptake of EEL through the formulation and implementation of favorable policies derived from the demonstration projects as well as lessons learned in other countries. This activity will also involve consultations with direct beneficiaries within the residential pilot areas, ensuring equal participation of women and men beneficiaries.

GEF supports the implementation of this pilot project to shift from conventional lighting to LED lamps in the selected residential applications.

Activity 3.2.2: Design, planning and conduct of training for local government municipalities and cities and pilot demonstration hosts This activity will involve capacity development trainings of local government municipalities and cities and pilot demonstration institution hosts about EEL systems and government procurement systems using the ESCO business models. During this activity women's active participation will be encouraged. This activity is intended for interested city governments to implement EE Street Lighting and to further expand the workability of the ESCO model for EE Street Lighting in the Smart City Design Concept. It will synthesize the lessons learned from pilot demonstration projects to guide the possible improvement of financial support schemes as necessary for the deployment of more replication projects. This activity also includes dissemination of the results of Activities *3.1.3.*

GEF will provide support for the design, planning and conduct of training, preparation of training manuals, evaluation and recommendations for improvement and sustainability.

- **Output 3.3: Implemented awareness and promotion program and information system explaining the benefits of high energy efficient lighting technologies, taking into account gender specific aspects in developing and implementing the programme**.

  Activity 3.3.1: Development of a gender-responsive EEL awareness and promotion program on the benefits of good quality LED lighting. This activity will involve the development of an awareness program, promotion packages and multimedia dissemination for general public, business sector and government sector in Indonesia on the benefits of energy efficient lighting technologies. These awareness raising activities are meant to raise attention as well as increase the knowledge about energy efficient lighting technologies. Target groups are government and non-government procurement agencies, city governments, gender equality-focused organizations, state-owned as well as commercial banks, investment banks and development banks as well as the general domestic market. This activity involves participation of regional office of *PLN* Regional Office), Regional House of Representative (*DPRD*), local universities, *MASKEEI* (*Masyarakat Konservasi dan Efisiensi Energi Indonesia*) or IECES (*Indonesia Energy Conservation* and Efficiency *Society*), and other government smart cities. This activity correlates back to Activity 1.1.2 regarding the market surveys of high efficient lamps in Indonesia, and is also linked to achievement of the results under Output 3.2.

  GEF will provide support for the dissemination including exhibition, 'green corners/desks for local EE lamps' in retailers, public and social media campaign on the benefits of innovative lighting technologies in several cities.

  Activity 3.3.2: Development and implementation of information support network system, database and website on EEL product distribution, sales and performance in the market for various consumer groups This activity will involve an information support network system and data base which are aimed to allow access by all stakeholders in EE lighting to vital information and data by providing an accredited database by which data on EE lighting product's distribution, sales, performance in market and other information is made available to support operations, management and decision-making in relation to the business transformation, information and awareness and market control in pursuit of the ADLIGHT goal and objectives. The project looks at an integrated and organized system or network for the collection, organization, storage and communication of information among key project partners and participants. This activity will involve the participation of the National Standardization Body (BSN), local manufacturers association (ALINDO, APERLINDO, GAMATRINDO), DG NREEC – MEMR, Lab test P3TEK of DG NREEC MEMR, State-Owned Utility Company/PLN, city(ies) government and line ministries (Ministries of Public Works, Transportation, Industry, Trade, Custom, OJK, BKF MoF) and other relevant agencies.

  GEF will provide support for the development and implementation of the information support network system, database and website.

IV. RESULTS AND PARTNERSHIPS

i. Expected Results

The following describes the expected results from the work that will be done through the ADLIGHT Project using the strategy described above which the project proposes to support Indonesia to advance the market for energy efficient lighting technologies. The project's planned key interventions, among others, include the following:

- Approval of *SNI*/MEPs and other policies
- Strengthened institutional structure through capacity building
- Successful demos and replication
- Capable and internationally accredited EEL laboratories
- Financing institutions and ESCOs continuous support to financial schemes
- Effective information and promotions
- Increased awareness around gender gaps and opportunities in the LED lighting market

The significant development that will mark the realization of the Alternative Scenario is the completion and attainment of the various targets under each success indicator which are further detailed in **Section VI: Project Result Framework** will include the following:

1. Overall results
   - Cumulative electricity savings by EOP of 77.45 GWh
   - Direct project GHG emissions mitigated by EOP of 62.58 kt CO_2
   - At least 10% of women and at least 10% of men, who previously lacked access to energy efficient lighting services and technologies in the residential areas, public buildings and streets in pilot areas, are benefiting from these services and technologies as influenced by the project
   - Women and men are benefiting equally from EEL technologies and increased access to knowledge about energy efficient lighting technologies

2. Improved quality, energy efficient and affordable locally-produced EEL products and systems
   - Efficiency of locally produced lighting systems increased, (lm/W) and Production cost decreased from baseline level
   - Six (6) lighting manufacturers who received technical assistance on the design and production of high efficient lighting products and systems
   - Two (2) investment grade proposals for business transformation plans approved by banks

3. Improved conditions for fair market competition of EE lighting products, informed by robust policy and institutional framework - *Component 2 is under UNEP as GEF Agency and captured in their Project Collaboration Agreement signed with the government of Indonesia*
   -

4. Increased market share for high quality and efficient lighting

- Full scale ESCO models incorporating innovative financial support instruments or schemes developed to accelerate penetration of EE lighting systems in public and commercial buildings and street lighting as well as in government-owned residential sector
- Five (5) pilot ESCO demonstrations (2 cities, 2 buildings, and a set of selected residential areas); With possible replication in 2 cities and 7 buildings and other residential areas
- Four (4) municipalities, six (6) clients (project developers/ building owners), five (5) technology providers and two (2) financial institutions stakeholders engaged.

The three-year ADLIGHT project is expected to reduce annual electricity through its direct pilot and immediate replications by 60.17 GWh/yr or cumulatively 77.45 GWh by the end of the project (EOP) in Year 3 (2021). It targets to contribute_378.31 GWh cumulatively in 2026 or 5 years after the EOP and 679.17 GWh in 2031 or 10 years after EOP. The equivalent reduction of CO_2 emissions is estimated at 48.62 kton CO_2 per year by 2021. Cumulatively, reduction of CO_2 emissions amounts to 62.58 kton CO_2 by 2021, 305.67 kton CO_2 by 2026 and 548.77 kton CO_2 by 2031. More details are in **Annex L**.

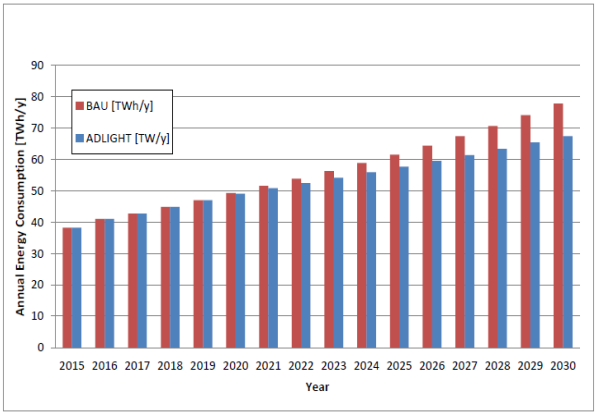
Table 4: Summary of Expected Results of Baseline and Alternative Scenarios

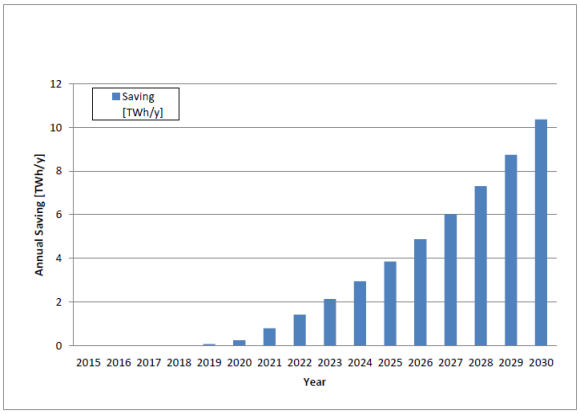
Particulars	2016 (Baseline)	2022 EOP Year 3	2027 5 yrs after EOP	2032 10 yrs after EOP
A. Direct and Direct-post project (Pilot and Replication)				
Energy Saving Total, GWh/yr	0	60.17	60.17	60.17
Cumulative Energy Saving, GWh	0	77.45	378.31	679.17
Emission Reduction per Year, kt CO_2/yr	0	48.62	48.62	48.62
Direct project GHG emissions mitigated, kt CO_2	0	62.58	305.67	548.77
B. Conservative penetration rate and increased efficiency Scenario				
Penetration rate for all sectors			10.0 %	20 %
Increased efficiency in all sectors*	R: 57.0% I,C,S,GO: 49.5% SL: 65.0%	R: 59.4% I,C,S,GO: 51.0% SL: 66.0%	R: 65.4% I,C,S,GO: 58.5% SL: 71.0%	R: 59.4% I,C,S,GO: 51.0% SL: 66.0%
Energy Saving Total, MWh/yr		241	3,851	10,357
Cumulative Energy Saving, MWh		310	11,461	48,757
Emission Reduction per Year, kt CO_2/yr		194	3,111	8,369
Cumulative Emission Reduction, kt CO_2		250	9,261	39,395
C. Optimistic penetration and increased efficiency Scenario				
Penetration rate for all sectors			9.2 %	32.2 %
Increased efficiency in all sectors*	R: 57.0% I,C,S,GO: 49.5% SL: 65.0%	R: 59.4% I,C,S,GO: 51.0% SL: 66.0%	R: 65.4% I,C,S,GO: 58.5% SL: 71.0%	R: 59.4% I,C,S,GO: 51.0% SL: 66.0%

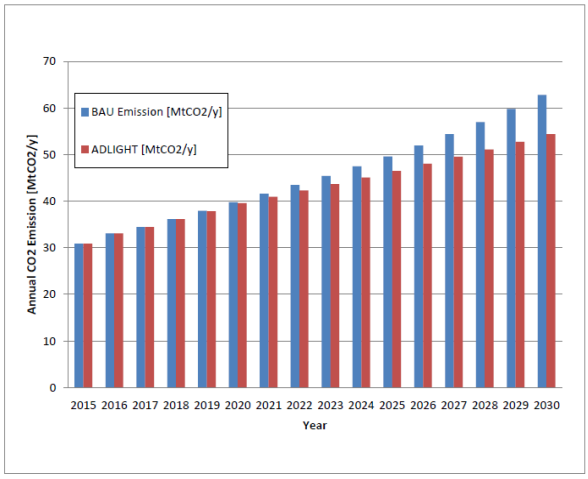
Particulars	2016 (Baseline)	2022 EOP Year 3	2027 5 yrs after EOP	2032 10 yrs after EOP
Energy Saving Total, MWh/yr		241	3,561	16,583
Cumulative Energy Saving, MWh		310	10,731	60,767
Emission Reduction per Year, kt CO_2/yr		194	2,877	13,399
Cumulative Emission Reduction, kt CO_2		250	8,671	49,100

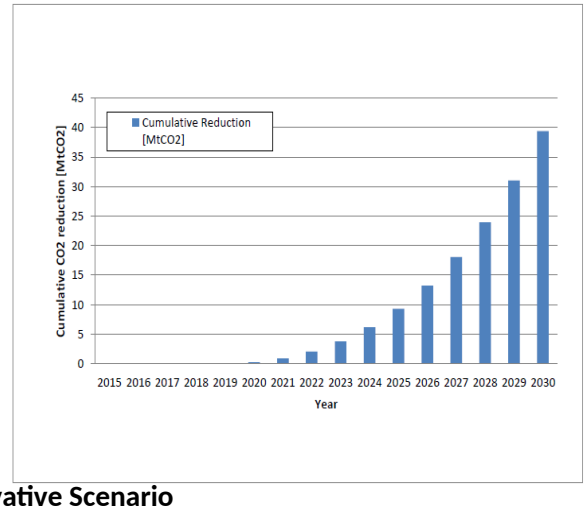
*R-Residential, I-Industry, C-Commercial, S-social, GO-Government Office, SL-Street Lighting

The following figures, illustrate BAU and alternative energy consumption, energy savings, equivalent CO_2 reduction under two (2) likely scenarios: (a) conservative penetration rate and improved energy efficiency and (b) optimistic penetration rate and improved energy efficiency as summarized in the estimated projections. Details are seen in **Annex L**.

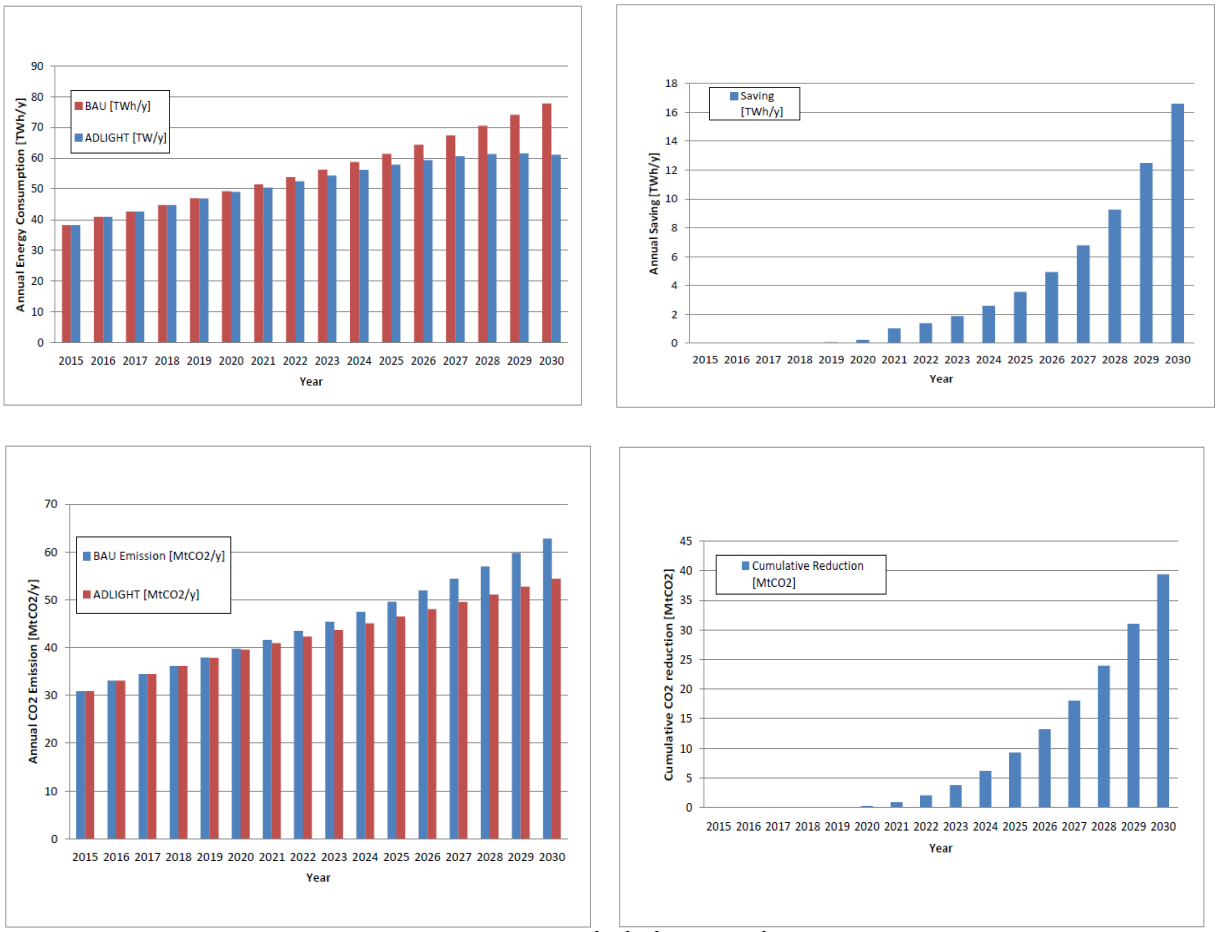








A. Conservative Scenario



B. Optimistic Scenario

Fig. 4: Conservative and optimistic penetration rate and increased efficiency Scenarios

ii. Partnerships

The project will be executed by the Ministry of Energy and Mineral Resources (MEMR). This project is part of the global programmatic framework for which UNEP is the lead GEF Agency. UNDP will be responsible for the implementation of Component 1 and Component 3 of the project in Indonesia as per the terms and conditions set out in this Project Document. UNEP will be responsible for the implementation of *Component 2 of the project, as per the terms and conditions set out in their Project Collaboration Agreement to be agreed and signed between UNEP and the government of Indonesia*). The initiative will follow the institutional structure as described in Section 6 of the Program Framework Document. The national project steering committee will be composed of the Ministry of Industry, Ministry of Finance, Ministry of Trade, Lighting Industry Association, Indonesia National Standards Organization BSN, national lamp testing laboratories, UNEP and UNDP. International project partners including international organizations, global manufacturers, utilities, and gender equality-focused organizations working in energy will support the project to achieve its objectives. This support will come in form of in-kind co-financing and parallel activities such as expertise

on setting MEPS, training on efficient lighting manufacturing, market surveillance activities, public awareness campaigns and regulatory making processes.

Partners include companies throughout the lighting value chain, financial institutions, academic institutions and laboratory and research centers who will work together in the three-year project which is co-implemented by UNDP and UNEP with the MEMR as the project executing agency.

Table 5: ADLIGHT Partners and Roles in the Project

Partners	Role in Project Implementation
Ministry of Energy and Mineral Resources (MEMR)	Provides overall direction and oversight as the Executing Agency; acts as Chairman of the National Steering Committee; provide the lead role in coordinating with other ministries and agencies in pushing forward the goals and objectives of the efficient lighting industry program and resolving inter-ministry barriers
Directorate General of New-Renewable Energy and Energy Conservation (DGNREEC)	Acts as the Responsible Party for the project and leads the Project Management Unit for the day-to-day operations and resolution of issues to ensure smooth implementation of the project; provides technical support for the technology programs; coordinates with other government agencies as well as UNDP and UNEP
MEMR P3TK EBTKE (Research and Development of DGNREEC)	Provides technical support and laboratory services for the needs of the LED industry; leads the capacity building and accreditation of laboratory facilities in meeting the MEPS/*SNI* standard requirement, benchmarking and round-robin testing
Min of Development Planning (*Badan Perencanaan Pembangunan Nasional or BAPPENAS)*	Acts as the integrator of the related government programs within the overarching objectives of the EEL program which crosses many sectoral boundaries and policy/organizational areas considering overall national interest
Public Procurement Agency – *Lembaga Kebijakan Pengadaan Barang/Jasa Pemerintah* (LKPP)	Acts as main resource agency for the development and adoption of government procurement policies in order to hasten the market penetration of locally produced high efficiency and affordable lighting products
Financial Service Authority (*Otoritas Jasa Keuangan* or *OJK*)	Coordinates among financial institutions and banks and provides overall policy direction and guidance in implementation of financing mechanisms that will be adopted by the project
PT Sarana Multi Infrastruktur - Persero (PT SMI)	Acts as resource group for green banking and green infrastructure as it applies to buildings and street lighting finance
City Government – Smart Cities	Act as pilot demonstration hosts in EEL application in street lighting and the evaluation of the impacts of the transformation toward EELs
Energy Service Company (ESCO)	Provides the main link between the EEL manufacturers and users and facilities project development, financing and EPC requirements
State Owned Banks & Financial Institution	Acts as catalysts in the development and implementation of the business transformation plans to transform the industry to produce and market EELs locally and potentially abroad
Perusahaan Listrik Negara (*PLN*) State Electricity Company)	Acts as resource organization on energy supply, pricing, and policy in support of the increased market for EELs and transition from the conventional lamps as well as possible participation in piloting LED distribution schemes and billing systems through

Partners	Role in Project Implementation
	ESCO model will be tested for residential sector LED programme with the cooperation of PLN, using its organization and network for the residential sector
Ministry of Trade	Act as resource ministry in the harmonization of policies and regulations regarding trading of locally produced and imported EELs; provides active support in the MVE schemes and market control to encourage efficient and reasonably priced lighting products that meet MEPS/*SNI*
Ministry of Industry	Act as resource ministry for the policies and regulations governing the lighting industry sector and the implementation of business transformation plans towards producing affordable, high quality lighting products and for determination of local content of locally manufactured products
Ministry of Public Works and Housing	Acts as resource ministry in the application of EELs in the street lighting and building consumer groups
Ministry of Finance/Bureau of Customs	Provides lead role in the application and regulation of policies regarding custom duties and importation of lighting product components as well exportation of standard products whenever viable; participated in the development and implementation of MVE schemes vis-à-vis MEPS/*SNI* adopted and the corresponding inspection software to discourage entry or usage of inefficient lighting products
Badan Standardisasi Nationa (*BSN*) or National Standardization Agency	Provides technical support and policy coordination in the development and adoption of the *SNI* and MEPS within the timeframe of the project considering ASEAN and other international commitments in MEPS and labeling
BPPT (Research Agency for Technology Development and Implementation)	Acts as resource agency in the R&R and product development for EEL technology and their application
Tingkat Kandungan Dalam Negeri (*TKDN*) or Llocal Content certifiers	Act as resource agency in the policy review and recommendations in enhancing the local content of locally manufactured lighting products
Credit Facility Guarantee/Credit Insurance Companies	Act as resource groups for determining appropriate financing support schemes in support of the full scale ESCO model regarding credit guarantee facility
Lighting Industry Associations (*GAMATRINDO*, *ALINDO* and *APERLINDO*)	Provides technical support and coordination among local manufacturers; leads the industry associations and its members in policy discussions to achieve timely progress in project implementation and performance targets; lead the transition to higher efficiency and cost-effective EEL production and acts as coordinator and focal point in the monitoring, verification and enforcement representing the EEL industry for measuring impact and corrective actions
Development Partners (MTRE3, ICED2, PFAN, and other bilateral projects)	Provide technical support and coordination regarding the implementation of relevant activities towards achievement of outputs that have common project objectives
Gender equality-focused organizations and ministry-designated focal points working on energy	Provide support in gender-mainstreaming activities in pursuit of the overall project objectives to be coordinated by the MEMR-designated gender focal point for the project

iii. **Risk Management**

As per standard UNDP requirements, the Project Manager will monitor risks quarterly and report on the status of risks to the UNDP Country Office. The UNDP Country Office will record progress in the UNDP ATLAS risk log. Risks will be reported as critical when the impact and probability are high (i.e. when impact is rated as 5, and when impact is rated as 4, and probability is rated at 3 or higher). Management responses to critical risks will also be reported to the GEF in the annual PIR. Project risks are summarized below in **Table 6**. A Project Risk Log is provided in **Annex H**.

Table 6: ADLIGHT Project Risk Log

Project Risks					
Description	**Type**	**Impact & Probability**	**Mitigation Measures**	**Owner**	**Status**
1. Prolonged delays in the implementation of MEPS due to Government procedures and processes	Political Regulatory	P = 2 I = 3	*BSN* to come up with *SNI* standards that can be used as reference in the government procurements and other EEL applications while MEPS is being developed	MEMR/DGNREEC BSN Industry associations Min of Trade and Industry	Reducing
2. Local lighting manufacturers may not be interested to upgrade the lighting production facilities due to increased investments, disruption to operation and business priorities	Financial Operational Strategic	P = 3 I = 3	Coordinate closely with manufacturers associations and local banks to facilitate investments; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
3. Petroleum products and electricity prices will be at levels that make energy efficiency projects, particularly in EEL, uncompetitive and not cost effective	Strategic Environmental Political	P = 3 I = 3	Government to be encouraged to adopt appropriate energy pricing policies and incentives to encourage EE project investments	MEMR/DGNREEC Min of finance Min of Planning Industry associations Min of Trade and Industry	Increasing
4. Manufacturers are not committed to improving the energy-efficiency and quality of products.	Organizational Operational	P = 2 I = 3	Coordinate closely with manufacturers associations; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
5. Manufacturers are not willing to share data on their operations and marketing for MVE	Financial Operational	P = 3 I = 3	Coordinate closely with manufacturers associations; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing

Project Risks					
Description	**Type**	**Impact & Probability**	**Mitigation Measures**	**Owner**	**Status**
6. Banks and financial institutions are not convinced about the business potential of manufacturing and market for LED systems	Financial Operational Organizational	P = 3 I = 3	Coordinate closely with manufacturers associations and local banks to facilitate investments; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry *OJK* Financial institutions/Banks	Reducing
7. Commitment to abide with regional and international EEL standards and harmonization is lacking	Political Regulatory	P = 2 I = 3	Implement Component 2 on policy and institutional support effectively	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
8. Policy framework may not attractive for manufacturers and users to invest in EEL projects.	Political Regulatory	P = 3 I = 3	Implement Component 2 on policy and institutional support effectively	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
9. Local EEL producers could not meet all technical standard specifications	Financial Operational Organizational	P = 3 I = 3	Coordinate closely with manufacturers associations; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
Overall		P = 3 I = 3			

*Probability on a scale from 1 (low) to 5 (high) and Impact on a scale from 1 (low) to 5 (high).

iv. **Stakeholder engagement plan**

The stakeholder process for the ADLIGHT project started during the stakeholder consultative meeting and logical framework analysis held in May 2017 where representatives from various sectors relevant to lighting products manufacturing and application were gathered for determining the barriers to transforming the market for efficient lighting devices. The manufacture and use of lighting product and systems cuts across many sectors: energy, transportation, industry, trade, financial, public works and housing, research and technology development, policy and regulation. The Project will be organized through a coordination mechanism that will involve representatives from the above-mentioned sectors.

At the national level, the project management team will build on the stakeholder consultation process that includes the inception workshop and other bi-lateral meetings. A broader consultation with a range of stakeholders will be held under the leadership of the Ministry of Energy and Mineral Resources. The consultation will be arranged via the Working groups set up under the Project Management Unit (**Figure 6**).

At the local level, stakeholders will be engaged through the UNDP's standard stakeholder engagement processes. The project management team and MEMR/DGNREEC will continue to

work closely with key project stakeholders such as project sponsors, co-financing institutions, community-based organizations and relevant NGOs.

The following civil society organizations will be closely involved in project implementation:

- Lighting Industry Associations (*GAMATRINDO*, *ALINDO* and *APERLINDO*)
- Tingkat *Kandungan Dalam Negeri* (*TKDN*) or Local Content certifiers
- *Masyarakat Konservasi dan Efisiensi Energi Indonesia MASKEEI* or IECES (Indonesia Energy Conservation and Efficiency *Society*)

Special consideration for the disenfranchised groups from the local communities, women, children, elderly, the poor, vulnerable and minorities are also being included because they will benefit greatly from greater access to and use of efficient and cost-effective energy efficient lighting products in terms of day-to-day need, safety and security. To coordinate various interests and needs, the project will have a National Steering Committee whose membership will come from major stakeholders to ensure that their needs are addressed through a broader range of national actors so that stakeholder involvement will widely be an inclusive and participative process.

v. <u>Gender equality and empowering women</u>

The ADLIGHT project provides unique opportunities to advance gender equality in interventions for advancing the manufacture and application of EELs systems in Indonesia. The ADLIGHT project has included specific activities and targets that advance and report on gender equality results, including more balanced participation between men and women in workshops and consultations, with particular focus on advancing gender equality through stimulating the LED market. Another particular focus is on the need for women's consultation through interventions related to sustainable street lighting solutions where EEL, particularly LEDs, have very important contributions.

During the project design, a Gender Analysis and Action Planning were conducted focusing on several areas for consideration in the ADLIGHT project (see **Annex N**). The main outcomes of the gender analysis are summarized below:

- The traditional energy sector is still one of the least gender-inclusive sectors to date globally. According to one estimate, women represent only 6 percent, 4 percent and less than 1 percent of the technical, decision-making and top management positions, respectively, in the energy sector.
- In Indonesia, women have a limited access to information related to energy, given the male-dominated industry, which in turn limits women's participation and contribution to shaping policy
- Gender gaps related to access to energy, finances, training, employment and entrepreneurship need to be addressed in Indonesia
- Women and men have different perceptions on what constitutes adequate illumination, as well as the sense of safety it provides on a particular street, with men having the tendency to perceive that the illumination had been adequate while the women tended to claim that it was not bright enough.

- Increasing the energy efficiency of street lighting is not only cost effective, it provides gender-sensitive results by improving nigh-time safety in cities
- Households where women are the predominant decision maker tend to have lower energy consumption

Deeper gendered assessments need to be conducted in order to better illustrate how women and men interact with the local economy differently with regard to LED lighting. Energy sectors are primarily male-dominated in Indonesia and globally, however women are an untapped resource, as key beneficiaries of lighting technologies, and consultants on how to most effectively raise awareness about new lighting technologies. Finally, addressing gender gaps at leadership, policy and industrial levels in the electricity sector is an opportunity to advance gender equality, and the economy of Indonesia at large.

The ADLIGHT project is committed to ensure that women's and men's different roles, perceptions, and opportunities in contributing to and benefiting from application of energy efficient LED lighting technologies in Indonesia will be considered. Gender mainstreaming will be included, as appropriate, to be gender transformative in project's implementation strategies and activities in advancing the application of EELs in the country.

v. **South-South and Triangular Cooperation (SSTrC)**

The ADLIGHT Project will pursue and promote South-South and triangular cooperation forms an integral part. Indonesia is very actively supporting this approach in its participation in many projects of the ASEAN on energy efficiency. The UNEP's Global Leapfrogging Program is also advocating this approach through global platform for knowledge sharing between participating countries, where ADLIGHT is part of this leveraging the experience of other countries on advancing high efficient lighting. Within the Global Leapfrogging Program, countries will cooperate in such activities as: knowledge exchange, technology transfer, financing, peer support, and country-to country or multi country initiatives, on high efficient lighting technology.

vi. Sustainability and Scaling Up

The project's multi-pronged barrier removal approach, covering policy and planning, commercial and technical viability, financing, and information and awareness, is designed to stimulate ongoing replication of the project demos and, thus, scale-up of project results.

The ADLIGHT project sustainability is well-founded within Indonesia's long drawn policies and program on energy efficiency and conservation (EE&C) as mandated in the country's *Government Regulation of the Republic of Indonesia No. 70/2009 on Energy Conservation*. To strengthen institutional support, the Directorate General of New-Renewable Energy and Energy Conservation (DGNREEC) was established in August 2010 to implement the EE&C program and continued to be reinforced by succeeding regulations and pronouncements placing EE&C an important component in the Presidential Regulation No. 61/2011 on National Action Plan to Reduce Green House Gases (*RAN-GRK*) as well as in the international commitment in the Nationally Determined Contribution (NDC) and regional agreements such as in the ASEAN. These are among the major policies that are supportive to energy efficient lighting program agenda. Lighting is a major energy end-use that accounts for 18.8% of the total electricity used for all sectors which gives a big reason

to pursue sustainable development in this area. The response of the private sector has been very encouraging and constitutes a major driver in achieving the ADLIGHT goal of transforming the market for efficient energy lighting towards its high potential and long lasting impact on energy saving and GHG reduction. The pilot EEL demonstrations in buildings and street lighting will lead to sustained growth and impact.

The strategy focuses on mitigating investment risks on production and increasing market penetration of high quality EELs not only in Indonesia but also in the region. The project will ensure that policy risks are addressed by working closely with the government and establishing the quality and performance standards for investors to invest in quality EEL products without fear of competition from low quality products. The project will support the relevant agencies and stakeholders to ensure technology, policy, institutional and human capacities are established for the realization of ADLIGHT-supported development pathway. With an increasing market and improved standards of EELs, the financial risk will be addressed in a sustainable manner. The ADLIGHT project activities are designed in a way that encourages replication. The project will support the Government of Indonesia in harmonization of its policies and legislative framework, particularly Minimum Energy Performance Standards (MEPS) for LED lighting products with international conventions and ASEAN standards. These activities will require updates and applications in similar related fields. This will increase the sustainability of the regulatory system.

The engagement of the local EEL manufacturing industry as the key stakeholder in this market transformation strategy will ensure its buy-in in demonstrating the benefits of quality production and creating conditions for them to benefit from sustainable application of EELs.

The project will develop business and financing models to implement large scale EEL production upgrade projects. Once financial institutions can see EE lighting investment as a bankable investment, replication will be easier to conduct. The state-owned bank building as one of main targets of pilot demonstration in the ADLIGHT project was selected to encourage banks to have first-hand experience and benefit (energy and cost saving per year) from EE lighting investment and projects. By including banks as project stakeholders and target recipient of pilot demos, the project will work towards improving the bankability perception of EE lighting investment in Indonesia and developing viable ESCO business models for a sustainable approach in future.

The project has included the establishment of an Efficient Lighting Knowledge Center for collecting market information on lighting products, quality standards of products, quality product listing (QPL) and database in cooperation with a network of qualified laboratories, consumer perception, including gender-disaggregated consumer data to be made available, and other related information regarding efficient lighting technology and technology providers. It will ensure that the knowledge management support is fully developed in line the active transition to EELs as a sustainable program.

V. PROJECT MANAGEMENT

i. Cost efficiency and effectiveness

The ADLIGHT project strategy of focusing on buidlings and street lighting is expected to maximize the energy savings possible within a three-year period through providing assistance to the lighting industry in order to modernize their facilities to produce highly efficient and cost-effective lighting products. With the estimated CO_2 emission reduction of around 1.65 million tons equivalent of the energy saving potential and project cost of US$ 3.89 million, the unit abatement cost is US$ 2.36 per ton CO_2.

ADLIGHT Project	**Value**
I - Direct project	
Direct post project CO_2 emission reductions (t CO_2)	548,766[5]
II – Indirect	
Indirect CO_2 emissions - top-down (tCO_2)	1,097,533[6]
GEF finance (US $)	3,895,872
Total emission reductions (t CO_2)	1,646,299[7]
Unit Abatement Cost (UAC) ($/t$CO_2$) - (Direct savings) as per GEF CO_2 emission reduction calculation methodology ($/t$CO_2$)	7.0
Unit Abatement Cost (UAC) ($/t$CO_2$) (Direct + Indirect)	2.36

Based on 2016 estimates, the projected 2032 (after 10 years from EOP) cumulative energy savings range from 48.7 GWh to 60.8 GWh in the conservative and optimistic EEL market penetration scenarios. This translates to estimated 4,187 to 5,227 tons of oil equivalent or in terms of foreign exchange equivalent of USD 1.79 million to 2.23 million.

The proposed project's design aims to deliver a high level of cost efficiency and effectiveness in use of GEF funds. Cost efficiency and effectiveness will be achieved by strategies that leverage limited inputs of GEF funding to stimulate much greater investments by other parties, namely the government and the private sector. The project is expecting to generate USD 18,115,000 in private sector investments within the project timeline and additional resources as the pilot demonstrations are replicated in the country.

ii. Project management

The Project will be implemented, subject to space availability, at the designated space for the Project Management Unit located at the MEMR. The project will be attended by Project Manager, Finance and Administrative Assistant and any other national expert hired for the execution of the project. The PMU will take care of day-to-day management and implementation of project activities leading to the Project Outputs and Outcomes, prepare annual work plans and build ownership for the project results.

[5] Must be considered for 10 years of direct CO_2 emission reductions from all pilot projects and replications. Since this project focuses entire sector, cumulative emission reductions during economic lifetime (considered 10 years including project duration of 3 years)

[6] Based on GEF CO_2 emission reduction calculation methodology, as it focuses entire sector. In this case it is the cumulative emission reduction for 10 year from best case of all sectors multiple by factor 0.5

[7] Considered for 10 years of economic lifetime as per GEF guidelines (Direct + Indirect).

iii. Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information

To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP and UNEP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[8] and the GEF policy on public involvement[9].

iv. Social and environmental safeguards

Environmental and social grievances will be reported to the GEF in the annual PIR. The Social and Environmental Screening Report is attached in **Annex F**.

As can be seen in the Social and Environmental Risk Screening Checklist, there are two risks identified under Principle 2, Gender Equality and Women's Empowerment and three risks under Standard 7, among which are Pollution Prevention and Resource Efficiency. The latter are regarding risks on potential pollution and risks from hazardous and non-hazardous waste materials from replaced fluorescent lamps/CFL and other conventional lighting systems and accessories, particularly mercury content and sharp glass bulb parts need to be managed and proper disposal system be implemented and monitored. The project's Activity 2.4.5: Design, planning and conduct of training on waste management and recycling of components and materials of conventional lighting replaced by LED has been included to address this issue and ensure effective waste management and disposal.

The Gender Analysis as described in **Annex N** includes a Gender Action Plan which was used in elaborating gender sensitive project activities throughout all Project Components.

The project risk categorization following from the Social and Environmental Screening Report as described in **Annex F** is ***Moderate***.

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[8] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[9] See https://www.thegef.org/gef/policies_guidelines

VI. PROJECT RESULTS FRAMEWORK

This project will contribute to the following Sustainable Development Goal (s): UNPDF/CPD Outcome 3 -- By 2020, Indonesia is sustainably managing its natural resources, on land and at sea, with an increased resilience to the effects of climate change, disaster and other shocks
This project will contribute to the following country outcome included in the UNDAF/Country Program Document: CPD Indicative Outputs -- 2.2.1. National energy policies and guideline developed and integrated into sub- national development plan.2.2.2. Sub-national authorities and key partners are able to implement programmes, mobilize resources and develop public-private partnership for RE/EE, which will contribute to the reduction of national greenhouse gases emission.
This project will be linked to the following output of the UNDP Strategic Plan: Output 1.4: Scaled up action on climate change adaptation and mitigation cross sectors which is funded and implemented. Output 1.5: Inclusive and sustainable solutions adopted to achieve increased energy efficiency and universal modern energy access (especially off-grid sources of renewable energy)

	Objective and Outcome Indicators	Baseline	Mid-term Target (2020)	End of Project (EOP) Target (2022)	Assumptions
Project Objective To increase the penetration of high quality energy efficient lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions.	• Cumulative electricity savings by EOP, GWh	• 0	• 17.27 GWh (62,172,000 MJ)	• 77.45 GWh (278,820,000 MJ)	• Petroleum products and electricity prices will be at levels that make energy efficiency projects, particularly energy efficient lighting (EEL), basically LED manufacturing and application, competitive and cost effective • Acceptance of the benefits and business opportunities resulting from the local manufacture and application of LED • Women's active participation in trainings is requested and encouraged
	• Direct project GHG emissions mitigated by EOP, kt CO2 (GEF Core Indicator 6.2)	• 0	• 13.96 kt $CO_{2\ (direct\ by\ EOP)}$ • 121.76 kt $CO_{2\ (Lifetime\ direct\ by\ EOP)}$	• 62.58 kt $CO_{2\ (direct\ by\ EOP)}$ • 548.77 kt $CO_{2\ (Lifetime\ direct\ by\ EOP)}$	
	• Number of women and men participating in capacity building trainings throughout the project (GEF Core Indicator 11)	• -N.A.	• 30/70	• 60/140	
Component/Outcome 1: Improved quality, energy efficient and affordable locally-produced EEL products and systems	• Efficiency of locally produced lighting systems increased, (lm/W) and Production cost decreased from baseline level (%) ○ Indoor Type	• Eff: **70** lm/W; • Supply chain cost: 100 % (Baseline supply chain cost to be determined at project inception)	• Eff: **120** lm/W; • Supply chain cost: **80%** of Baseline	• Eff: **150** lm/W; • Supply chaincost: **70%** of Baseline	• Local lighting manufacturers will take up the transformation of their facilities as viable investment in production techniques for affordable and highly efficient lighting products • Manufacturers remain committed to improving the energy-efficiency and quality of products. • Manufacturers remain willing to share data.

	Objective and Outcome Indicators	Baseline	Mid-term Target (2020)	End of Project (EOP) Target (2022)	Assumptions
	○ Outdoor Type	• Eff: **100** lm/W; • Supply chain cost:100% (at project inception)	• Eff: **130** lm/W; • Supply chain Cost: **80%** of Baseline	• **170** lm/W; • Supply chain cost: **70%** of Baseline	
	• Cumulative number of lighting manufacturers who received technical assistance to upgrade production facilities	• 0	• 3 manufacturers (1 for each association)	• 6	• Commitment and active participation of manufacturers and associations with adequate financial resources to participate in the program.
	• Investment grade proposal for business transformation plans ○ Submitted by manufacturers ○ Approved by banks.	• 0 • 0	• 3 • 3	• 6 • 6	• Banks and financial institutions[10] are convinced about the business potential of manufacturing and market for LED systems.
	• Ratio of women and men who believe they have the capacity to ○ Submit (local manufacturers) ○ Approve (banks) investment grade proposal for business transformation plans	• 0 • 0	• 30/70 • 30/70	• 30/70 • 30/70	• Women's active participation in trainings has been requested and encouraged and women and men employees from relevant government institutions have actively participated in the capacity building activities on business transformation plans
Component/ Outcome 2: Improved conditions for fair market competition of EE lighting products informed by robust policy and institutional framework - ***Component 2 is under UNEP as GEF Agency and captured in their Project Collaboration Agreement***	•				

[10]Various banks and financing institutions in Indonesia, including Bank Mandiri, BRI, BNI, BCA, Bank Muamalat, BRI Syariah, BJB and Bank Artha Graha are mobilized and introduced by OJK on financing EEL business opportunities

	Objective and Outcome Indicators	Baseline	Mid-term Target (2020)	End of Project (EOP) Target (2022)	Assumptions
signed with the government of Indonesia					
Component/ Outcome 3: Increased market penetration of high quality and efficient lighting	• Cumulative no. of innovative financial support schemes developed to accelerate penetration of EE lighting systems	• 0	• 2	• 4[11]	• Targeted users and key stakeholders of EEL (basically LED) products and systems are well informed about the long-term benefits and costs and are convinced to invest in such efficient technology application projects
	• Cumulative no. of pilot demonstrations[12] completed and replication plans developed and approved for implementation[13]	• 0	• 3 (1 cities and 2 buildings)	• 5 (2 cities, 2 buildings and a residential area) With possible replication in 2 cities and 7 buildings.	• Policy framework would allow private investments to invest in energy efficiency street lighting projects. • Local EEL producers can meet all technical standard specifications and ESCOs can develop investment-grade proposals that will convince banks to finance the EEL projects within acceptable insurance and guarantee provisions required by banks for sustainability.
	• Number of stakeholders engaged o municipalities o clients (project developers/ building owners) o technology providers	 • 0 • 0 • 0	 • 2 • 3 • 3	 • 4 • 6 • 5	• Pilot demonstrations will use production from local manufacturers

[11] Proposed innovative financial support schemes to accelerate penetration of advanced lighting systems, may include: (a) PPP mechanism for smart city(ies) LED street lighting projects; (b) ESCO model Feasibility Study (FS) Template and Financial Modeling Software; (c) Credit Guarantee Facility for Energy Efficiency LED lighting investment; and (d) Bundling LED lighting projects as accepted collateral for banking sector

[12] Initially identified pilot demonstrations during the PPG stage include the following: (a) smart lighting and smart monitoring system LED, (b) ESCO model for state-owned bank building, (c) ESCO model for LKPP (Public Procurement Agency) for government LED implementation, and (d) public-private partnership scheme model for LED projects.

[13] Assistance in completing the pending existing ESCO model in Smart Cities in Solo and Bandung as the 2 cities which have on-going project development, being assisted by Bappenas and LKPP, but it requires next stage of mobilization towards final financial arrangement

	Objective and Outcome Indicators	Baseline	Mid-term Target (2020)	End of Project (EOP) Target (2022)	Assumptions
	○ financial institutions	• 0	• 1	• 2	
	• Women's and men's level of satisfaction with EEL systems provided (reliability, affordability, convenience, efficiency)	• N.A.	• At least 60 % of women and men beneficiaries are highly satisfied with the EEL systems provided (minimum score 7/10)	• At least 70 % of women and men beneficiaries are highly satisfied with the EEL systems provided (minimum score 7/10)	• The design and installation of the EEL systems is well informed on women's and men's needs, preferences and habits in requirement, consumption and disposal of EEL technologies

VII. MONITORING AND EVALUATION (M&E) PLAN

The project results as outlined in **Section VII: Project Results Framework** will be monitored annually and evaluated periodically during project implementation to ensure the project effectively achieves these results.

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP and UNDP Evaluation Policy
The UNDP Country Office will work with the relevant project stakeholders to ensure M&E requirements are met in a timely fashion and to high quality standards. Additional mandatory GEF-specific M&E requirements (as outlined below) will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[14]. The costed M&E plan included below, and the Monitoring plan in Annex, will guide the GEF-specific M&E activities to be undertaken by this project.

In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed during the Project Inception Workshop and will be detailed in the Inception Report. This will include the exact role of project target groups and other stakeholders in project M&E activities including the GEF Operational Focal Point and national/regional institutes assigned to undertake project monitoring. The GEF Operational Focal Point will strive to ensure consistency in the approach taken to the GEF-specific M&E requirements (notably the GEF-7 Core Indicators) across all GEF-financed projects in the country. This could be achieved for example by using one national institute to complete the GEF-7 Core Indicators for all GEF-financed projects in the country, including projects supported by other GEF Agencies.

M&E Oversight and monitoring responsibilities:

Project Manager: The Project Manager is responsible for day-to-day project management and regular monitoring of project results and risks, including social and environmental risks. The Project Manager will ensure that all project staff maintain a high level of transparency, responsibility and accountability in M&E and reporting of project results. The Project Manager will be responsible for providing all the regular reports as agreed under the M&E to the UNDP. The Project Manager will inform the Project Board, the UNDP (Country Office and the UNDP-GEF RTA) of any delays or difficulties as they arise during implementation so that appropriate support and corrective measures can be adopted.

The Project Manager will develop annual work plans based on the multi-year work plan included in Annex A, including annual output targets to support the efficient implementation of the project. The Project Manager will ensure that the standard UNDP and GEF M&E requirements are fulfilled to the highest quality. This includes quarterly reporting on the progress and expenditure reports as per the requirements of UNDP. This includes, but is not limited to, ensuring the results framework indicators are monitored annually in time for evidence-based reporting in the GEF PIR, and that the monitoring of risks and the various plans/strategies developed to support project implementation (e.g. gender strategy, KM strategy etc..) occur on a regular basis.

Project Board: The Project Board will take corrective action as needed to ensure the project achieves the desired results. The Project Board will hold project reviews to assess the performance of the project and appraise the Annual Work Plan for the following year. In the project's final year, the Project Board will hold an end-of-project review to capture lessons learned and discuss opportunities for scaling up and to

[14] See https://www.thegef.org/gef/policies_guidelines

highlight project results and lessons learned with relevant audiences. This final review meeting will also discuss the findings outlined in the project terminal evaluation report and the management response.

Project Implementing Partner (Executing Agency): The Implementing Partner is responsible for providing any and all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary and appropriate. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes, and is aligned with national systems so that the data used by and generated by the project supports national systems.

UNDP Country Office: The UNDP Country Office will support the Project Manager as needed, including through annual supervision missions. The annual supervision missions will take place according to the schedule outlined in the annual work plan. Supervision mission reports will be circulated to the project team and Project Board within one month of the mission. The UNDP Country Office will initiate and organize key GEF M&E activities including the annual GEF PIR, the *independent mid-term review* and the independent terminal evaluation. The UNDP Country Office will also ensure that the standard UNDP and GEF M&E requirements are fulfilled to the highest quality.

The UNDP Country Office is responsible for complying with all UNDP project-level M&E requirements as outlined in the UNDP POPP. This includes ensuring the UNDP Quality Assurance Assessment during implementation is undertaken annually; that annual targets at the output level are developed, and monitored and reported using UNDP corporate systems; the regular updating of the ATLAS risk log; and, the updating of the UNDP gender marker on an annual basis based on gender mainstreaming progress reported in the GEF PIR and the UNDP ROAR. Any quality concerns flagged during these M&E activities (e.g. annual GEF PIR quality assessment ratings) must be addressed by the UNDP Country Office and the Project Manager.

The UNDP Country Office will retain all M&E records for this project for up to seven years after project financial closure in order to support ex-post evaluations undertaken by the UNDP Independent Evaluation Office (IEO) and/or the GEF Independent Evaluation Office (IEO).

UNDP-GEF Unit: Additional M&E and implementation quality assurance and troubleshooting support will be provided by the UNDP-GEF Regional Technical Advisor and the UNDP-GEF Directorate as needed.

Additional GEF monitoring and reporting requirements:

Inception Workshop and Report:

A project inception workshop will be held within two months after the project document has been signed by all relevant parties to, amongst others:

a. Re-orient project stakeholders to the project strategy and discuss any changes in the overall context that influence project implementation;
b. Discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms;
c. Review the results framework and finalize the indicators, means of verification and monitoring plan;

d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP in M&E;
e. Update and review responsibilities for monitoring the various project plans and strategies, including the risk log; Environmental and Social Management Plan and other safeguard requirements; the gender strategy; the knowledge management strategy, and other relevant strategies;
f. Review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; and
g. Plan and schedule Project Board meetings and finalize the first year annual work plan.

The Project Manager will prepare the inception report no later than one month after the inception workshop. The inception report will be cleared by the UNDP Indonesia Country Office and the UNDP-GEF Regional Technical Adviser based in Bangkok, in coordination with the designated UNEP GEF team and its Task Manager in Bangkok office and will be approved by the Project Board. The final Inception Report will be shared with project participants to formalize various agreements and plans decided during the meeting.

GEF Project Implementation Report (PIR): The ADLIGHT Project Manager, the UNDP Indonesia Country Office and the UNDP-GEF Regional Technical Advisor, will provide objective input to the annual GEF PIR covering the reporting period July (previous year) to June (current year) for each year of project implementation. Inputs for Component 2 for the PIR report will be prepared by the Component 2 coordinator (UNEP). The Project Manager will ensure that the indicators included in the project results framework are monitored annually in advance of the PIR submission deadline so that progress can be reported in the PIR. Any environmental and social risks and related management plans will be monitored regularly, and progress will be reported in the PIR.

The PIR submitted to the GEF will be shared with the Project Board. The UNDP Indonesia Country Office will coordinate the input of the GEF Operational Focal Point for Indonesia and other stakeholders to the PIR as appropriate. The quality rating of the previous year's PIR will be used to inform the preparation of the subsequent PIR.

Lessons learned and knowledge generation:

Results from the implementation of the ADLIGHT project will be disseminated within the project duration through the various knowledge-management-related activities under the three components of the project and beyond the project intervention area through existing information sharing networks and forums and new knowledge sharing schemes initiated by the project. The project will identify and participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may be of benefit to the project. The project will identify, analyze and share lessons learned that might be beneficial to the design and implementation of similar projects and disseminate these lessons widely. There will be continuous information exchange between this project and other projects of similar focus in the same country, region and globally.

Results from the project will be disseminated within and beyond the project intervention area through a number of existing information sharing networks and forums that may be held in ASEAN and the UNEP's global Leapfrogging Program, U4E and similar networks. The project will participate, as relevant and appropriate, in other UNDP/GEF and UNEP/GEF sponsored projects.

The project has budgeted ample amounts to support the knowledge generation and sharing in pursuit of the project objective. It will also identify, analyze, and share lessons learned that might be beneficial in the design and implementation of similar future projects.

GEF Focal Area Core indicators:

The prescribed GEF Core Indicators will be used to monitor global environmental benefit results. The baseline/CEO Endorsement GEF Focal Area Core Indicators (s) – submitted in **Annex D** to this ADLIGHT project document – will be updated by the Project Manager/Team and shared with the mid-term review consultants and terminal evaluation consultants before the required review/evaluation missions take place. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website. The updated GEF Core Indicators (s) will be submitted to the GEF along with the completed Mid-term Review report and Terminal Evaluation report.

Independent Mid-term Review (MTR): An independent mid-term review process will begin after the second PIR has been submitted to the GEF, and the MTR report will be submitted to the GEF in the same year as the 2nd PIR. The MTR findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the project's duration. The terms of reference, the review process and the MTR report will follow the standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the UNDP Evaluation Resource Center (ERC). As noted in this guidance, the evaluation will be 'independent, impartial and rigorous'. The consultants that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project under review. The GEF Operational Focal Point for Indonesia and other stakeholders will be involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/GEF Directorate. The final MTR report and MTR TOR will be available in English and will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser, and approved by the Project Board and will be posted on the UNDP ERC. A management response to MTR recommendations will be posted in the ERC within six weeks of the MTR report's completion.

Terminal Evaluation (TE): An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terminal evaluation process will begin three months before operational closure of the project allowing the evaluation mission to proceed while the project team is still in place, yet ensuring the project is close enough to completion for the evaluation team to reach conclusions on key aspects such as project sustainability. The Project Manager will remain on contract until the TE report and management response have been finalized. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the UNDP Evaluation Resource Center. As noted in this guidance, the evaluation will be 'independent, impartial and rigorous'. The consultants that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated. The GEF Operational Focal Point and other stakeholders will be involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/GEF Directorate. The final TE report and TE TOR will be cleared by the UNDP Indonesia Country Office, and the UNDP-GEF

Regional Technical Adviser based in Bangkok, and will be approved by the Project Board. The TE report will be publicly available in English and posted on the UNDP ERC. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report's completion.

The UNDP Indonesia Country Office will include the planned project terminal evaluation in the UNDP Country Office evaluation plan, and will upload the final terminal evaluation report in English and the corresponding management response to the UNDP Evaluation Resource Centre (ERC). Once uploaded to the ERC, the UNDP IEO will undertake a quality assessment and validate the findings and ratings in the TE report, and rate the quality of the TE report. The UNDP IEO assessment report will be sent to the GEF IEO along with the project terminal evaluation report.

Final Report: The project's terminal PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information**:** To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[15] and the GEF policy on public involvement[16].

Mandatory GEF M&E Requirements and M&E Budget:

GEF M&E requirements	Primary responsibility	Indicative costs to be charged to the Project Budget[17] (US$)		Time frame
		GEF grant	Co-financing	
Inception Workshop	UNDP Country Office UNEP-GEF Task Manager Project Manager (PM)	USD 11,000		Within two months of project document signature
Inception Report	Project Manager	(included in routine project staff activity)	None	Within two weeks of inception workshop
Standard UNDP monitoring and reporting requirements as outlined in	UNDP Country Office	(included in routine project	None	Quarterly, annually

[15] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[16] See https://www.thegef.org/gef/policies_guidelines

[17] Excluding project team staff time and UNDP staff time and travel expenses.

GEF M&E requirements	Primary responsibility	Indicative costs to be charged to the Project Budget (US$)		Time frame
		GEF grant	Co-financing	
the UNDP POPP and UNEP Requirements	Project Manager	staff activity		
Risk management	Project Manager Country Office UNEP Task Manager			Quarterly, annually
Monitoring of indicators in project results framework	Project Manager	USD 12,000 (Per year: USD 4,000)		Annually
GEF Project Implementation Report (PIR)	Project Manager and UNDP Country Office and UNDP-GEF team, and UNEP Task Manager	(included in routine project staff activity	None	Annually
NIM Audit as per UNDP audit policies	UNDP Country Office External auditors	USD 21,000 (Per year: USD 7,000)		Annually or other frequency as per UNDP Audit policies
Lessons learned and knowledge generation	Project Manager	USD 9,000		Annually
Monitoring of environmental and social risks, and corresponding management plans as relevant	Project Manager UNDP CO UNEP Task Manager	(included in routine project staff activity		On-going
Stakeholder Engagement Plan	Project Manager UNDP Country Office			On-going
Gender Action Plan	Project Manager UNDP Country Office UNDP GEF team			On-going
Addressing environmental and social grievances	Project Manager UNDP Country Office BPPS as needed	None for time of project manager, and UNDP CO		
Project Board meetings	Project Board UNDP Country Office UNEP Task Manager Project Manager	USD 6,000		At minimum annually
Supervision missions	UNDP Country Office	None[18]		Annually
Oversight missions	UNDP-GEF team and UNEP Task Manager	None[27]		Troubleshooting as needed
GEF Secretariat learning missions/site visits	UNDP Country Office and Project Manager and UNDP-GEF team, and UNEP GEF team	USD 6,000		To be determined.
Mid-term GEF Core Indicators **to be**	Project Manager	(included in		Before mid-term

[18] The costs of UNDP Country Office and UNDP-GEF Unit's participation and time are charged to the GEF Agency Fee.

GEF M&E requirements	Primary responsibility	Indicative costs to be charged to the Project Budget (US$)		Time frame
		GEF grant	Co-financing	
updated by (add name of national/regional institute if relevant)		routine project staff activity		review mission takes place.
Independent Mid-term Review (MTR) and management response	UNDP Country Office and Project team and UNDP-GEF team, UNEP Task Manager	USD 20,000		Between 2nd and 3rd PIR.
Terminal GEF Core indicators to be updated by (add name of national/regional institute if relevant)	Project Manager	(included in routine project staff activity		Before terminal evaluation mission takes place
Independent Terminal Evaluation (TE) included in UNDP evaluation plan, and management response	UNDP Country Office and Project team and UNDP-GEF team, and UNEP GEF team	USD 50,000		At least three months before operational closure
Translation of MTR and TE reports into English	UNDP Country Office	None		
TOTAL indicative COST Excluding project team staff time, and UNDP staff and travel expenses		USD 135,000		

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Roles and responsibilities of the project's governance mechanism:

UNDP, as the Implementing Agency of GEF for Components 1 and 3 will implement components 1 and 3 and manage the funding allocated by GEF to UNDP in relation to components 1 and 3, under the implementing modalities based on its own policies and procedures. Management of the funds allocated by GEF to UNEP in relation to the implementation of Component 2 shall be the exclusive responsibility of UNEP and shall be managed by UNEP in accordance with the terms and conditions set out in the Project Cooperation Agreement to be entered into between UNEP and the Government of Indonesia. UNDP shall therefore not be held accountable in relation to such funds allocated by GEF to UNEP in relation to Component 2.

One PMU, consisting of Project Manager and Financial and Administrative Support, will be established to manage the whole project. The PMU will be responsible for the overall management and monitoring of project implementation and results in relation to Component 1 and 3, pursuant to the terms set out in this Project Document with UNDP, and in relation to Component 2, pursuant to the terms set out in the Project Cooperation Agreement with UNEP.

The **Implementing Partner (Executing Agency)** for this project is the Ministry of Energy and Mineral Resource (MEMR) with the Directorate General of New Renewable Energy and Energy Conservation (DG-NREEC), under the same Ministry as the key technical partner. For UNDP, in relation to Components 1 and 3 of the Project, the Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Risk management as outlined in this Project Document;
- Procurement of goods and services, including human resources;
- Financial management, including overseeing financial expenditures against project budgets;
- Approving and signing the multiyear workplan;
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

The Implementing Partner (or EA) will designate a high-ranking official as the National Project Director. He/She will assume responsibility for the Project on behalf of the National Government. The National Project Manager will report to the NPD.

UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes oversight of project execution to ensure that the project is being carried out in accordance with agreed standards and provisions. UNDP is responsible for delivering GEF project cycle management services comprising

project approval and start-up, project supervision and oversight, and project completion and evaluation. UNDP is responsible for the Project Assurance role of the Project Board/Steering Committee.

The project organization structure is as follows:

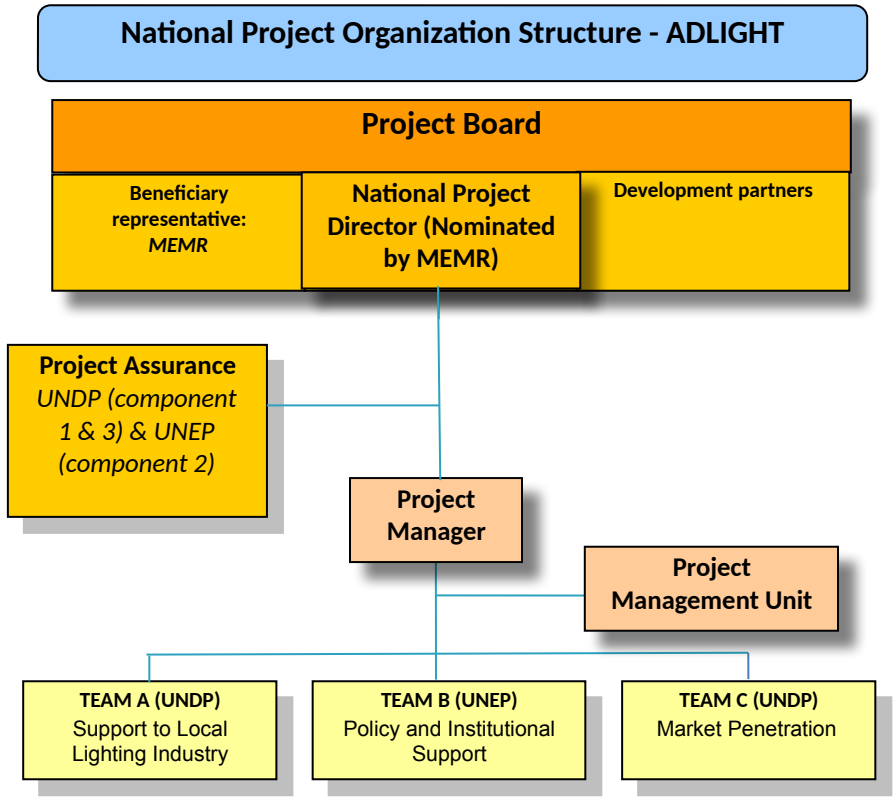


Fig. 5: ADLIGHT Project Management Structure

Project Board: The Project Board (also called Project Steering Committee) is responsible for taking corrective action as needed to ensure the project achieves the desired results. In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.

In case consensus cannot be reached within the Board, the UNDP Resident Representative (or their designate) will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

Specific responsibilities of the Project Board include:

- Provide overall guidance and direction to the project, ensuring it remains within any specified constraints;
- Address project issues as raised by the project manager;
- Provide guidance on new project risks, and agree on possible mitigation and management actions to address specific risks;
- Agree on project manager's tolerances as required, within the parameters set by UNDP-GEF, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded;
- Advise on major and minor amendments to the project within the parameters set by UNDP-GEF;

- Ensure coordination between various donor and government-funded projects and programmes;
- Ensure coordination with various government agencies and their participation in project activities;
- Track and monitor co-financing for this project;
- Review the project progress, assess performance, and appraise the Annual Work Plan for the following year;
- Appraise the annual project implementation report, including the quality assessment rating report;
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project;
- Review combined delivery reports prior to certification by the implementing partner;
- Provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily according to plans;
- Address project-level grievances;
- Approve the project Inception Report, Mid-term Review and Terminal Evaluation reports and corresponding management responses;
- Review the final project report package during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

The composition of the Project Board includes the following roles:

Project Executive: The Project Executive is an individual who represents ownership of the project and chairs the Project Board. For the ADLIGHT Project, the Project Executive is the Director of Energy Conservation (DEC) of DG NREEC, Ministry of Energy and Mineral Resources (MEMR)/DG-NREEC. In daily project implementation, the MEMR/DG-NREEC is represented by National Project Director as project executive.

Beneficiary Representative(s): The beneficiary Representatives are individuals or groups representing the interests of those who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. The Beneficiary representative is the Ministry of Energy and Mineral Resource (MEMR).

Development Partner(s): The Development Partners are individuals or groups representing the interests of the parties concerned that provide funding and/or technical expertise to the project. For the ADLIGHT project, the Development Partners are expected to be GIZ and USAID. The final membership of the Project Board will be finalized during the Inception workshop for the project.

Project Assurance: UNDP performs the quality assurance role and supports the Project Board and Project Management Unit by carrying out objective and independent project oversight and monitoring functions in relation to Components 1 and 3. This role ensures appropriate project management milestones are managed and completed. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. UNDP provides a three – tier oversight services involving the UNDP Country Offices and UNDP at regional and headquarters levels. Project assurance is totally independent of the Project Management function.

Project Manager: The Project Manager has the authority to run the project on a day-to-day basis on behalf of the Implementing Partner within the constraints laid down by the Project Board. The

Implementing Partner appoints the Project Manager, who must be different from the Implementing Partner's representative in the Project Board.

The Project Manager's primary responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost. The Project Manager will inform the Project Board and the Project Assurance roles of any delays or difficulties as they arise during implementation so that appropriate support and corrective measures can be adopted. The Project Manager will remain on contract until the Terminal Evaluation report and the corresponding management response have been finalized and the required tasks for operational closure and transfer of assets are fully completed.

Specific responsibilities include:

- Manage the overall conduct of the project.
- Plan the activities of the project and monitor progress against the approved workplan.
- Execute activities by managing personnel, goods and services, training and low-value grants, including drafting terms of reference and work specifications, and overseeing all contractors' work.
- Monitor events as determined in the project monitoring plan, and update the plan as required.
- Provide support for completion of assessments required by UNDP, spot checks and audits.
- Manage requests for the provision of UNDP financial resources through funding advances, direct payments or reimbursement using the FACE form.
- Monitor financial resources and accounting to ensure the accuracy and reliability of financial reports.
- Monitor progress, watch for plan deviations and make course corrections when needed within project board-agreed tolerances to achieve results.
- Ensure that changes are controlled and problems addressed.
- Perform regular progress reporting to the project board as agreed with the board, including measures to address challenges and opportunities.
- Prepare and submit financial reports to UNDP on a quarterly basis.
- Manage and monitor the project risks – including social and environmental risks - initially identified and submit new risks to the Project Board for consideration and decision on possible actions if required; update the status of these risks by maintaining the project risks log;
- Capture lessons learned during project implementation.
- Prepare revisions to the multi-year workplan, as needed, as well as annual and quarterly plans if required.
- Prepare the inception report no later than one month after the inception workshop.
- Ensure that the indicators included in the project results framework are monitored annually in advance of the GEF PIR submission deadline so that progress can be reported in the GEF PIR.
- Prepare the GEF PIR;
- Assess major and minor amendments to the project within the parameters set by UNDP-GEF;
- Monitor implementation plans including the gender action plan, stakeholder engagement plan, and any environmental and social management plans;
- Monitor and track progress against the GEF Core indicators.
- Support the Mid-term review and Terminal Evaluation process.

Delineation of Roles and Responsibilities of UNDP and UNEP

Considering that ADLIGHT is a national as well as a multi-country project being implemented by UNDP (for Components 1 and 3, as per the terms of this Project Document) and UNEP (for Component 2, as per the terms of the Project Cooperation Agreement to be signed between UNEP and the Government of Indonesia), the oversight roles and responsibilities shall be carried out in the different aspects of governance and project management as follows.

	UNDP	**UNEP**
Outcomes and Outputs Coverage	Outcome 1 and 3	Outcome 2
Budgeting/Financial	USD 2,633,372	USD 1,262,500
Implementation	NIM	NIM
M&E	Standard reporting, e.g. APR/PIR, Quarterly Reports, Mid-term Review, Terminal Evaluation for assigned component, to be consolidated by PMU	Standard reporting, e.g. APR/PIR, Quarterly Reports, Mid-term Review, Terminal Evaluation for assigned component, to be consolidated by PMU
Post-project arrangements	As determined in the project plans for Outcome 1 and 3	As determined in the project plans for Outcome 2

IX. FINANCIAL PLANNING AND MANAGEMENT

The total resources for project implementation are USD 40,990,209. This is financed through a GEF grant USD 2,633,372 administered by UNDP representing the incremental resources needed achieve the Alternative Scenario under ADLIGHT. This will be supported by in-kind co-financing of USD 80,000 to be administered by UNDP and in-kind co-financing of USD 60,000 to be administered by UNEP. Complementing these GEF, UNDP and UNEP grants will be parallel co-financing valued at USD 37,094,337 as detailed below representing related project resources from the Government of Indonesia and the private sector. A GEF Grant of USD 1,262,500 is administered by UNEP. UNDP, as the lead GEF Implementing Agency, is responsible for the execution of the GEF resources and the cash co-financing transferred to UNDP for Component 1 and 3, while UNEP will be responsible likewise for Component 2. Financial reporting to GEF will be the overall responsibility of UNDP through the PIR to be assisted by the PMU, with inputs from UNEP for Component 2.

Parallel co-financing: The actual realization of project co-financing will be monitored during the mid-term review and terminal evaluation process and will be reported to the GEF. The planned parallel co-financing will be used as follows:

Co-financing Source	Co-financing Type	Co-financing Amount ($)	Planned Activities/ Outputs	Risks	Risk Mitigation Measures
Government of Indonesia					
1. Ministry of Energy and Mineral Resource (DG-NREEC)	In-kind	268,000	All Outputs	Changes in policy direction, leaders and priorities	Close coordination and effective MEMR/DGNREEC/PMU project management and M&E
	Cash	13,277,000			
	Subtotal	13,545,000			
2. Ministry of Energy and Mineral Resource (P3TEK – R&D Dept)	In-kind (lab)	4,167,300	Outputs 1.1, 2.1, 2.3, 2.5	Budgeting constraints	Coordination with MEMR and PMU
3. City of Solo/Surakarta (Mayor) Government	In-kind (Pilot EE Street light)	10,015,037	Outputs 3.1, 3.2, 3.3	Budgeting constraints	Close coordination
4. National Standardization Agency (*Badan Standardisasi Nasional*, *BSN*)	In-kind	112,000	Output 2.3	Delays in the development and enforcement of MEPS	Close follow-up and facilitation as well as developing and issuing the SNI
	cash	50,000			
	Subtotal	162,000			
Sub-total (GOI)		**27,889,337**			
Private Sector					

Co-financing Source	Co-financing Type	Co-financing Amount ($)	Planned Activities/ Outputs	Risks	Risk Mitigation Measures
5. Indonesian Lighting Manufacturers Association – GAMATRINDO	Equity	750,000	All Outputs	Delays in implementation of Business Transformation Plans	Close coordination and active participation in all components
6. Indonesian Lighting Manufactures Association - APERLINDO	Equity	5,000,000	All Outputs	Delays in implementation of Business Transformation Plans	Close coordination and active participation in all components
7. Solarens (local LED manufacture)	Equity	1,115,000	Outputs 3.1, 3.2, 3.3	Delays in implementation of Business Transformation Plans	Close coordination and active participation in all components
8. Adyawinsa (local LED manufacture)	Equity	2,000,000			
9. Global Efficient Lighting Centre	In-kind	200,000	Output 2.3	Delays in the development and enforcement of MEPS	Close follow-up and facilitation as well
Sub-total Private Sector		**9,065,000**			
GEF Agency					
10. UNEP	In-kind	60,000	All Outputs	None	
11. UNDP	In-kind	80,000	All Outputs	None	
Sub-total GEF Agency		**140,000**			
Total Co-financing		**37,094,337**			
	(select)				
	(select)				

Budget Revision and Tolerance (for the UNDP administered part of the GEF grant): As per UNDP requirements outlined in the UNDP POPP, the Project Board will agree on a budget tolerance level for each plan under the overall annual work plan allowing the Project Manager to expend up to the tolerance level beyond the approved project budget amount for the year without requiring a revision from the Project Board. Should the following deviations occur, the Project Manager and UNDP Country Office will seek the approval of the UNDP-GEF team as these are considered major amendments by the GEF:

a. Budget re-allocations among components in the project with amounts involving 10% of the total project grant or more;
b. Introduction of new budget items/or components that exceed 5% of original GEF allocation.

Any over expenditure incurred beyond the available GEF grant amount will be absorbed by non-GEF resources (e.g. UNDP TRAC or cash co-financing).

Audit: The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop.

Project Closure (for the UNDP administered part of the GEF grant): Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

Project extensions: The BPPS/GEF Executive Coordinator must approve all project extensions. All extensions incur costs and the GEF project budget cannot be increased. A single extension may be granted on an exceptional basis only if the following conditions are met: one extension only for a project for a maximum of six months; the project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs will be covered by non-GEF resources; the UNDP Country Office oversight costs during the extension period must be covered by non-GEF resoruces.

Operational completion (for the UNDP administered part of the GEF grant): The project will be operationally completed when the last UNDP-financed inputs have been provided and the related activities have been completed. This includes the final clearance of the Terminal Evaluation Report (that will be available in English) and the corresponding management response, and the end-of-project review Project Board meeting. Operational closure must happen with 3 months of posting the TE report to the UNDP ERC. The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. At this time, the relevant parties will have already agreed and confirmed in writing on the arrangements for the disposal of any equipment that is still the property of UNDP.

Transfer or disposal of assets (for the UNDP administered part of the GEF grant): In consultation with the NIM Implementing Partner and other parties of the project, UNDP programme manager (UNDP Resident Representative) is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project. In all cases of transfer, a transfer document must be prepared and kept on file[19]. The transfer should be done before Project Management Unit complete their assignments.

Financial completion (for the UNDP administered part of the GEF grant): The project will be financially closed when the following conditions have been met:

a. The project is operationally completed or has been cancelled;
b. The Implementing Partner has reported all financial transactions to UNDP;
c. UNDP has closed the accounts for the project;
d. UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

[19] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

The project will be financially completed within 6 months of operational closure or after the date of cancellation. Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to the BPPS/GEF Unit for confirmation before the project will be financially closed in Atlas by the UNDP Country Office.

Refund to Donor (for the UNDP administered part of the GEF grant): Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/GEF Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF Trustee.

X. TOTAL BUDGET AND WORK PLAN

Total Budget and Work Plan			
Atlas Proposal or Award ID:	00098585	Atlas Primary Output Project ID:	00101814
Atlas Proposal or Award Title:	**Advancing Indonesia's Lighting Market to High Efficient Technologies (ADLIGHT)**		
Atlas Business Unit	IDN10		
Atlas Primary Output Project Title	**Advancing Indonesia's Lighting Market to High Efficient Technologies (ADLIGHT)**		
UNDP-GEF PIMS No.	5721		
Implementing Partner	Ministry of Energy and Mineral Resources (MEMR)		

GEF Outcome/Atlas Activity	Responsible Party/ Implementing Agent	Fund ID	Donor Name	Atlas Budgetary Account Code	ATLAS Budget Description	Year 1	Year 2	Year 3	Total	See Budget Note:
Component/Outcome 1: Improved quality, energy efficient and affordable locally-produced EE Lighting (EEL) products and systems	**MEMR**	**62000**	**GEF**	71200	International consultants	90,000	34,000	34,000	158,000	1
				71300	National consultants	100,000	18,000	18,000	136,000	2
				72100	Contractual services-companies	155,300	100,000	70,000	325,300	3
				71600	Travel	119,440	33,250	33,250	185,940	4
				75700	Training, workshop and conference	55,015	5,050	5,050	65,115	5
				74200	Audio Visual & Print Prod Costs	110,000	15,000	50,000	175,000	6
				74500	Miscellaneous	3,300	3,300	3,400	10,000	7
					Sub-total Outcome 1	**633,055**	**208,600**	**213,700**	**1,055,355**	
Component/Outcome 3: Increased penetration of high quality and efficient lighting	**MEMR**	**62000**	**GEF**	71200	International consultants	31,500	-	-	31,500	8
				71300	National consultants	35,000	35,000	35,000	105,000	9
				71400	Contractual services-Individual	-	-	-	-	
				72100	Contractual services-Companies	125,000	85,000	85,000	295,000	10
				71600	Travel	35,000	34,750	34,750	104,500	11
				74100	Professional services	-	-	-	-	
				75700	Training, workshop and conference	50,000	50,000	50,000	150,000	12

				74200	Audio Visual & Print Prod Costs	20,000	20,000	20,000	60,000	13
				72200	Equipment and furniture	533,500	50,000	50,000	633,500	14
				72800	Information Technology Equipmt	1,000	1,000	1,000	3,000	15
				72500	Supplies	-	-	-	-	
				73300	Rental of Information Technology Equipment	-	-	-	-	
				74500	Miscellaneous	3,300	3,300	3,400	10,000	16
					Sub-total Outcome 3	**834,300**	**279,050**	**279,150**	**1,392,500**	
ALL COMPONENTS					**TOTAL COMPONENT 1 and 3**	**1,467,355**	**487,650**	**492,850**	**2,447,855**	
Project Management Cost										
Project Management Cost	**MEMR**	**62000**	**GEF**	71200	International consultants	23,000	22,000	22,000	67,000	17
				71300	National consultants	21,005.20	22,005.20	22,005.20	65,015.60	18
				74100	Professional services	7,000	7,000	7,000	21,000	19
				71600	Travel	12,000	2,500	2,000	16,500	20
				72500	Supplies	2,000	1,500	1,500	5,000	21
				75700	Training, Workshop & Conference	7,500	-	3,501.40	11,001.40	22
					Sub-total PMC	**72,505.20**	**55,005.20**	**58,006.60**	**185,517**	
					TOTAL PROJECT (including PMC)	**1,539,860.20**	**542,655.20**	**550,856.60**	**2,633,372**	

Budget Notes:

No.	Budget details
	Outcome 1
1	International consultancy services for needs analysis and provision of technical assistance to selected local EEL product manufacturers on the design and production of high efficiency lighting products and systems, preparation of business transformation plans and related capacity development and establishment of a EEL knowledge center and for allocated costs for Project Management (including allocated independent consultancy costs for MTR and TE for the Component).
2	Local consultancy services to support the international consultant for needs analysis and provision of technical assistance to selected local EEL product manufacturers on the design and production of high efficiency lighting products and systems, preparation of business transformation plans and related capacity development and waste disposal/recycling of retrofit lamps and for allocated costs for Project Management (including Project Manager, Financial Officer and other support staff as well as allocated MTR and TE costs for the Component)..
3	Periodic market surveys, assessment of needs for technical assistance and provision of technical advisory support to manufacturers and ESCOs
4	Travel costs for surveys, capacity development and technical support services
5	Capacity development, training/workshops for manufacturers and banking/financial institutions and business forums
6	Awareness programs and knowledge management for manufacturers, banks, and users for EEL
7	Operation and maintenance of project equipment, report preparation costs, reproduction of reports, sundry expenses, mailing and incidental supplies and expenses
	Outcome 3
8	International consultancy services for innovative financial model enabling accelerated penetration of advanced lighting systems, focusing on the development of ESCO business models and for allocated costs for Project Management (including allocated independent consultancy costs for MTR and TE for the Component).
9	Local consultants for technical and financial advisory services on innovative financial support schemes; development and implementation of appropriate ESCO model; development and implementation of EEL pilot demonstrations; awareness and promotional programs and for allocated costs for Project Management (including Project Manager, Financial Officer and other support staff as well as allocated MTR and TE costs for the Component).
10	Contractual services-Companies for support services in the implementation of pilot demos using ESCO business models to be contracted with professional groups in performance of energy consumption pre-audits, investment grade energy audits, preparation and legal services for EPC Contracts, MVE, metering and other professional services by service providers
11	Travel costs for field visits, FGDs, meetings, surveys, capacity development and technical support services
12	Training workshops on capacity development and implementation of pilot demonstrations using ESCO business model for local government municipalities and cities and pilot demonstration hosts; development and operationalization of an EEL awareness and promotional program and information support network system, database and website
13	Awareness programs and knowledge management for manufacturers, banks, and users for active market penetration campaigns, promotional activities, and information dissemination
14	Procurement of equipment and furniture for EEL lighting systems including the design, approval, capacity development and implementation of pilot demonstrations using ESCO business models as applied to buildings and street lighting
15	Support for hardware and software and other IT requirements in developing and implementing innovative financial support schemes, accelerated LED lamp deployment for buildings and for street lighting
16	Operation and maintenance of project equipment, report preparation costs, reproduction of reports, sundry expenses, mailing and incidental supplies and expenses
	Project Management

17	Portion of independent consultant costs to undertake the project inception preparations, mid-term review and the terminal evaluation with remaining project management costs on international consultants allocated among the Components
18	Portion of the independent national consultant costs to undertake the project inception preparations, mid-term review and the terminal evaluation including that for Project Manager and Financial Officer with remaining project management costs for national consultants allocated among the Components.
19	Consultancy services for audit of project implementation
20	Supplementary travel cost for M&E and knowledge management for PMU
21	Supplies and other consumables for the PMU office
22	Inception Workshop and End-of-Project Conference on project accomplishment and related knowledge management and dissemination of results

Summary of Funds: [20]

	Amount Year 1	Amount Year 2	Amount Year 3	Total
GEF (administered by UNDP)	$1,539,860.20	$542,655.20	$550,856.60	$2,633,372
GEF (administered by UNEP)	$517,460	$425,410	$319,630	$1,262,500
Government of Indonesia	$12,136,039	$12,136,038	$3,617,260	$27,889,337
Private Sector	$5,339,000	$1,863,000	$1,863,000	$9,065,000
UNDP	$26,667	$26,667	$26,666	$80,000
UNEP	$20,000	$20,000	$20,000	$60,000
TOTAL	**$19,579,026.20**	**$15,013,770**	**$6,397,413**	**$40,990,209.00**

Note:

In accordance with Regulation of the Minister of Finance on Grant Administration Management (Regulation of the Minister of Finance No.99/PMK.05/2017), only grant provided by the donor shall be registered in the system of the Ministry of Finance. Therefore, the total grant that will be registered under the Ministry of Finance system for ADLIGHT project is that provided by the GEF in the amount of $3,895,872. While co-financing amount from government of Indonesia, private sector, UNDP and UNEP will not be registered as it will be used by each of co-financier for conducting parallel activities to ADLIGHT project. The reporting of co-financing commitment will follow donor's requirement and procedures.

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XI. LEGAL CONTEXT

The project document shall be the instrument envisaged and defined in the Supplemental Provisions (**Annex R**) to the Project Document, attached hereto and forming an integral part hereof, as "the Project Document".

This project will be implemented by Ministry of Energy and Mineral Resources/ Directorate General of New-Renewable Energy and Energy Conservation (MEMR/DGNREEC) as "Implementing Partner" in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

XII. RISK MANAGEMENT

1. Consistent with the Article III of the Supplemental Provisions to the Project Document, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

   (a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

   (b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual

exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

v. Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental

Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

10. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

11. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

12. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

13. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

14. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

15. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

16. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

XIII. MANDATORY ANNEXES:

A. Multi-year Work Plan
B. Monitoring Plan
C. Evaluation Plan
D. GEF Core Indicators (s) at baseline
E. Terms of Reference for Project Board, National Project Director, Project Manager and Component Coordinators
F. UNDP Social and Environmental and Social Screening Template (SESP)
G. UNDP Project Quality Assurance Report
H. UNDP Risk Log
I. Results of the capacity assessment of the project implementing partner and HACT micro assessment
J. Additional agreements
    - Co-financing Letter
    - Country Office Support Service Agreement
K. Descriptions of Demonstration Projects
L. GHG Emission Reduction Estimates
M. Annual Targets
N. Gender Analysis and Action Plan
O. Knowledge Management Strategy
P. Stakeholder Engagement Plan
Q. ADLIGHT Problem Tree
R. Supplemental Provisions to the Project Document: The Legal Context

Annex A: Multi Year Work Plan

The schedule of the ADLIGHT project outputs which are not shown in the Project Results Framework in **Section VI** is further detailed in the multi-year work plan as shown below.

Outcomes//Outputs	Responsible Party	Year 1	Year 2	Year 3
Component/Outcome 1: Improved quality, energy efficient and affordable locally-produced EE Lighting (EEL) products and systems				
Output 1.1: Establishment of knowledge center and systems that helps manufacturers in their production planning and policy makers in reviewing enabling environment	MEMR/DG-NREEC			
Output 1.2: Adopted and implemented business transformation plans of selected local lighting manufacturers to produce high quality energy efficient LED lamps that can meet future MEPS	MEMR/DG-NREEC			
Output 1.3: Completed capacity development program for banking/financial institutions on the evaluation and financing of lighting industry modernization projects	MEMR/DG-NREEC			
Component/Outcome 3: Increased penetration of high quality and efficient lighting				
Output 3.1: Development of an innovative financial model enabling accelerated penetration of advanced lighting systems, focusing on the development of ESCO business models	MEMR/DG-NREEC			
Output 3.2: Pilot demonstrations for accelerated LED lamp deployment in buildings and for street lighting in context of sustainable cities as well as in residential sector	MEMR/DG-NREEC			
Output 3.3: Implemented awareness and promotion program and information system explaining the benefits of high energy efficient lighting technologies taking into account gender specific aspects in developing and implementing the programmes	MEMR/DG-NREEC			

Annex B: Monitoring Plan

The Project Manager will collect results data according to the following monitoring plan.

Monitoring	Indicators	Description	Data source/Collection Method	Frequency	Responsible for data collection	Means of verification	Assumptions and Risks
Project objective To increase the penetration of high quality energy-efficient lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions	• Cumulative electricity savings by EOP, GWh	Calculated as the difference between the business as usual energy consumption on lighting and the actual electricity consumption	• MEMR/DGNREEC and PNU reporting through M&E	Annually Reported in DO tab of the GEF PIR	MEMR/ DGNREEC and PMU	• Monitoring reports from the industry and M&E reports through the UNDP-GEF APR/PIR system by MEMR and PMU in coordination with the LED manufacturers associations • Mid-term review and Terminal Evaluation Reports	• Petroleum products and electricity prices will be at levels that make energy efficiency projects, particularly energy efficient lighting (EEL), basically LED manufacturing and application, competitive and cost effective • Acceptance of the benefits and business opportunities resulting from the local manufacture and application of LED
	• Cumulative CO_2 emissions reduction, kt	Calculated from the cumulative electricity savings multiplied by the applicable emission factor for the year.	MEMR/DGNREEC and PNU reporting through M&E	Annually Reported in DO tab of the GEF PIR	MEMR/ DGNREEC and PMU	• Monitoring reports from the industry and M&E reports through the UNDP-GEF APR/PIR system by MEMR and PMU in coordination with the LED manufacturers associations • Mid-term review and Terminal Evaluation Reports	• Manufacturers remain committed to improving the energy-efficiency and quality of products. • Manufacturers remain willing to share data.
	• Number of women and men participating in capacity building trainings throughout the project	Calculated from survey results using the typical data regarding the pilot projects	MEMR/DGNREEC and PNU reporting through M&E	Annually Reported in DO tab of the GEF PIR	MEMR/ DGNREEC and PMU	Reports on survey results and analysis at inception, mid-term and EOP	The design and installation of the EEL systems is well informed on women's and men's needs, preferences and habits in consumption and disposal of EEL technologies
Component/ Outcome 1: Improved quality, energy efficient	• Efficiency of locally produced lighting systems increased, (lm/W) and	Calculated from the efficacy of lamps in terms of lumens/watt; Cost is calculated based on	Through industry association monitoring reports	Annually Reported in DO tab of the GEF PIR	MEMR/ DGNREEC and PMU	Annual progress report from PMU and industry associations	• Local lighting manufacturers will take up the transformation of their facilities as viable investment

Monitoring	Indicators	Description	Data source/Collection Method	Frequency	Responsible for data collection	Means of verification	Assumptions and Risks
and affordable locally-produced EEL products and systems	Production cost decreased from baseline level (%) o Indoor Type o Outdoor Type	average unit cost of the product during the start of the project and in the succeeding 2 years of the project					in production techniques matching efficiency and affordability of locally produced EELs • Manufacturers remain committed to improving the energy-efficiency and quality of products. • Manufacturers remain willing to share data.
	• Cumulative number of lighting manufacturers who received technical assistance on the design and production of high efficient lighting products and systems	Registered lighting manufacturers which were selected and qualified for support by the project	Through industry association monitoring reports	Annually	MEMR/ DGNREEC and PMU	Annual progress report from PMU and industry associations	• Manufacturers remain committed to improving the energy-efficiency and quality of products. • Manufacturers remain willing to share data.
	• Investment grade proposal for business transformation plans o Submitted by manufacturers o Approved by banks.	Documents on investment proposal submitted and approved with the necessary quality and completeness	Through reports by banks and verified by PMU	Reported in DO tab of the GEF PIR	MEMR/ DGNREEC and PMU	Annual progress report from PMU and industry associations	• Manufacturers remain committed to improving the energy-efficiency and quality of products. • Manufacturers remain willing to share data.
	• Ratio of women and men who believe they have the capacity to o Submit (local manufacturers) o Approve (banks) investment grade	Data collected through the market surveys	MEMR/DGNREEC and PMU reporting through M&E	Annually	MEMR/ DGNREEC and PMU	Reports on survey results and analysis at inception, mid-term and EOP	• Periodic market surveys that are designed and to be implemented to target women and men separately and are distributed to both women and men gain the cooperation of the people and the industry associations

Monitoring	Indicators	Description	Data source/Collection Method	Frequency	Responsible for data collection	Means of verification	Assumptions and Risks
	proposal for business transformation plans ○						
	•						
Component/ Outcome 3: Increased penetration of high quality and efficient lighting	Cumulative no. of innovative financial support schemes developed to accelerate penetration of EE lighting systems	Counting of documents on financial support mechanisms that have been developed	MEMR/DGNEERC and validated by PMU	Reported in DO tab of the GEF PIR	MEMR/ DGNREEC and PMU	Copy of the endorsed/approved policy and guidelines on financial support schemes	Banks and financial institutions[21] are convinced about the business potential of manufacturing and market for LED systems
	• Cumulative no. of pilot demonstrations[22] completed and replication plans developed and approved for implementation[23]	Counting of pilot demonstration and replication plans that were developed and approve	MEMR/DGNEERC and validated by PMU	Annually	MEMR/ DGNREEC and PMU	MEMR Reports on construction and operation of the demo units an dplans for relicaiton.	Banks and financial institutions[24] are convinced about the business potential of manufacturing and market for LED systems
	• Number of stakeholders engaged ○ municipalities ○ clients (project developers/ building owners) ○ technology providers	Calculated from survey results using the typical data regarding the pilot projects	MEMR/DGNEERC and validated by PMU	Annually	MEMR/ DGNREEC and PMU	Reports on survey results and analysis at inception, mid-term and EOP	The design and installation of the EEL systems is well informed on women's and men's needs, preferences and habits in consumption and disposal of EEL technologies The design and implementation of the awareness raising program

[21]Various banks and financing institutions in Indonesia, including Bank Mandiri, BRI, BNI, BCA, Bank Muamalat, BRI Syariah, BJB and Bank Artha Graha are mobilized and introduced by OJK on financing EEL business opportunities

[22] Initially identified pilot demonstrations during the PPG stage include the following: (a) smart lighting and smart monitoring system LED, (b) ESCO model for state-owned bank building, (c) ESCO model for LKPP (Public Procurement Agency) for government LED implementation, and (d) public-private partnership scheme model for LED projects.

[23] Assistance in completing the pending existing ESCO model in Smart Cities in Solo and Bandung as the 2 cities which have on-going project development, being assisted by Bappenas and LKPP, but it requires next stage of mobilization towards final financial arrangement

[24]Various banks and financing institutions in Indonesia, including Bank Mandiri, BRI, BNI, BCA, Bank Muamalat, BRI Syariah, BJB and Bank Artha Graha are mobilized and introduced by OJK on financing EEL business opportunities

Monitoring	Indicators	Description	Data source/Collection Method	Frequency	Responsible for data collection	Means of verification	Assumptions and Risks
	○ financial institutions						has considered women's and men's information needs on EEL systems and preferred channels for information sharing
	○ Women's and men's level of satisfaction with EEL systems provided (reliability, affordability, convenience, efficiency)	Calculated from survey results using the typical data regarding the pilot projects	MEMR/DGNEERC and validated by PMU	Reported in DO tab of the GEF PIR	MEMR/ DGNREEC and PMU	• Reports on workshops and promotional events conducted and on participants training evaluation. • Letters of interest from FI, technology providers and municipalities	Banks and financial institutions[25] are convinced about the business potential of manufacturing and market for LED systems
Mid-term GEF Core Indicators (if FSP project only)	N/A	N/A	Standard GEF Core Indicators available at www.thegef.org Baseline GEF Core Indicators Tool included in Annex.	After 2nd PIR submitted to GEF	MEMR/ DGNREEC and PMU)	Completed GEF Core Indicators	Delays in implementation Delays in hiring international and local consultants
Terminal GEF Core Indicators	N/A	N/A	Standard GEF Core Indicators available at www.thegef.org Baseline GEF Core Indicators included in Annex.	After final PIR submitted to GEF	MEMR/ DGNREEC and PMU)	Completed GEF Core Indicators	Reports from the industry and consuming sectors are not submitted nor analyzed for the targeted energy saving and GHG reduction
Mid-term Review (if FSP project only)	N/A	N/A	To be outlined in MTR inception report	Submitted to GEF same year as 3rd PIR	Independent evaluator	Completed MTR	Delays in implementation Delays in hiring international and local consultants
Environmental and Social risks and management plans, as relevant.	N/A	N/A	Updated SESP and management plans	Annually	Project Manager UNDP CO	Updated SESP	Lack of data and information on actual experience

[25]Various banks and financing institutions in Indonesia, including Bank Mandiri, BRI, BNI, BCA, Bank Muamalat, BRI Syariah, BJB and Bank Artha Graha are mobilized and introduced by OJK on financing EEL business opportunities

ANNEX C: EVALUATION PLAN

Evaluation Title	Planned start date Month/year	Planned end date Month/year	Included in the Country Office Evaluation Plan	Budget for consultants	Other budget (i.e. travel, site visits etc…)	Budget for translation
Mid Term review	September 2019	December 2019	Yes	USD 35,000	USD 5,000	Not needed
Terminal Evaluation	September 2021	December 2021	Yes	USD 35,000	USD 5,000	Not needed
Total evaluation budget				USD 70,000		

Annex D: GEF-7 Core Indicators (s) at baseline

Interim Offline Reporting Template for GEF-7 Core Indicators

5721 Indonesia
Baseline // Sept 2018

Contents

Core Indicator 6: Greenhouse gas emissions mitigated (metric tons of carbon dioxide equivalent).....105
Core Indicator 11: Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment.....107

Core Indicator 6: Greenhouse gas emissions mitigated (metric tons of carbon dioxide equivalent)

GHG emission type	Metric tons CO_2-eq (expected at PIF)	Metric tons CO_2-eq (expected at CEO ER)	Metric tons CO_2-eq (expected at MTR)	Metric tons CO_2-eq (expected at TE)
Lifetime direct project GHG emissions mitigated	n/a	548,776		
Lifetime direct post-project emissions mitigated		N/A (no financial mechanisms)		
Lifetime indirect GHG emissions mitigated		1,097,533		

Figure at a given stage must be the sum of all figures reported under the first two sub-indicators (6.1 and 6.2) for that stage.

6.1 Carbon sequestered or emissions avoided in the sector of Agriculture, Forestry and Other Land Use

GHG emission type	Ha (expected at PIF)	Metric tons CO_2-eq (baseline at PIF)	Ha (expected at CEO ER)	Metric tons CO_2-eq (baseline at CEO ER)	Ha (expected at MTR)	Metric tons CO_2-eq (above baseline at MTR)	Ha (expected at TE)	Metric tons CO_2-eq (above baseline at TE)
Lifetime direct project GHG emissions mitigated	n/a							
Lifetime direct post-project emissions mitigated								
Lifetime indirect GHG emissions mitigated								
Anticipated year	---		---		---		---	

6.2 Emissions avoided

GHG emission type	Metric tons CO_2-eq (baseline at PIF)	Metric tons CO_2-eq (baseline at CEO ER)	Metric tons CO_2-eq (above baseline at MTR)	Metric tons CO_2-eq (above baseline at TE)
Lifetime direct project GHG emissions mitigated	n/a	548,776		
Lifetime direct post-project emissions mitigated		N/A (no financial mechanisms)		
Lifetime indirect GHG emissions mitigated		1,097,533		
Anticipated year		2020		

6.3 Energy saved (megajoules)

Type of Intervention	MJ (expected at PIF)	MJ (expected at CEO Endorsement)	MJ (achieved at MTR)	MJ (achieved at TE)
Cumulative electricity savings by EOP	n/a	278,820,000 (77.45 GWh)		

Add rows as needed.

6.4 Increase in installed renewable energy capacity per technology (megawatts).

Type of Renewable Energy	MW (entered at PIF)	MW (entered at CEO Endorsement)	MW (entered at MTR)	MW (entered at TE)
[biomass, geothermal, ocean, small hydro, solar photovoltaic, solar thermal, wind power, and storage]	n/a			

Add rows as needed.

Core Indicator 11: Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment

	Total number (expected at PIF)	Total number (expected at CEO Endorsement)	Total number (achieved at MTR)	Total number (achieved at TE)
Women	n/a	60		
Men		140		
Total		200		

Annex E: Terms of Reference for Project Manager, Component Coordinators, Finance and Administrative Associate

1. Project Manager (PM)

Duty Station:	Jakarta with travel to other places as required.
Duration of assignment:	12 person-months with possible extension (Full-time).
Direct Supervisor:	National Project Director

Responsibilities and Duties:

The primary task of the PM is to plan, organize and implement the ADLIGHT project under the direction of the National Project Director. He/she will be responsible for in management of input mobilization and day-to-day project operations including timely resolution of issues, problems or bottlenecks. He/she will supervise the technical and administrative staff under the PMU to be established under supervision of National Project Director designated by MEMR/DGNREEC, in close coordination with the UNDP GEF Regional Technical Advisor in Regional Office in Bangkok, UNDP Indonesia CO and UNEP Task Manager in Bangkok. The PM should have multidisciplinary capabilities in energy efficient lighting (EEL) policy and institutional support development, communication and awareness; technical capacity development; energy efficiency services provision support; financing; and demonstration. To assume this overall responsibility, the specific tasks of the PM are as follows:

1. Prepare, update, and submit the required work plan and reports to UNDP Indonesia CO Director for clearance timely and with satisfactory quality, including: Annual Work Plan (AWP), Quarterly Work Plan (QWP), Annual Project Reports (APR), GEF Project Implementation Review (PIR) Report, the Quarterly Progress Reports (QPR), and the Final Project Reports as scheduled;
2. Finalize detailed ToRs for key outputs (i.e. personnel, consultants, sub-contractors, training, and procurements), within assistance from the International and National Senior Technical Advisers, and submit to NPD and UNDP Indonesia CO for clearance and approval;
3. Take responsibility for monitoring the overall progress and use of the resources of all outputs and initiative corrective actions if necessary;
4. Assume direct responsibility for managing the project budget on behalf of the NPD, ensuring that:
    - Project funds are made available when needed, and are disbursed properly;
    - Accounting records and supporting documents are kept;
    - Required financial reports are prepared;
    - Financial operations are transparent and financial procedures/regulations, in coordination with PMU, under a National Implementation Modality (NIM) as maybe adopted for ADLIGHT project;
    - Project Management is ready to stand up audit at any time.
5. Assume direct responsibility for managing the physical resources (e.g. vehicles, office, equipment, furniture, stationary…) provided to the project by UNDP;
6. Keep record and update the Project issues and risks and recommend necessary actions to NPD to effectively cope with identified risks, and coordinate the implementation of the approved risk mitigation actions;
7. Liaison and coordinate the implementation of all project partnership agreements to ensure that partners implement the project components in a collaborative manner with the Project quarterly and annual work plan;
8. Liaison with UNDP Indonesia CO or designated quality assurance personnel on the project quality management issues;

9. Supervise the project staff and local or international experts/consultants working for the project;
10. Provide back-stopping support for administrative work of the Project Management Unit if necessary;
11. Implement, monitor and report on project's Gender Action Plan and Stakeholder Engagement Plan and
12. Organize the Inception Workshop, tripartite meetings and NSC meetings as well as evaluation missions in coordination with UNDP/UNEP.

Qualifications:

- Minimum of Master's Degree in management, administration, economics, or any other relevant fields with engineering background.
- At least 5 years of proven track record on national or international project management experience, and at least three years of experience in energy project;
- Track record of, or proven experience in successful contract management, supervision and reporting skills;
- Excellent English proficiency, both in written and spoken,
- Knowledgeable about Indonesia's energy policies, EE programs and Indonesia's administrative structures;
- Familiar with technical assistance projects and GEF/UNDP/UNEP projects in Indonesia; and
- Highly self-motivated, with good leadership skills.
- Previous experience in analyzing gender sensitive data and incorporating gender aspects in energy project is considered as an asset

2. Finance and Administrative Associate

Duty Station:	Jakarta with travel to other places as required.
Duration of assignment:	12 person-months with possible extension (full time).
Direct Supervisor:	Project Manager

The Project Management Unit (PMU) will be established and be supervised by UNDP CO, in close coordination with the MEMR/DGNREEC. It will be composed of technical and administrative staff to be hired by combined resources of GEF and co-financing under the ADLIGHT Project.

Responsibilities and Duties:
The Finance and Administrative Associate shall be responsible for staff support in (i) financial management, (ii) procurement, (iii) administrative and operational backstopping, (iv) facilitation of knowledge management, document storage and sharing, and (v) other support activities to the operations of PMU towards ensuring the smooth functioning of administrative systems under the project. Specifically, he/she will be responsible for the following tasks:

1. Prepare requests for advance of funds and/or direct payments as maybe instructed by the Deputy Project Manager and approved by the Project Manager
2. Monitor budget expenditures and maintain a proper record of approved project budgets and their revisions;
3. Prepare proposals for budget revisions;
4. Prepare and submit expenditure and program budget status reports;
5. Respond to queries with respect to financial aspects of the program, liaise with UNDP appointed and external auditors wherever required;
6. Undertake other financial and administrative tasks on an ad hoc basis.

7. Support management of procurement of equipment, supplies and services and arrange for equipment maintenance and insurance as required and asset management as per UNDP guidelines
8. Maintain contacts with UNDP CO and Implementing Partners on a variety of the operational issues;
9. Maintain records on all project personnel/national consultants and their respective status (contracts, ToRs, time and attendance – if appropriate, etc.) in accordance with accepted policies and procedures
10. Carry out the administrative system and procedures consistent with UNDP's National Implementation Modality
11. Provide secretariat support to workshops, seminars and trainings, monitoring, evaluation and information dissemination.

Qualification and Experience:

- Bachelor Degree in accounting or financing,
- At least 5-years of experience as accountant, administration, preferably with UNDP/GEF project;
- Experience in project management such as in meeting deadlines, timely submission of acceptable deliverables, etc.;
- Good English communication skills, both written and verbal;
- Knowledge of the Atlas or any similar systems;
- With satisfactory skills on the use of office software packages such as MS Word, Excel and Power Point, etc.; and
- Good interpersonal skills.

3. Component Coordinator

Duty Station:	Jakarta with travel to other places as required.
Duration of assignment:	12 person-months with possible extension (full time).
Direct Supervisor:	Project Manager

The primary role of the Component Coordinator is to manage and coordinate the implementation of various capacity development, enabling environment and technical assistance activities that will be carried out under each project component. He/She will assist the PM in ensuring quality and timeliness of activities and delivery of outputs. He/She will be based at the MEMR/DGNREEC.

Responsibilities and Duties:

The Component Coordinator shall report directly to the Project Manager (PM) and shall be responsible for:

1. Managing and coordinating the implementation of all component activities to ensure the maintenance of quality and timeliness, and delivery of outputs under the component and in consonance with linked activities in other project components in producing common outputs.
2. Liaising and working closely with the project partners and beneficiaries under the component consistent with the agreed partnership arrangements towards common project goals
3. Assist the NPM in reporting regularly to the Project Board/NSC on the project's progress
4. Liaising with designated focal points from UNDP and other stakeholders, through the PM, indicating any estimated changes to the work plan, and proposing a budget revision when appropriate

5. Ensuring that the requisite allocations for the component are available in accordance with the agreed budget and established schedules of payment, if any, in consultation with the PM, MEMR/DGNREEC and UNDP
6. Working closely with key stakeholders in the drafting and preparation of relevant Terms of
7. Reference for local consultants.
8. Monitoring the project component implementation performance, particularly on the delivery of outputs.
9. Preparing progress and financial reports of the project component when required.
10. Make use of the project's accounting system to ensure accuracy and reliability of financial
11. Reporting for the component
12. Active involvement in the preparation of relevant knowledge products (including publications
13. and reports) involved in the component
14. Provision of the necessary inputs for the application of the ATLAS External User system, assisting in creating requisitions and vouchers, and other relevant ATLAS processes
15. Coordinating the management and implementation of activities of the project component as set out in the project document and recommending any such modifications/revisions as may be necessary to the PMU through the PM;
16. Provision of inputs in the preparation of the annual work plan (AWP) and budget of the
17. project component and its timely submission to the PM;
18. Provision of inputs and assistance in the preparation of regular progress reports for the component to the local implementing agency, MEMR/DGNREEC and UNDP; and,
19. Assisting the PM in reporting to the NSC the status and progress of the project component and seek appropriate recommendation if required.

Qualification and Experience:

- Advanced Degree in Mechanical/Electrical/Civil Engineering or Architecture or any other science-based background
- Possess a professional qualification or a member of an acknowledged professional organization or society.
- At least 5 years extensive working experience in the field of energy efficiency and possesses a strong knowledge of the local BSEEP industry and scenario.
- Extensive experience in project management with adequate exposure to financial management.
- Have a good command of the English language.

Annex F: UNDP Social and Environmental and Social Screening Template (SESP)

Project Information

Project Information	
1. Project Title	Advancing Indonesia's Lighting Market to High Efficient Technologies (ADLIGHT)
2. Project Number	PIMS 5721
3. Location (Global/Region/Country)	INDONESIA

Part A. Integrating Overarching Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Overarching Principles in order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the Project mainstreams the human-rights based approach
The project was designed and will be implemented with due consideration of human rights principles, through engagement of the concepts of equality in the dissemination of knowledge and sharing of benefits. ADLIGHT will implement activities that will be promoting energy efficiency activities in high-energy consuming sector, such as public buildings, thus more energy can be saved and distributed to more households/industries. In the process of selection and implementation of energy efficiency projects that will be demonstrated in the pilot province, participatory approach and free, prior and informed consent (FPIC) process will be conducted by involving stakeholders that will be affected by the projects. The implementing partner as well as the project partners acknowledge human rights practices under international law and the application of human rights-related standards.
Briefly describe in the space below how the Project is likely to improve gender equality and women's empowerment
The ADLIGHT project includes gender mainstreaming, as appropriate, to be gender responsive or gender transformative, in its implementation strategies and activities in advancing the application of EELs in the country, by means of: • Increase in participation and involvement of women (compared to the baseline level at inception) in the training, surveys, capacity building and dissemination activities for the component's activities in the business transformation plans in the manufacturers and the support groups in terms of communication, information and educational activities on energy efficiency for both women and men. • Increase of women (compared to the baseline level at inception) in public participation as a result of the training, capacity building and awareness activities from the planning stage through implementation, monitoring and evaluation, e.g., reporting malfunction in street lighting, buildings and other applications. • Considering that women have more security concerns at night, they should be involved in the project particularly in the benefits and environmental impacts of EELs. For the pilot demonstration of the RE technologies, particularly in the outer islands, the project will make sure that sufficient consultations are conducted with women and that their participation in working groups will be encouraged. The proposed GEF project will also involve women working in both management and technical departments of the MEMR and related stakeholder agencies/institutions who can play important roles in the design, development and implementation of various activities of ADLIGHT. The ADLIGHT project considered the government National Gender Policy. UNDP Indonesia in its capacity as the co-implementor of the ADLIGHT project has adopted its Gender Strategy and Action Plan that guides its gender programs for the period 2017-2020. During the project implementation, updates on relevant gender mainstreaming policy and guidelines shall be incorporated in the Project's action plans and strategies to be reviewed annually.
Briefly describe in the space below how the Project mainstreams environmental sustainability
Sustainability of the project impact will be ensured through the leadership of MEMR in cooperation with all the project stakeholders and with the provincial government which

were selected to host the project's pilot demonstrations. Their mitigation programs and administrative functions are geared towards promoting and supporting energy efficiency implementation as a key for environmental and development agenda. Enforcement of clear supporting regulatory and institutional system for EE investment, engagement of private sector, presence of local capacity to operate and maintain the EE lighting infrastructure and operational EE project financing mechanism are also other key factors ensuring sustainability of project impact. The project activities are designed in a way that encourages scale up and replication building on the experiences in the pilot demonstrations for buildings and street lighting. The project will support the Government of Indonesia in integrating the mitigation program as part of development agenda through Local Energy Planning (RUED). Thus, annually the mitigation actions will be planned and budgeted. Secondly, the scale up is also expected to result from establishment of enabling environment for effectively implementing the Business Transformation Plans, capacity building, MEPS/SNI development and promotional activities to bolster the market penetration of EE Lighting in the target sectors. The scale up and replication in other provinces are also expected to result from the piloting experience replication with the leadership of MEMR/DGNEERC.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Describe briefly potential social and environmental risks identified in Attachment 1 – Risk Screening Checklist (based on any "Yes" responses). If no risks have been identified in Attachment 1 then note "No Risks Identified" and skip to Question 4 and Select "Low Risk". Questions 5 and 6 not required for Low Risk Projects.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5 below before proceeding to Question 6*			**QUESTION 6: What social and environmental assessment and management measures have been conducted and/or are required to address potential risks (for Risks with Moderate and High Significance)?**
Risk Description	***Impact and Probability (1-5)***	***Significance (Low, Moderate, High)***	***Comments***	***Description of assessment and management measures as reflected in the Project design. If ESIA or SESA is required note that the assessment should consider all potential impacts and risks.***
Potential pollution results in the generation of waste (both hazardous and non-hazardous) from replaced fluorescent lamps/CFL and other conventional lights and particularly, mercury content and sharp cladding and bulb parts of these lamps. *(SES Principle 3, standard 7 Pollution, q1, q2, q3)*	I = 3 P = 5	Moderate	Proper waste management actions will be followed to ensure obsolete energy lamp parts as well as mercury contained therein is properly disposed of.	During project formulation (PPG phase), an assessment of available energy equipment and extent of replacement was undertaken. A strategy to manage waste/obsolete energy equipment and lamps will be developed so as to avoid potential pollution from waste lamps, and particularly mercury from replaced fluorescent lamps/CFL. The project's Activity 2.4.5: Design, planning and conduct of training on waste management and recycling of components and materials of conventional lighting replaced by LED has been included to address this issue.
The project can potentially have adverse impacts on gender equality and/or the	I = 3	Low		The professional job capacity development interventions of the project will be designed in such a way that equal opportunities

situation of women and girls in case the activities related to LED promotion and training/capacity building are skewed towards male participation. (*SES Principle 2 Gender, q1, q2)*	P = 1			exist for all genders and ages. By doing this, qualified men and women will benefit and this will contribute to the improvement and promotion of gender equality. This will also improve the number of qualified and capable women in Indonesia in technical areas such as the energy efficient lighting sector. The monitoring and evaluation of the project activities will include tracking a number of human development indicators, and among them would be gender equity, like the number of trained and employed women in energy efficient lighting sector.

	QUESTION 4: What is the overall Project risk categorization?		
	Select one (see SESP for guidance)		**Comments**
	Low Risk	☐	
	Moderate Risk	√	The project's Activity 2.4.5: Design, planning and conduct of training on waste management and recycling of components and materials of conventional lighting replaced by LED has been included to address this issue and ensure effective waste management and disposal. Gender considerations are included throughout all project activities.
	High Risk	☐	
	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are relevant?		
	Check all that apply		**Comments**
	Principle 1: Human Rights	☐	
	Principle 2: Gender Equality and Women's Empowerment	√	Participation of women need to be encouraged in the project's gender mainstreaming, as appropriate, to be gender responsive or gender transformative, in its implementation strategies and activities in advancing the application of EELs in the country. The activity areas include surveys, information dissemination, planning, demonstration, monitoring and reporting or almost in all areas of the project.
	1. Biodiversity Conservation and Natural Resource Management	☐	
	2. Climate Change Mitigation and Adaptation	☐	
	3. Community Health, Safety and Working Conditions	☐	
	4. Cultural Heritage	☐	
	5. Displacement and Resettlement	☐	

	6. Indigenous Peoples	☐	
	7. Pollution Prevention and Resource Efficiency	√	Potential pollution and risks from hazardous and non-hazardous waste materials from replaced fluorescent lamps/CFL and other conventional lighting systems and accessories particularly mercury content and sharp glass bulb parts need to be managed and proper disposal system be implemented

Final Sign Off

Signature	***Date***	***Description***
QA Assessor		UNDP staff member responsible for the Project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD)**,** Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks	
Principles 1: Human Rights	**Answer (Yes/No)**
1. Could the Project lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
2. Is there a likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups? [26]	No
3. Could the Project potentially restrict availability, quality of and access to resources or basic services, in particular to marginalized individuals or groups?	No
4. Is there a likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them?	No
5. Is there a risk that duty-bearers do not have the capacity to meet their obligations in the Project?	No
6. Is there a risk that rights-holders do not have the capacity to claim their rights?	No
7. Have local communities or individuals, given the opportunity, raised human rights concerns regarding the Project during the stakeholder engagement process?	No
8. Is there a risk that the Project would exacerbate conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Principle 2: Gender Equality and Women's Empowerment	
1. Is there a likelihood that the proposed Project would have adverse impacts on gender equality and/or the situation of women and girls?	Yes
2. Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
3. Have women's groups/leaders raised gender equality concerns regarding the Project during the stakeholder engagement process and has this been included in the overall Project proposal and in the risk assessment?	No
4. Would the Project potentially limit women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
Principle 3: Environmental Sustainability: Screening questions regarding environmental risks are encompassed by the specific Standard-related questions below	
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management	
1.1 Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	No
1.2 Are any Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3 Does the Project involve changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4 Would Project activities pose risks to endangered species?	No
1.5 Would the Project pose a risk of introducing invasive alien species?	No
1.6 Does the Project involve harvesting of natural forests, plantation development, or reforestation?	No
1.7 Does the Project involve the production and/or harvesting of fish populations or other aquatic species?	No
1.8 Does the Project involve significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.9 Does the Project involve utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)	No
1.10 Would the Project generate potential adverse trans-boundary or global environmental concerns?	No
1.11 Would the Project result in secondary or consequential development activities which could lead to adverse social and environmental effects, or would it generate cumulative impacts with other known existing or planned activities in the area? *For example, a new road through forested lands will generate direct environmental and social impacts (e.g. felling of trees, earthworks, potential relocation of inhabitants). The new road may also facilitate encroachment on lands by illegal settlers or generate unplanned commercial development along the route, potentially in*	No

[26] Prohibited grounds of discrimination include race, ethnicity, gender, age, language, disability, sexual orientation, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender people and transsexuals.

	sensitive areas. These are indirect, secondary, or induced impacts that need to be considered. Also, if similar developments in the same forested area are planned, then cumulative impacts of multiple activities (even if not part of the same Project) need to be considered.	
Standard 2: Climate Change Mitigation and Adaptation		
2.1	Will the proposed Project result in significant[27] greenhouse gas emissions or may exacerbate climate change?	No
2.2	Would the potential outcomes of the Project be sensitive or vulnerable to potential impacts of climate change?	No
2.3	Is the proposed Project likely to directly or indirectly increase social and environmental vulnerability to climate change now or in the future (also known as maladaptive practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
Standard 3: Community Health, Safety and Working Conditions		
3.1	Would elements of Project construction, operation, or decommissioning pose potential safety risks to local communities?	No
3.2	Would the Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	No
3.3	Does the Project involve large-scale infrastructure development (e.g. dams, roads, buildings)?	No
3.4	Would failure of structural elements of the Project pose risks to communities? (e.g. collapse of buildings or infrastructure)	No
3.5	Would the proposed Project be susceptible to or lead to increased vulnerability to earthquakes, subsidence, landslides, and erosion, flooding or extreme climatic conditions?	No
3.6	Would the Project result in potential increased health risks (e.g. from water-borne or other vector-borne diseases or communicable infections such as HIV/AIDS)?	No
3.7	Does the Project pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during Project construction, operation, or decommissioning?	No
3.8	Does the Project involve support for employment or livelihoods that may fail to comply with national and international labor standards (i.e. principles and standards of ILO fundamental conventions)?	No
3.9	Does the Project engage security personnel that may pose a potential risk to health and safety of communities and/or individuals (e.g. due to a lack of adequate training or accountability)?	No
Standard 4: Cultural Heritage		
4.1	Will the proposed Project result in interventions that would potentially adversely impact sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: Projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.2	Does the Project propose utilizing tangible and/or intangible forms of cultural heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
5.1	Would the Project potentially involve temporary or permanent and full or partial physical displacement?	No
5.2	Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3	Is there a risk that the Project would lead to forced evictions?[28]	No
5.4	Would the proposed Project possibly affect land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
6.1	Are indigenous peoples present in the Project area (including Project area of influence)?	No
6.2	Is it likely that the Project or portions of the Project will be located on lands and territories claimed by indigenous peoples?	No
6.3	Would the proposed Project potentially affect the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the Project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to the screening question 6.3 is "yes" the potential risk impacts are considered potentially severe and/or critical and the Project would be categorized as either Moderate or High Risk.*	No
6.4	Has there been an absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	Does the proposed Project involve the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No

[27] In regards to CO_2, 'significant emissions' corresponds generally to more than 25,000 tons per year (from both direct and indirect sources). [The Guidance Note on Climate Change Mitigation and Adaptation provides additional information on GHG emissions.]

[28] Forced evictions include acts and/or omissions involving the coerced or involuntary displacement of individuals, groups, or communities from homes and/or lands and common property resources that were occupied or depended upon, thus eliminating the ability of an individual, group, or community to reside or work in a particular dwelling, residence, or location without the provision of, and access to, appropriate forms of legal or other protections.

6.6	Is there a potential for forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources?	No
6.7	Would the Project adversely affect the development priorities of indigenous peoples as defined by them?	No
6.8	Would the Project potentially affect the physical and cultural survival of indigenous peoples?	No
6.9	Would the Project potentially affect the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices?	No
Standard 7: Pollution Prevention and Resource Efficiency		
7.1	Would the Project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or trans-boundary impacts?	Yes[29]
7.2	Would the proposed Project potentially result in the generation of waste (both hazardous and non-hazardous)?	Yes
7.3	Will the proposed Project potentially involve the manufacture, trade, release, and/or use of hazardous chemicals and/or materials? Does the Project propose use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Stockholm Conventions on Persistent Organic Pollutants or the Montreal Protocol*	Yes
7.4	Will the proposed Project involve the application of pesticides that may have a negative effect on the environment or human health?	No
7.5	Does the Project include activities that require significant consumption of raw materials, energy, and/or water?	No

Annex G: UNDP Project Quality Assurance Report

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

OVERALL PROJECT

EXEMPLARY (5) ◉◉◉◉◉	HIGHLY SATISFACTORY (4) ◉◉◉◉○	SATISFACTORY (3) ◉◉◉○○	NEEDS IMPROVEMENT (2) ◉◉○○○	INADEQUATE (1) ◉○○○○
At least four criteria are rated Exemplary, and all criteria are rated High or Exemplary.	All criteria are rated Satisfactory or higher, and at least four criteria are rated High or Exemplary.	At least six criteria are rated Satisfactory or higher, and only one may be rated Needs Improvement. The SES criterion must be rated Satisfactory or above.	At least three criteria are rated Satisfactory or higher, and only four criteria may be rated Needs Improvement.	One or more criteria are rated Inadequate, or five or more criteria are rated Needs Improvement.

DECISION

- **APPROVE** – the project is of sufficient quality to continue as planned**.** Any management actions must be addressed in a timely manner.
- **APPROVE WITH QUALIFICATIONS** – the project has issues that must be addressed before the project document can be approved. Any management actions must be addressed in a timely manner.
- **DISAPPROVE** – the project has significant issues that should prevent the project from being approved as drafted.

RATING CRITERIA

STRATEGIC

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1	

Evidence

1. **Does the project's Theory of Change specify how it will contribute to higher level change? (Select the option from 1-3 that best reflects the project):**
   - **3:** The project has a theory of change with explicit assumptions and clear change pathway describing how the project will contribute to outcome level change as specified in the programme/CPD, backed by credible evidence of what works effectively in this context. The project document clearly describes why the project's strategy is the best approach at this point in time.
   - **2:** The project has a theory of change. It has an explicit change pathway that explains how the project intends to contribute to outcome-level change and why the project strategy is the best approach at this point in time, but is backed by limited evidence.
   - **1:** The project does not have a theory of change, but the project document may describe in generic terms how the project will contribute to development results, without specifying the key assumptions. It does not make an explicit link to the programme/CPD's theory of change.

*Note: Management Action or strong management justification must be given for a score of 1

Option 3. The project has a theory of change that considers critical factors for achieving an increased market penetration of energy efficient lighting systems, thus contributing to the UNPDF/CPD Outcome 3: By 2020, Indonesia is sustainably managing its natural resources, on land and at sea, with an increased resilience to the effects of climate change, disasters and other shocks (Project Document's Strategy chapter).

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

2. Is the project aligned with the thematic focus of the UNDP Strategic Plan? (select the option from 1-3 that best reflects the project):

- **3:** The project responds to one of the three areas of development work[1] as specified in the Strategic Plan; it addresses at least one of the proposed new and emerging areas; an issues-based analysis has been incorporated into the project design; and the project's RRF includes all the relevant SP output indicators. *(all must be true to select this option)*
- **2:** The project responds to one of the three areas of development work[1] as specified in the Strategic Plan. The project's RRF includes at least one SP output indicator, if relevant. *(both must be true to select this option)*
- **1:** While the project may respond to one of the three areas of development work[1] as specified in the Strategic Plan, it is based on a sectoral approach without addressing the complexity of the development issue. None of the relevant SP indicators are included in the RRF. This answer is also selected if the project does not respond to any of the three areas of development work in the Strategic Plan.

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Evidence

Option 3. The project responds to the development area Sustainable development pathways in the Strategic Plan and addresses the new and emerging area Access to modern energy services and energy efficiency. This will be achieved by promoting the increased deployment of high efficiency lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions.

Strategic Plan Areas of Development Work	Strategic Plan New and Emerging Issues
☒ Sustainable Development Pathways	☐ Sustainable production technologies
☐ Inclusive and effective democratic governance	☒ Access to modern energy services
☐ Resilience building	☐ Natural resources management
	☐ Extractive industries
	☐ Urbanisation
	☐ Citizen security
	☐ Social protection
	☐ Risk management for resilience

.

RELEVANT

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1	

[1] 1. Sustainable development pathways; 2. Inclusive and effective democratic governance; 3. Resilience building

[29] Potential pollution and risks from hazardous and non-hazardous waste materials from replaced fluorescent lamps/CFL and other conventional lighting systems and accessories particularly mercury content and sharp glass bulb parts need to be managed and proper disposal system be implemented and monitored.

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

3. Does the project have strategies to effectively identify, engage and ensure the meaningful participation of targeted groups/geographic areas with a priority focus on the excluded and marginalized? (select the option from 1-3 that best reflects this project):

- **3:** The target groups/geographic areas are appropriately specified, prioritizing the excluded and/or marginalized. Beneficiaries will be identified through a rigorous process based on evidence (if applicable.) The project has an explicit strategy to identify, engage and ensure the meaningful participation of specified target groups/geographic areas throughout the project, including through monitoring and decision-making (such as representation on the project board) *(all must be true to select this option)*
- **2:** The target groups/geographic areas are appropriately specified, prioritizing the excluded and/or marginalized. The project document states how beneficiaries will be identified, engaged and how meaningful participation will be ensured throughout the project. *(both must be true to select this option)*
- **1:** The target groups/geographic areas are not specified, or do not prioritize excluded and/or marginalized populations. The project does not have a written strategy to identify or engage or ensure the meaningful participation of the target groups/geographic areas throughout the project.

*Note: Management Action must be taken for a score of 1

Option 2. The main target groups/geographic areas of the project have been identified in the project design. Ways to ensure their participation in the project have been identified in the Stakeholder Engagement Plan. A more detailed specification of beneficiaries and pilot areas will be carried out during project implementation.

ROAR Target Groups:	ROAR Target Geographical Areas:
☐ Extreme poor	☒ Urban (large cities)
☐ Women	☐ Urban (small towns)
☐ Female Headed Households	☐ Peri-urban (surround metropolitan areas & cities)
☐ Youth	☐ Rural
☒ Others (specify): Energy efficiency lighting sector: local EEL manufacturers, financing institutions, government agencies, and end users who will benefit from LED lighting in buildings and on the streets of pilot areas	☐ Others (specify): _____

.

Select (all) targeted groups: (drop-down)
Evidence

4. Have knowledge, good practices, and past lessons learned of UNDP and others informed the project design? (select the option from 1-3 that best reflects this project):

- **3:** Knowledge and lessons learned (gained e.g. through peer assist sessions) backed by credible evidence from evaluation, corporate policies/strategies, and monitoring have been explicitly used, with appropriate referencing, to develop the project's theory of change and justify the approach used by the project over alternatives.
- **2:** The project design mentions knowledge and lessons learned backed by evidence/sources, which inform the project's theory of change but have not been used/are not sufficient to justify the approach selected over alternatives.
- **1:** There is only scant or no mention of knowledge and lessons learned informing the project design. Any references that are made are not backed by evidence.

*Note: Management Action or strong management justification must be given for a score of 1

Option 3. The ADLIGHT project is built upon knowledge and relevant ongoing energy efficiency lighting projects/programs as well as already completed projects. The added value of ADLIGHT to the contributions of ongoing/completed initiatives in the area of application of local efficient lighting products is summarized in the project document (Project Document's Development Challenge chapter).

3 | 2 | 1

Evidence

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

5. Does the project use gender analysis in the project design and does the project respond to this gender analysis with concrete measures to address gender inequities and empower women? (select the option from 1-3 that best reflects this project):

- **3:** A participatory gender analysis on the project has been conducted. This analysis reflects on the different needs, roles and access to/control over resources of women and men, and it is fully integrated into the project document. The project establishes concrete priorities to address gender inequalities in its strategy. The results framework includes outputs and activities that specifically respond to this gender analysis, with indicators that measure and monitor results contributing to gender equality. *(all must be true to select this option)*
- **2:** A gender analysis on the project has been conducted. This analysis reflects on the different needs, roles and access to/control over resources of women and men. Gender concerns are integrated in the development challenge and strategy sections of the project document. The results framework includes outputs and activities that specifically respond to this gender analysis, with indicators that measure and monitor results contributing to gender equality. *(all must be true to select this option)*
- **1:** The project design may or may not mention information and/or data on the differential impact of the project's development situation on gender relations, women and men, but the constraints have not been clearly identified and interventions have not been considered.

*Note: Management Action or strong management justification must be given for a score of 1

Option 2. A gender analysis has been conducted during the project design (Annex N). Based on the gender analysis, the project has included gender mainstreaming, as appropriate, in its implementation strategies, outputs, activities, results framework and indicators.

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1	

Evidence

6. Does UNDP have a clear advantage to engage in the role envisioned by the project vis-à-vis national partners, other development partners, and other actors? (select from options 1-3 that best reflects this project):

- **3:** An analysis has been conducted on the role of other partners in the area where the project intends to work, and credible evidence supports the proposed engagement of UNDP and partners through the project. It is clear how results achieved by relevant partners will contribute to outcome level change complementing the project's intended results. If relevant, options for south-south and triangular cooperation have been considered, as appropriate. *(all must be true to select this option)*
- **2:** Some analysis has been conducted on the role of other partners where the project intends to work, and relatively limited evidence supports the proposed engagement of and division of labour between UNDP and partners through the project. Options for south-south and triangular cooperation may not have not been fully developed during project design, even if relevant opportunities have been identified.
- **1:** No clear analysis has been conducted on the role of other partners in the area that the project intends to work, and relatively limited evidence supports the proposed engagement of UNDP and partners through the project. There is risk that the project overlaps and/or does not coordinate with partners' interventions in this area. Options for south-south and triangular cooperation have not been considered, despite its potential relevance.

*Note: Management Action or strong management justification must be given for a score of 1

Option 3. Relevant partners have been identified and their role in project implementation has been outlined in the project document. The ADLIGHT Project will also pursue and promote South-South and triangular cooperation (SSTC) forms, such as knowledge exchange, technology transfer, financing, peer support, and country-to country or multi country initiatives, on high efficient lighting technology. The SSTC activities will be facilitated by using global platform under UNEP Leapfrogging global programme (Results and Partnerships chapter)

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Evidence

SOCIAL & ENVIRONMENTAL STANDARDS

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

7. Does the project seek to further the realization of human rights using a human rights based approach? (select from options 1-3 that best reflects this project):

- **3:** Credible evidence that the project aims to further the realization of human rights, upholding the relevant international and national laws and standards in the area of the project. Any potential adverse impacts on enjoyment of human rights were rigorously identified and assessed as relevant, with appropriate mitigation and management measures incorporated into project design and budget. *(all must be true to select this option)*
- **2:** Some evidence that the project aims to further the realization of human rights. Potential adverse impacts on enjoyment of human rights were identified and assessed as relevant, and appropriate mitigation and management measures incorporated into the project design and budget.
- **1:** No evidence that the project aims to further the realization of human rights. Limited or no evidence that potential adverse impacts on enjoyment of human rights were considered.

*Note: Management action or strong management justification must be given for a score of 1

Option 2. The project was designed and will be implemented with due consideration of human rights principles, through engagement of the concepts of equality in the dissemination of knowledge and sharing of benefits. In the process of selection and implementation and energy efficiency projects that will be demonstrated in the pilot province, participatory approach and free, prior and informed consent (FPIC) process will be conducted by involving stakeholders that will be affected by the projects (Annex P)

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Evidence

8. Did the project consider potential environmental opportunities and adverse impacts, applying a precautionary approach? (select from options 1-3 that best reflects this project):

- **3:** Credible evidence that opportunities to enhance environmental sustainability and integrate poverty-environment linkages were fully considered as relevant, and integrated in project strategy and design. Credible evidence that potential adverse environmental impacts have been identified and rigorously assessed with appropriate management and mitigation measures incorporated into project design and budget. *(all must be true to select this option)*.
- **2:** No evidence that opportunities to strengthen environmental sustainability and poverty-environment linkages were considered. Credible evidence that potential adverse environmental impacts have been identified and assessed, if relevant, and appropriate management and mitigation measures incorporated into project design and budget.
- **1:** No evidence that opportunities to strengthen environmental sustainability and poverty-environment linkages were considered. Limited or no evidence that potential adverse environmental impacts were adequately considered.

*Note: Management action or strong management justification must be given for a score of 1

Option 3. The project will contribute to carbon emission reduction through promotion and deployment of energy efficient lighting systems. The project's environmental effects involve pollution from hazardous and non-hazardous waste from replaced fluorescent lamps/CFL and other conventional lights and particularly, mercury content and sharp cladding and bulb parts of these lamps. As a mitigation measure, the project will develop a strategy to manage waste/obsolete energy equipment and lamps. (Annex H)

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Evidence

9. Has the Social and Environmental Screening Procedure (SESP) been conducted to identify potential social and	Yes	No
	SESP Not Required	

MANAGEMENT & MONITORING

10. Does the project have a strong results framework? (select from options 1-3 that best reflects this project):	3	2
	1	

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

• **3:** The project's selection of outputs and activities are at an appropriate level and relate in a clear way to the project's theory of change. Outputs are accompanied by SMART, results-oriented indicators that measure all of the key expected changes identified in the theory of change, each with credible data sources, and populated baselines and targets, including gender sensitive, sex-disaggregated indicators where appropriate. *(all must be true to select this option)* • **2:** The project's selection of outputs and activities are at an appropriate level, but may not cover all aspects of the project's theory of change. Outputs are accompanied by SMART, results-oriented indicators, but baselines, targets and data sources may not yet be fully specified. Some use of gender sensitive, sex-disaggregated indicators, as appropriate. *(all must be true to select this option)* • **1:** The results framework does not meet all of the conditions specified in selection "2" above. This includes: the project's selection of outputs and activities are not at an appropriate level and do not relate in a clear way to the project's theory of change; outputs are not accompanied by SMART, results-oriented indicators that measure the expected change, and have not been populated with baselines and targets; data sources are not specified, and/or no gender sensitive, sex-disaggregation of indicators. *Note: Management Action or strong management justification must be given for a score of 1 Option 3. The project's Results Framework relates well to the theory of change and includes SMART indicators and sex-disaggregated data where appropriate. One of the baselines is not yet populated and will be established through market surveys during project implementation. (Project Results Framework)	**Evidence**
11. Is there a comprehensive and costed M&E plan in place with specified data collection sources and methods to support evidence-based management, monitoring and evaluation of the project? Yes, the project has a M&E plan including M&E budget in the project document (Monitoring and Evaluation Plan).	Yes (3) / No (1)
12. Is the project's governance mechanism clearly defined in the project document, including planned composition of the project board? (select from options 1-3 that best reflects this project): • **3:** The project's governance mechanism is fully defined in the project composition. Individuals have been specified for each position in the governance mechanism (especially all members of the project board.) Project Board members have agreed on their roles and responsibilities as specified in the terms of reference. The ToR of the project board has been attached to the project document. *(all must be true to select this option)*. • **2:** The project's governance mechanism is defined in the project document; specific institutions are noted as holding key governance roles, but individuals may not have been specified yet. The prodoc lists the most important responsibilities of the project board, project director/manager and quality assurance roles. *(all must be true to select this option)* • **1:** The project's governance mechanism is loosely defined in the project document, only mentioning key roles that will need to be filled at a later date. No information on the responsibilities of key positions in the governance mechanism is provided. *Note: Management Action or strong management justification must be given for a score of 1 Option 2. Roles and responsibilities of the project's governance mechanism are outlined in the project document (Governance and Management Arrangements chapter and Annex F). The ToR of the Project Board are not attached to the project document.	3 / 2 1 **Evidence**
13. Have the project risks been identified with clear plans stated to manage and mitigate each risks? (select from options 1-3 that best reflects this project): • **3:** Project risks related to the achievement of results are fully described in the project risk log, based on comprehensive analysis drawing on the theory of change, Social and Environmental Standards and screening, situation analysis, capacity assessments and other analysis. Clear and complete plan in place to manage and mitigate each risk. *(both must be true to select this option)* • **2:** Project risks related to the achievement of results identified in the initial project risk log with mitigation measures identified for each risk.	3 / 2 1 **Evidence**

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

- **1:** Some risks may be identified in the initial project risk log, but no evidence of analysis and no clear risk mitigation measures identified. This option is also selected if risks are not clearly identified and no initial risk log is included with the project document.

*Note: Management Action must be taken for a score of 1

Option 2. UNDP risk log describes the project's risks and mitigation measures and is attached to the project document. (Annex H)

EFFICIENT

14. Have specific measures for ensuring cost-efficient use of resources been explicitly mentioned as part of the project design? This can include: i) using the theory of change analysis to explore different options of achieving the maximum results with the resources available; ii) using a portfolio management approach to improve cost effectiveness through synergies with other interventions; iii) through joint operations (e.g., monitoring or procurement) with other partners.

Yes (3) | No (1)

Yes. Cost efficiency and effectiveness will be achieved by strategies that leverage limited inputs of GEF funding to stimulate much greater investments by other parties, namely the government and the private sector. The project is expecting to generate USD 18,115,000 in private sector investments within the project timeline and additional resources as the pilot demonstrations are replicated in the country. (Feasiblity chapter)

15. Are explicit plans in place to ensure the project links up with other relevant on-going projects and initiatives, whether led by UNDP, national or other partners, to achieve more efficient results (including, for example, through sharing resources or coordinating delivery?)

Yes (3) | No (1)

Yes. The project is in line with the planned interventions of the national government in the development of standards for energy efficient lighting according to the ASEAN Plan of Action Energy Cooperation 2016-2025 (APAEC), which highlights regional harmonization of LED lighting standards. The ADLIGHT project is also in line with the Solo city government's planned installation of LED street lighting. The linkages between the ADLIGHT project and these interventions have been considered throughout the project document.

16. Is the budget justified and supported with valid estimates?

3 | 2

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Evidence

- **3:** The project's budget is at the activity level with funding sources, and is specified for the duration of the project period in a multi-year budget. Costs are supported with valid estimates using benchmarks from similar projects or activities. Cost implications from inflation and foreign exchange exposure have been estimated and incorporated in the budget.
- **2:** The project's budget is at the activity level with funding sources, when possible, and is specified for the duration of the project in a multi-year budget. Costs are supported with valid estimates based on prevailing rates.
- **1:** The project's budget is not specified at the activity level, and/or may not be captured in a multi-year budget.

Option 3. The project's budget is at the activity level presented in the Total Budget and Work Plan chapter.

17. Is the Country Office fully recovering the costs involved with project implementation?

3 | 2

1

Evidence

- **3:** The budget fully covers all project costs that are attributable to the project, including programme management and development effectiveness services related to strategic country programme planning, quality assurance, pipeline development, policy advocacy services, finance, procurement, human resources, administration, issuance of contracts, security, travel, assets, general services, information and communications based on full costing in accordance with prevailing UNDP policies (i.e., UPL, LPL.)

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

- **2:** The budget covers significant project costs that are attributable to the project based on prevailing UNDP policies (i.e., UPL, LPL) as relevant.
- **1:** The budget does not adequately cover project costs that are attributable to the project, and UNDP is cross-subsidizing the project.

*Note: Management Action must be given for a score of 1. The budget must be revised to fully reflect the costs of implementation before the project commences.

Option 3. The budget fully covers all project costs.

EFFECTIVE

18. Is the chosen implementation modality most appropriate? (select from options 1-3 that best reflects this project):

- **3:** The required implementing partner assessments (capacity assessment, HACT micro assessment) have been conducted, and there is evidence that options for implementation modalities have been thoroughly considered. There is a strong justification for choosing the selected modality, based on the development context. *(both must be true to select this option)*
- **2:** The required implementing partner assessments (capacity assessment, HACT micro assessment) have been conducted and the implementation modality chosen is consistent with the results of the assessments.
- **1:** The required assessments have not been conducted, but there may be evidence that options for implementation modalities have been considered.

*Note: Management Action or strong management justification must be given for a score of 1

Assessment of the implementing partner has been conducted by another project under the same implementing partner – Directorate General of New Renewable Energy and Energy Conservation, within the Ministry of Energy and Mineral Resources (Annex I). Based on the assessment conducted, the proposed implementation modality for project components under UNDP's responsibility is NIM.

3	2
1	

Evidence

19. Have targeted groups, prioritizing marginalized and excluded populations that will be affected by the project, been engaged in the design of the project in a way that addresses any underlying causes of exclusion and discrimination?

- 3: Credible evidence that all targeted groups, prioritising marginalized and excluded populations that will be involved in or affected by the project, have been actively engaged in the design of the project. Their views, rights and any constraints have been analysed and incorporated into the root cause analysis of the theory of change which seeks to address any underlying causes of exclusion and discrimination and the selection of project interventions.
- 2: Some evidence that key targeted groups, prioritising marginalized and excluded populations that will be involved in the project, have been engaged in the design of the project. Some evidence that their views, rights and any constraints have been analysed and incorporated into the root cause analysis of the theory of change and the selection of project interventions.
- 1: No evidence of engagement with marginalized and excluded populations that will be involved in the project during project design. No evidence that the views, rights and constraints of populations have been incorporated into the project.

Option 1. During the project design the target groups engaged were national and local government institutions, lighting manufacturers and financing institutions. Marginalized and excluded population will be engaged through surveys and stakeholder consultations before, during and after implementation of energy efficiency projects in pilot areas.

3	2
1	

Evidence

	Yes	No
20. Does the project conduct regular monitoring activities, have explicit plans for evaluation, and include other lesson learning (e.g. through After Action Reviews or Lessons Learned Workshops), timed to inform course corrections if needed during project implementation? Yes. The project will be monitored and evaluated periodically during project implementation according to the Monitoring and Evaluation plan.	(3)	(1)
21. The gender marker for all project outputs are scored at GEN2 or GEN3, indicating that gender has been fully mainstreamed into all project outputs at a minimum.	(3)	(1)

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

*Note: Management Action or strong management justification must be given for a score of "no" Yes. The project's outputs score GEN2.	**Evidence**
22. Is there a realistic multi-year work plan and budget to ensure outputs are delivered on time and within allotted resources? (select from options 1-3 that best reflects this project): • **3:** The project has a realistic work plan & budget covering the duration of the project *at the activity* level to ensure outputs are delivered on time and within the allotted resources. • **2:** The project has a work plan & budget covering the duration of the project at the output level. • **1:** The project does not yet have a work plan & budget covering the duration of the project. Option 3. The project has a budget and work plan at the activity level.	3 | 2 1 **Evidence**

SUSTAINABILITY & NATIONAL OWNERSHIP

23. Have national partners led, or proactively engaged in, the design of the project? (select from options 1-3 that best reflects this project): • **3:** National partners have full ownership of the project and led the process of the development of the project jointly with UNDP. • **2:** The project has been developed by UNDP in close consultation with national partners. • **1:** The project has been developed by UNDP with limited or no engagement with national partners. Option 2. The project has been designed in close consultation with the Ministry of Energy and Mineral Resources.	3 | 2 1 **Evidence**
24. Are key institutions and systems identified, and is there a strategy for strengthening specific/ comprehensive capacities based on capacity assessments conducted? (select from options 0-4 that best reflects this project): • **3:** The project has a comprehensive strategy for strengthening specific capacities of national institutions based on a systematic and detailed capacity assessment that has been completed. This strategy includes an approach to regularly monitor national capacities using clear indicators and rigorous methods of data collection, and adjust the strategy to strengthen national capacities accordingly. • **2.5:** A capacity assessment has been completed. The project document has identified activities that will be undertaken to strengthen capacity of national institutions, but these activities are not part of a comprehensive strategy to monitor and strengthen national capacities. • **2:** A capacity assessment is planned after the start of the project. There are plans to develop a strategy to strengthen specific capacities of national institutions based on the results of the capacity assessment. • **1.5:** There is mention in the project document of capacities of national institutions to be strengthened through the project, but no capacity assessments or specific strategy development are planned. • **1:** Capacity assessments have not been carried out and are not foreseen. There is no strategy for strengthening specific capacities of national institutions. Option 2.5. A capacity assessment of government institutions has been conducted during project design and is part of the development challenge and strategy of the project. Capacity development for institutions on of EE lighting systems is part of the project's outputs	3 | 2.5 2 | 1.5 1 **Evidence**
25. Is there is a clear strategy embedded in the project specifying how the project will use national systems (i.e., procurement, monitoring, evaluations, etc.,) to the extent possible? Yes. The project is proposed to be implemented under NIM modality for project components under UNDP's responsibility. In this case, if requested by the government, Country Office Support Services will be adopted by UNDP.	Yes (3) | No (1)
26. Is there a clear transition arrangement/ phase-out plan developed with key stakeholders in order to sustain or scale up results (including resource mobilisation strategy)? Yes. The project activities are designed to encourage scaling up of pilot demonstrations for buildings and street lighting. The scale up is also expected to result from establishment of enabling environment for	Yes (3) | No (1)

effectively implementing the Business Transformation Plans, capacity building, MEPS/SNI development and promotional activities to bolster the deployment of EE Lighting in the target sectors.		

ANNEX H: UNDP Risk Log

Project risks					
Description	**Type**	**Impact & Probability**	**Mitigation Measures**	**Owner**	**Status**
1. Prolonged delays in the implementation of MEPS due to Government procedures and processes	Political Regulatory	P = 2 I = 3	*BSN* to come up with *SNI* standards that can be used as reference in the government procurements and other EEL applications while MEPS is being developed	MEMR/DGNREEC BSN Industry associations Min of Trade and Industry	Reducing
2. Local lighting manufacturers may not be interested to upgrade the lighting production facilities due to increased investments, disruption to operation and business priorities	Financial Operational Strategic	P = 3 I = 3	Coordinate closely with manufacturers associations and local banks to facilitate investments; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
3. Petroleum products and electricity prices will be at levels that make energy efficiency projects, particularly in EEL, uncompetitive and not cost effective	Strategic Environmental Political	P = 3 I = 3	Government to be encouraged to adopt appropriate energy pricing policies and incentives to encourage EE project investments	MEMR/DGNREEC Min of finance Min of Planning Industry associations Min of Trade and Industry	Increasing
4. Manufacturers are not committed to improving the energy-efficiency and quality of products.	Organizational Operational	P = 2 I = 3	Coordinate closely with manufacturers associations; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
5. Manufacturers are not willing to share data on their operations and marketing for MVE	Financial Operational	P = 3 I = 3	Coordinate closely with manufacturers associations; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
6. Banks and financial institutions are not convinced about the business potential of manufacturing and market	Financial Operational Organizational	P = 3 I = 3	Coordinate closely with manufacturers associations and local banks to facilitate investments; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry *OJK*	Reducing

Project risks					
Description	**Type**	**Impact & Probability**	**Mitigation Measures**	**Owner**	**Status**
for LED systems				Financial institutions/Banks	
7. Commitment to abide with regional and international EEL standards and harmonization is lacking	Political Regulatory	P = 2 I = 3	Implement Component 2 on policy and institutional support effectively	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
8. Policy framework may not attractive for manufacturers and users to invest in EEL projects.	Political Regulatory	P = 3 I = 3	Implement Component 2 on policy and institutional support effectively	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
9. Local EEL producers could not meet all technical standard specifications	Financial Operational Organizational	P = 3 I = 3	Coordinate closely with manufacturers associations; facilitate through ESCOs and TWG	MEMR/DGNREEC Industry associations Min of Trade and Industry	Reducing
Overall		P = 3 I = 3			

Annex I: Results of the capacity assessment of the project implementing partner and HACT micro assessment

Micro Assessment of the Ministry of Energy & Mineral Resources (Directorate General of EBTKE) - Implementing Partner of the MTRE3 project
Commissioned by UNDP Indonesia
Individual Consultant : Osa Hartoyo
Date : 2 October 2017

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1. Background, Scope and Methodology
2. Summary of Risk Assessment Results
3. Detailed Internal Control Findings and Recommendations

Annex I. Implementing Partner and Programme Information
Annex II. Organisational Chart of the Implementing Partner
Annex III. List of persons met
Annex IV. Micro Assessment Questionnaire

1. Background, Scope and Methodology

Background

The micro assessment is part of the requirements under the Harmonized Approach to Cash Transfers (HACT) Framework. The HACT framework represents a common operational framework for UN agencies' transfer of cash to government and non-governmental implementing partners.

The micro-assessment assesses the IP's control framework. It results in a risk rating (low, moderate, significant or high). The overall risk rating is used by the UN agencies, along with other available information (e.g. history of engagement with the agency and previous assurance results), to determine the type and frequency of assurance activities as per each agency's guideline and can be taken into consideration when selecting the appropriate cash transfer modality for an IP.

Scope

The micro-assessment provides an overall assessment of the Implementing Partner's programme, financial and operations management policies, procedures, systems and internal controls. It includes:

- A review of the IP legal status, governance structures and financial viability; programme management, organizational structure and staffing, accounting policies and procedures, fixed assets and inventory, financial reporting and monitoring, and procurement;

- A focus on compliance with policies, procedures, regulations and institutional arrangements that are issued both by the Government and the Implementing Partner.

It takes into account results of any previous micro assessments conducted of the Implementing Partner.

Methodology

We performed the micro-assessment at 19 May and 6 June 2017 in Jakarta.

Through discussion with management, observation and walk-through tests of transactions, we have assessed the Implementing Partner's and the related internal control system with emphasis on:

- The effectiveness of the systems in providing the Implementing Partner's management with accurate and timely information for management of funds and assets in accordance with work plans and agreements with the United Nations agencies;
- The general effectiveness of the internal control system in protecting the assets and resources of the Implementing Partner.

We discussed the results of the micro assessment with applicable UN agency personnel and the IP prior to finalization of the report. The list of persons met and interviewed during the micro-assessment is set out in Annex III.

2. Summary of Risk Assessment Results

It is highly recommended that this project manages activities related to Market Transformation for Renewable Energy and Energy Efficiency through Design and Implementation of Appropriate Mitigation Actions in Energy Sector. The scope and constraints related to the inherent risk can be explained as follows:

- We only discussed, analysed, verified data and documents with the NPM, project associate and later with Kasubdit Keuangan at Dirjen EBTKE. Therefore, we do not visit the field and/or ask for confirmation to the local government or other third parties.
- The program activities must prioritize the effort transform the market for renewable energy and energy efficiency through design and implementation of appropriate mitigation actions in energy sector. Further, these activities should be in line with the policy stipulated by the Ministry of ESDM, Bappenas and UNDP.

The table below summarizes the results and main internal control gaps found during application of the micro-assessment questionnaire (in Annex IV). Detailed findings and recommendations are set out in section 3. below.

Tested subject area	Risk assessment*	Brief justification for rating (main internal control gaps)
1. Implementing partner	Low risk	There is no specific gap in this subject area, except that MTRE3 needs to describe the internal policy for fraud, waste and misuse.
2. Programme Management	Low risk	We noted there is no internal control gap in this area, because the program has just started.

Tested subject area	Risk assessment*	Brief justification for rating (main internal control gaps)
3. Organizational structure and staffing	Low risk	We noted that the finance staff will be hired later to support the project financial transaction. More than that, there is no significant gap.
4. Accounting policies and procedures	Low risk	In addition to the statement in the PMIG, it is suggested that the SOP elaborates also the definition related to conflict of interest in the finance and accounting, procurement of goods/services.
5. Fixed Assets and Inventory	Low risk	We noted there is no gap in this subject area.
6. Financial Reporting and Monitoring	Low risk	During the assessment, this subject area is still managed under UNDP finance unit. To recognize the gap, it is now to be ensured that the project prepares it accordingly.
7. Procurement	Low risk	Except to elaborate the definition of conflict of interest, there is no significant gap found.
Overall Risk Assessment	**Low risk**	We notice that in overall, after the project is completely in operation, the possible gaps in internal control shall be resolved well.

**High, Significant, Moderate, Low*

3. Detailed Internal Control Findings and Recommendations

No.	Description of Finding	Recommendation
1.	To maintain adequate safeguard the SOP shall include such condition.	The SOP in project management shall include the description to protect organization from safeguard requirement.
2.	To elaborate definition of conflict of interest in finance, accounting, procurement of goods.	The SOPs shall include such definition mentioned here.

Assessment and recommendation report is prepared by:

Osa Hartoyo _____ Signature and date: 2 October 2017

I here with agree with the above justification and approve the recommendations for Directorate General of EBTKE, Ministry of ESDM

Approved by: Christophe Bahuet
Country Director, UNDP Indonesia

Signature and date 26/10/17

Annex J: Additional Agreements
(Co-financing Letters)
(in separate file)

Commitment Letter -

Ministry of Energy and Mineral Resource (DG-NREEC)

Commitment Letter -

Ministry of Energy and Mineral Resource (P3TEK – R&D Dept)

Commitment Letter -
National Standardization Agency (*Badan Standardisasi Nasional*, *BSN*)

Commitment Letter -

City Government of Solo/Surakarta (Mayor)

Commitment Letter -
Indonesian Lighting Manufacturers Association – GAMATRINDO

Commitment Letter -
Indonesian Lighting Manufacturers Association – APERLINDO

Commitment Letter -
Global Efficient Lighting Centre

Commitment Letter -
Solarens (local LED manufacturer)

Commitment Letter -
Adyawinsa (local LED manufacturer)

Commitment Letter -
UNDP

Annex K: Description of Demonstrations

Annex K1: Pilot Demonstration Concept Note #1 – Guaranteed EPC: Government Building

A. Title	**Guaranteed Energy Performance Contracting Pilot Demonstration in Government Buildings**	
B. Contract Model	Guaranteed Saving	
C. Proposed Location	Government Building: LKPP Building	
D. Brief Description		
E. Estimated Number of Lamps Involved	7,391 pieces	
F. Estimated Total Load	124 kW	
G. Planned Activities	Phase 1: Project Development Phase 2: Implementation Phase 3: Performance Contracting Period Phase 4: Replication Preparation	
H. Expected Results	Energy Saving	223 – 292 MWh p. a
	Emission Reduction	181 – 236 tCO_2eq p. a
	Cost Saving	USD 26,713 – 34,970 p. a
I. Monitoring, Verification and Enforcement Plan	Lighting Retrofit	Product Standard Compliance
		Product Warranty
		Installation Safety
		Commissioning
		Waste Management
	Saving Performance	Basic Functionality
		Electric Current
		Electric Voltage
		Luminous Flux
	Financial Settlement	Proper Financial Accounting
		Service Payment
		Below/beyond-standard compensation
		Installment Payment
		Project Close-Out
J. Host selection criteria	1. Energy Utilization Index 2. Potential Savings 3. Co-financing commitment 4. Replication Impact 5. Technical Compliance	
K. ESCO selection criteria	1. Qualification 2. Project History 3. Pre-Energy Audit 4. Contracting Proposal Compliance 5. Bankability	
L. Responsibility Matrix	ADLIGHT	Responsible for host and ESCO selection, and work closely with ESCO in monitoring, verification, enforcement

		activities in the project. ADLIGHT will also remain providing technical assistance during the replication phase.
	Financial Institution	Responsible for providing loans, channeling the fund, collecting it back based on installment scheme, if applicable
	Loan Guarantee Agency, e.g. IIGF	Responsible for sharing the risks involved by providing access to loan guarantee funds, if applicable
	Energy User (host)	Responsible for the investment and for paying the service received from ESCO based on the contract, supporting ESCO in giving access to do audit and installation activities, procuring the lights, and paying back the loan received from financial institution for the up-front cost
	ESCO	Responsible for designing the project, implementing it, and reporting the progress and ultimately ensuring the savings expected from the project.
M. Estimated Budget	Project Development	USD 12,500
	Implementation	USD 71,500
	Performance Contracting	USD 19,000
	Replication Preparation	USD 15,000
	Total	**USD 118,000**
N. Implementation Timeline	Project Development	July to August 2018
	Implementation	September to December 2018
	Contracting Period	January 2019 to December 2020
	Replication Preparation	July to September 2019

Introduction

The Minister of Energy and Mineral Resources (MEMR) in Regulation Nr.13/2012 has made a clear statement that all government building in national, provincial and municipal level must implement the national energy efficiency program, including in lighting system. The energy efficiency projects conducted in government buildings becomes the initial attempt for energy management that will support the reduction of energy cost in all buildings. It can motivate other government building in the same or in different regions.

However, only few of the government building managements have the adequate capability to implement this program. Besides of the lack of knowledge in energy efficiency, the limited budget resources, the

hesitation in relying on new technology, the procurement is also facing various restrictions from national and regional procurement regulations. The project demonstration attempts to lift these barriers by delivering the energy efficiency project by the model of Energy Performance Contracting business model (EPC).

EPC shifts the focus away from selling energy efficient lights towards the desired benefits and services derived from the use of the technology. In this project demonstration, the said benefit is the energy saving gained from retrofitting lighting in buildings to LED lamps.

Due to its complexity, energy performance contracting project will involve a third-party company which is well known as the Energy Services Company (ESCO). ESCO provides a broad range of energy solutions including designs and implementation of energy savings projects, retrofitting, and risk management.

The MEMR through in Regulation Nr.14/2016 regulated two kinds of Energy Performance Contracting involving ESCO; the Guaranteed Saving Performance Contract, and the Shared Saving Performance Contract. For government building, the guaranteed saving model is more acceptable and less conflicting with public procurement regulations.

Guaranteed Energy Saving Performance Contract

The ESCO mainly provides technical services: design, installation and operation for which they are paid for by the client (e.g. on monthly basis). In the Guaranteed Savings model the client is making the investment in the (lighting) equipment. The client can pay the ESCO for its technical services from the energy cost savings. In this case, the savings will be guaranteed by the ESCO, during the contract period that has been agreed beforehand. For efficiency project that requires high amount of funding, the energy user will most likely rely on a financial institution.

ESCO will guarantee the energy saving to pay back the investment that the energy user has invested. If the energy saving turns out to be less than guaranteed, ESCO is obliged to compensate the amount of the loss the energy user suffers. Conversely, if the energy saving exceeds the expectation, the energy user should share the surplus saving based on the proportion that has been agreed in the contract.

Financial Mechanism and Sustainability

The demo host government agency building owner, *LKPP*, has been strategically pre-selected because of its role in government procurement and its involvement in the project will lead to its own appreciation of the EE lighting program and show other similarly situated government agencies the benefits of the program. The project will provide investment support in terms of procurement and installation of EE lighting devices co-shared with *LKPP* as user and investor In close coordination with *LKPP*, The project will also provide technical assistance in doing the investment grade energy audits, formulation of EPC and financial arrangements, MVE and preparation and start-up implementation of replication plans. The project will provide technical assistance in developing the procurement policy and guidelines for EE lighting projects in Component 2 (UNEP), drawing from the experience of LKPP in this demo which is expected to be sustainable owing to the energy savings and GHG reduction in shifting to EE lighting systems. As described above, the scheme will follow the “Guaranteed Savings” ESCO model, with the energy user as the government agency owner of the building investing on the lamp replacements and efficient lighting systems and the ESCO being contracted for its services to realize the potential economic and environmental benefits. For purposes in demonstrating this scheme by the ADLIGHT project, the

investment support and technical assistance will involve such up-front costs that usually become hurdles for coming to EE investment initiatives even while there are potential benefits that have been identified. The project will remove these barriers and pave the way attract numerous government agency building owners to follow this developmental route for more efficient and sustainable operations. Once these developmental concerns are addressed, the EE lighting program for government buildings will be properly launched and the sustaining mechanisms will be put in place for MEMR to follow through and sustain through the effective implementation of the replication plan that will be initiated within the project period. After addressing the program-related up-front costs that will be supported by GEF project, the EE lighting program will continue to be supported by the MEMR as part of its regular administrative and monitoring function while the targeted government agencies as building owners and energy users can already budget for its own EE lighting transformation requirement that are sufficiently justified and implemented under the approved procurement plans and guidelines resulting from these project barrier removal activities.

Proposed Location

The Policy Institute for Public Procurement (*Lembaga Kebijakan Pengadaan Publik*/LKPP) is a suitable and potential government building where this project demonstration can take place. It has a significant number of lamps, with almost none of efficient lights. As the national body that regulates public procurement, it has high replication impact when it successfully become the role model of ESCO Guaranteed Savings model for other government buildings.

Expected Results

Annual Energy Saving
Energy efficiency project is using technology that requires less energy to perform the similar function. By replacing lamps into LED lights that consumes lower energy but still give at least the same lighting, the total annual energy saving is expected to reach the range 223 to 292 MWh.

Annual Emission Reduction

Based on the data released by PLN the emission factor for electricity production in Indonesia in year 2016 is 0.808 tons CO_2eq/MWh. Thus, the annual emission reduction is estimated in the following to reach the range 181 to 236 tCO_2eq.

Annual Cost Saving
The LKPP Building is categorized as P2 (Government Level 2) in PLN (the national utility company) customer classification system. It has to pay IDR 1,614 per kWh (include tax) electric usage. The annual cost saving for both buildings are estimated to reach the range USD 26,713 to 34,970.

Expected Annual Cost Saving

Monitoring, Verification and Enforcement Plan
During the implementation and performance contract period, ADLIGHT will carry out monitoring, verification, and enforcement to make sure the project runs as expected and befits to be a role model demonstration for future replications.

Estimated Budget
The costs in the project development will cover activities in selecting demonstration host and the ESCO, subsidizing the pre-energy audit and pre-feasibility study. The project will also subsidize the cost to

conduct the investment grade energy audit and full feasibility study, and legalization process of contracts and agreements. The cost for project development phase activities is estimated to reach USD 12,500.

The project investment which consists of procurement of lighting and devices, installation and commissioning will be provided by energy user or through a financial institution. The project cost to implement the project is estimated to reach USD 71,500.

In the Performance Contracting Period, the ADLIGHT team as a form of assistance during the demonstration period will help the energy user who will be involved in every monitoring activity conducted by the ESCO in a periodical basis. In the verification activities, this project will also cover the cost to provide technical assistance of the financial accounting. The cost to run activities in this phase is estimated to reach USD 19,000.

In the period of replication preparation, the project will cover the cost to conduct seminars to disseminate results of project demonstration to identify potential replications and conduct workshops to formulate next steps recommendations. The cost to prepare the replication is estimated to reach USD 15,000.

The total cost to run this project demonstration is estimated to reach USD 118,000 consisting of USD 35,000 for GEF investment support, USD 46,500 for GEF technical assistance support and USD 36,500 for investment from the user.

Activity	Estimated cost	GEF Investment Support	GEF TA Support	Investment from User
PHASE 1: PROJECT DEVELOPMENT				
Investment Grade Energy Audit	4,000		4,000	
Energy Saving Performance Contract	1,500		1,500	
Financing Agreement	2,000		2,000	
Sub Total	**7,500**		**7,500**	
PHASE 2: IMPLEMENTATION				
Light and devices procurement	60,000	30,000		30,000
Installation	10,000	5,000		5,000
Commissioning	1,500			1,500
Sub Total	**71,500**	**35,000**		**36,500**
PHASE 3: PERFORMANCE CONTRACTING PERIOD				
Operation and Maintenance	3,000		3,000	
Monitoring	8,000		8,000	
Verification	5,000		5,000	
Enforcement	3,000		3,000	
Sub Total	**19,000**		**19,000**	
PHASE 4: REPLICATION PREPARATION				
Replication potential identification	17,000		17,000	
Next steps recommendation for replication	3,000		3,000	
Sub Total	**20,000**		**20,000**	

Activity	Estimated cost	GEF Investment Support	GEF TA Support	Investment from User
Total	**118,000**	**35,000**	**46,500**	**36,500**

Implementation Timeline

The host stipulation and ESCO selection is the main priority activity as soon as the project demonstration begins, in the first quarter of 2019. The audit, contract, and the agreements will be signed right after it, at the fourth quarter of the first year, followed by the implementation of the project at the same term. Thus, the performance contracting period can begin right at the beginning of the second year and could last until the end of the project period or even earlier, depends on the feasibility study and contract agreement later. After six months operating, the saving from the LED light is expected to show an indication of a successful business model. Thus, the replication preparation by disseminating the result of the project demonstration can already be started in the fourth quarter of 2019.

Annex K2: Pilot Demonstration Concept Note #2 – Shared Saving EPC: Commercial Building

A. Title	Shared Energy Performance Contracting Project Demonstration in Commercial Buildings	
B. Contract Model	Shared Saving	
C. Proposed Location	Commercial Building: BRI Makassar Building	
D. Brief Description		
E. Estimated Number of Lamps Involved	1,367 pieces	
F. Estimated Total Load	44 kW	
G. Planned Activities	Phase 1: Project Development Phase 2: Implementation Phase 3: Performance Contracting Period Phase 4: Replication Preparation	
H. Expected Results	Energy Saving	69 – 89 MWh p. a
	Emission Reduction	56 – 71 tCO_2eq p. a
	Cost Saving	USD 8,225 – 10,541 p. a
I. Monitoring, Verification and Enforcement Plan	Lighting Retrofit	Product Standard Compliance
		Product Warranty
		Installation Safety
		Commissioning
		Waste Management
	Saving Performance	Basic Functionality
		Electric Current
		Electric Voltage
		Luminous Flux
	Financial Settlement	Proper Financial Accounting
		Service Payment
		Below/beyond-standard compensation
		Installment Payment
		Project Close-Out
J. Host selection criteria	1. Energy Utilization Index 2. Potential Savings 3. Co-financing commitment 4. Replication Impact 5. Technical Compliance	
K. ESCO selection criteria	1. Qualification 2. Project History 3. Pre-Energy Audit 4. Contracting Proposal Compliance 5. Bankability	
L. Responsibility Matrix	ADLIGHT	Responsible for host and ESCO selection, and work closely with ESCO in monitoring, verification, enforcement activities in the project. ADLIGHT will also remain providing technical assistance

		during the replication phase.
	Financial Institution	Responsible for providing loans, collecting repayments back based on installment scheme, if applicable.
	Loan Guarantee Agency, e.g. IIGF	Responsible for sharing the risks involved by providing access to loan guarantee funds, if applicable.
	Energy User (host)	Responsible for paying the service received from ESCO based on the contract, supporting ESCO in giving access to do audit and installation activities, procuring the lights, and paying back the loan received from financial institution for the up-front cost
	ESCO	Responsible for designing the project, implementing it, and reporting the progress.

M. Estimated Budget	Project Development	USD 10,500
	Implementation	USD 16,500
	Performance Contracting	USD 15,500
	Replication Preparation	USD 8,000
	Total	**USD 50,000**
N. Implementation Timeline	Project Development	July to August 2018
	Implementation	January to March 2019
	Contracting Period	April 2019 to March 2021
	Replication Preparation	September to December 2019

Introduction

Commercial buildings have high energy needs and can put great strain on the nation's power grids during peak periods. Developing more efficient buildings helps ensure a steady supply of affordable power and significantly lowers operating costs for business.

Energy saving in commercial buildings, including lighting system, can be implemented through Shared Savings ESCO business model. This shifts the focus away from selling energy efficient lights towards the desired benefits and services derived from the use of the technology. In this project demonstration, the said benefit is the energy saving gained from retrofitting lighting in buildings to LED lamps.

The project will involve a third-party company, the Energy Services Company (ESCO)to provide a broad range of energy solutions including designs and implementation of energy savings projects, retrofitting, and risk management.

Besides of the lack of technical knowledge to implement this program, the lack of schemes from financial institution to fund this kind of project remains the biggest issue.

The MEMR through in Regulation Nr.14/2016 regulated two kinds of Energy Performance Contracting involving ESCO; the Guaranteed Saving Performance Contract, and the Shared Saving Performance Contract. In this pilot the ESCO Shared Saving model will be tested.

Shared Energy Saving Performance Contract

Shared Energy Saving is a financing agreement in which a private company offers to implement an energy efficiency program, including capital improvements (like lighting upgrades) in exchange for a portion of the energy cost saving. In negotiating a successfully shared energy performance contract (SEPC), a third-party financing entity will provide funding to cover the initial upfront costs of the energy-efficient lights. The ESCO providing the investment and LED lamps will return the borrowed investment to the ESCO from the monthly saving. Over the term of the shared savings agreement, after the balance is paid off in entirety, the energy user retains future energy saving and incurs lower energy expenditures for the lifetime of their buildings.

Financial Mechanism and Sustainability

The energy user as a commercial entity and the ESCO share the risk and provide the fund for the investment to replace the lamps and cover other necessary costs.

The project investment can be in a form of equity or loan by the ESCO with a certain interest rate that applies within the financial institution with a loan guarantee back up that can be provided by a loan guarantee agency such as the IIGF. The loan has to be paid back through an installment scheme from the savings earned post-retrofit of the lamps based on the agreement. The financial institution will collect the loan back every month or based on a period agreed in the contract. Eventually, the financial institution is expected to receive the fund back with additional income generated from the interest. It will then provide the cost recovery to finance other replication EPC projects by the bank on a sustained basis.

In this financing model, the project will demonstrate that ESCOs can fit in the EE lighting transformation program and ESCOs can be the investor for the projects to be financed through the financial institutions in partnership with building owners who are both convinced of the EE lighting energy and environmental benefits. The ADLIGHT project will provide significant support to develop ESCOs attain their financial and technical capability, as part of the project's barrier removal technical assistance initiatives and therefore enhance the trust in ESCOs for them to be qualified and able in developing and implementing EE lighting projects as part of their overall portfolio.

It should be noted that this mechanism is not yet in place but will be part of the project considering that the project itself cannot provide loans or guarantees per UNDP policy.

Proposed Location

The BRI Makassar building is a suitable and potential building where this project demonstration for commercial building can be conducted. The intensive and frequently extended working hours up to midnight causes a significant increase in the energy consumption, especially the use of lighting system which consists of mostly conventional lights. Therefore, conducting this pilot project in a financial institution building can give the empirical understanding of the financial benefits of retrofitting conventional lighting system to efficient ones. It is expected that more financial institutions open up their

facilities to finance energy efficiency project based on EPC model business as an outcome of this project demonstration.

Expected Results

Annual Energy Saving
Energy efficiency project is using technology that requires less energy to perform the similar function. By replacing lamps into LED lights that consumes lower energy but still give at least the same lighting, the total annual energy saving is expected to reach the range of 69 to 88 MWh.

Annual Equivalent CO_2 Reduction
Based on the data released by PLN the emission factor for electricity production in Indonesia in year 2016 is 0.808 tons CO_2eq/MWh. Thus, the annual emission reduction is estimated in the following table to reach the range of 56 to 71 tCO_2eq.

Annual Cost Saving
The BRI Makassar Building is categorized as B2 (Business Level 2) in PLN (the national utility company) customer classification system. It has to pay IDR 1,614 per kWh (include tax) electric usage. The annual cost saving for both buildings are estimated to reach the range of USD 8,225 to 10,541.

Monitoring, Verification and Enforcement Plan

During the implementation and performance contract period, ADLIGHT will carry out monitoring, verification, and enforcement to make sure the project runs as expected and befits to be a role model demonstration for future replications.

Estimated Budget

The costs in the project development will cover activities in selecting demonstration host and the ESCO, subsidizing the pre-energy audit and pre-feasibility study. The project will also subsidize the cost to conduct the investment grade energy audit and full feasibility study, and legalization process of contracts and agreements. The cost for project development phase activities is estimated to reach USD 10,500.

The project investment which consists of procurement of lighting and devices, installation and commissioning will be shouldered by the ESCO at a cost estimated to reach USD 16,500.

In the Performance Contracting Period, the ADLIGHT team will be involved in every monitoring activity conducted by the ESCO in a periodical basis. In the verification activities, this project will also cover the cost to provide technical assistance of the financial accounting. The cost to run activities in this phase is estimated to reach USD 15,500.

In the period of replication preparation, the project will cover the cost to conduct seminars to disseminate results of project demonstration to identify potential replications and conduct workshops to formulate next steps recommendations. The cost to prepare the replication is estimated to reach USD 8,000.

The total cost to run this project demonstration to be supported by GEF is estimated to reach USD 18,000 consisting of USD 18,000 for technical assistance support energy audit, EPC and financial agreements documentation. The companies as users are expected to invest USD 15,000 to support operation, maintenance and MVE. The ESCO will invest USD 16,500 for procurement, installation and commissioning of EE lighting.

	Estimated Cost	ESCO Investment	GEF TA Support	Investment from User
PHASE 1: PROJECT DEVELOPMENT				
Investment Grade Energy Audit	2,000		2,000	
Energy Saving Performance Contract	1,500		1,500	
Financing Agreement	2,000		2,000	
Sub Total	**5, 000**		**5,000**	
PHASE 2: IMPLEMENTATION				
Light and devices procurement	12,000	12,000		
Installation	3,000	3,000		
Commissioning	1,500	1,500		
Sub Total	**16,000**	**16,500**		
PHASE 3: PERFORMANCE CONTRACTING PERIOD				
Operation and Maintenance	1,500	1,500		
Monitoring	6,000	6,000		
Verification	5,000	5,000		
Enforcement	3,000	3,000		
Sub Total	**15,500**	**15,500**		
PHASE 4: REPLICATION PREPARATION				
Replication potential identification and planning	10,000		10,000	
Next steps recommendation for replication	3,000		3,000	
Sub Total	**13,000**		**13,000**	
Total	**50,000**	**32,000**	**18,000**	

Implementation Timeline

The host stipulation and ESCO selection is the main priority activity soon as the project demonstration begins, in the first quarter of 2019. The audit, contract, and the agreements will be signed right after it, at the fourth quarter of the first year, followed by the implementation of the project at the same term. Thus, the performance contracting period can begin right at the beginning of the second year and could last until the end of the project period or even earlier, depends on the feasibility study and contract agreement later. After six months operating, the saving from the LED light is expected to show an indication of a successful business model. Thus, the replication preparation by disseminating the result of the project demonstration can already be started in the fourth quarter of 2019.

Annex K3: Pilot Demonstration Concept Note #3– Guaranteed EPC: Street Lighting

Title	**PPP Project Demonstration for City of Surakarta (+ City to be determined)– Guaranteed Savings**	
Proposed Location	City of Surakarta (+ City to be determined)	
Brief Description	The city of Surakarta (+ City to be determined) is retrofitting their streetlight from conventional streetlight to LED streetlight using a PPP scheme. There are 21,222 light points, 6,045 are under metering contract and 15,177 are under lump sum contract. The PPP scheme besides retrofitting the existing lights, it will include installing meter on the light points under the lump sum contract.	
Number of Lamps Involved	± 21,222 light points	
Estimated Total Load	77,816,862.50 KW	
Planned Activities	Phase 1: Project Development Phase 2: Implementation Phase 3: Performance Contracting Period Phase 4: Replication Preparation	
Expected Results	Annual Energy Saving	± 22,465.98 MWh
	Annual Emission Reduction	16,659.85 tons CO_2
	Annual Cost Saving	IDR 35,035,006,078
Monitoring, Verification and Enforcement Plan	Lighting Retrofit	Product Standard Compliance
		Product Warranty
		Installation Safety
		Commissioning
		Waste Management
	Saving Performance	Basic Functionality
		Electric Current
		Electric Voltage
		Luminous Flux
		Below Standard Compensation
		Project Close Out
Host selection criteria	1. Energy Utilization Index 2. Potential Savings 3. Co-financing commitment 4. Replication Impact 5. Technical Compliance 6. Pre-FS readiness	
PPP selection criteria	1. Political Will of Executive and Legislative 2. Partnership with PLN 3. Adequate Budget Location 4. Bankability 5. Streetlight Management Skill	
Responsibility Matrix	ADLIGHT	Responsible for host and PPP selection, provide the fund for finalizing PPP final business case and to assist on the tendering, contracting, and supervision of the deployment of the project.
	Financial Institution	Responsible for providing the fund through loans by the energy user, collecting it back based on installment scheme and later channel

		the cost recovery for replicative projects by itself on a sustained basis, if applicable.
	City (host)	Responsible to prepare all legal requirements in order for the contract to be awarded and executed to the PPP contract winner as stated on the Presidential Regulation no. 38/2015. The city must assist in the coordination between PLN and the PPP contract winner to make sure on the transparency of the energy billing.
	PPP Contractor (ESCO or project developer)	Responsible for designing the project, providing the funds through equity or loans with loan guarantee back up by a Loan guarantee facility, such as IIGF, implementing it, reporting the progress, and maintenance of the project.
O. Estimated Budget	Installing metering for street lighting	USD 100,000
	Project Final Business Case and Contracting Supervision	USD 100,000
	Investment Assistance & Installation	USD 591,500
	Contracting, O&M and Replication Preparation	USD 28,000
	Total	**USD 819,500**
P. Implementation Timeline	Project Final Business Case and Contracting Supervision	January 2019 to June 2019
	Investment Assistance	January 2019 to End of Contract
	Replication Preparation	June 2019 to January 2021

Introduction

As provided in government issuances and ministerial decrees regarding the PPP rules in infrastructure (to which energy efficiency projects are included), the National and Regional Governments must follow the relevant regulations if they want to develop and implement a PPP project for energy efficiency project infrastructure.

With the Presidential Decree No. 38 year 2015, it enables regional government to do PPP in energy conservation infrastructure; many cities in Indonesia want to retrofit their old conventional street lighting to high energy efficiency street lighting. In line with this decree, the Minister of Energy and Mineral Resources issued a Ministry Regulation No.14 Year 2016 about Energy Conservation Service Company (ESCO). This regulation covers two kinds of ESCO modalities on treatment of energy savings: (a) the Guaranteed Energy Saving Performance Contract (Guaranteed ESPC) and (b) Shared Energy Saving Performance Contract (Shared ESPC). However, the Ministry of Finance stated that for government, it can only follow a Guaranteed ESPC for the PPP. In Guaranteed ESPC, the ESCO guarantees to the amount of attainable energy saving achievements.

Proposed Financial Mechanism and Sustainability

ADLIGHT plans to support this demonstration project by providing the fund for partial investment to retrofit the streetlight since this will involve new designs and protection systems that need to be developed and to assist the PPP project until contract signing. The EELs for street lighting are quite different from regular LEDs used in buildings.

The project will shoulder the up-front cost in terms of the initial procurement and installation costs of the EE lighting systems and will provide technical assistance in establishing a municipality-based cost-recovery scheme. The municipality can use the fund utilized for the investment support and recovered from the savings to be pooled as seed money to develop a facility that will be used to accumulate and further create more funds for future street lighting projects. Through the facility, the municipality will be allowed to keep the energy cost savings in order to pool the savings in a "revolving fund". With the means in the revolving fund, the municipalities will then be able to continue implementing future EE street lighting projects.

The city of Surakarta Streetlight Billing

The electricity bills for the streetlight have increased from year to year for the City of Surakarta, the billing for electricity has increased rapidly. The rapid increase is caused by two factors, the energy price to generate electricity has gone up and the Government of Indonesia has lifted subsidy for streetlight bills. As of 2015, the City of Surakarta streetlight bill was Rp. 44,035,439,005 or around USD 3,310,935. In 2017, the city estimated the streetlight bill will be Rp. 44,400,000,000 or around USD 3,338,346.

Monitoring, Verification and Enforcement Plan

During the implementation and performance contract period, the ADLIGHT project will carry out monitoring, verification, and enforcement (MVE) to make sure the demo project can perform as expected and can become a role model demonstration for future replication.

On March 20, 2015, the Indonesian President issued a Presidential Decree no. 38 year 2015, the presidential decree is about *Kerjasama Pemerintah dengan Badan Usaha dalam penyediaan infrastruktur* (*KPBU*)/Public Private Partnership in the provision of infrastructure (PPP). In the Presidential Decree, there are nineteen (19) infrastructures that can be partnership, and one of the infrastructures is energy conservation. The Presidential Decree required a derivative regulation from Ministry or Institution that handles national development planning, finance and states assets, policy on government goods and services, and domestic government. As instructed by the presidential decree the Ministry and Institution issues the necessary regulation.

The total cost to run this project demonstration is estimated to reach USD 819,500 which is composed of GEF Investment support of up to USD 598,500 (resources to be shared over 2 cities) for the procurement of EE lighting system and installation of new power line and commissioning which could be acquired through regular procurement procedures and USD 60,000 for technical assistance support. The city government as user will invest USD 463,750 as counterpart in terms of costs of metering, investment costs for lighting equipment, consultant costs and operation, maintenance and MVE.

Activity	Estimated Cost PER CITY	GEF Investment Support CITY A	GEF Investment Support CITY B	GEF TA Support	Investment from CITY A	Investment from CITY B
PHASE 0: PREPARATIONS FOR REVOLVING FUND SET-UP						

Installing metering for street lighting	**100,000**				**100,000**	**100,000**
PHASE 1: OUTLINE & FINAL BUSINESS CASE						
Final business case consultant, financial and legal consultant	100,000			50,000	50,000	50,000
Subtotal	**100,000**			**50,000**	**50,000**	**50,000**
PHASE 2: IMPLEMENTATION						
Light and devices procurement	354,900	177,450	177,450		177,450	177,450
Installation new power line & commissioning	236,600	118,300	118,300		118,300	118,300
Subtotal	**591,500**	**295,750**	**295,750**		**295,750**	**295,750**
PHASE 3: PPP CONTRACTING PERIOD						
Operation and Maintenance	2,000				2,000	2,000
Monitoring	8,000				8,000	8,000
Verification	6,000				6,000	6,000
Enforcement	2,000				2,000	2,000
Subtotal	**18,000**				**18,000**	**18,000**
PHASE 4: REPLICATION PREPARATION						
Replication potential identification	8,000			8,000		
Next steps recommended for replication	2,000			2,000		
Subtotal	**10,000**			**10,000**		
TOTAL BUDGET	**819,500**	**295,750**	**295,750**	**60,000**	**463,750**	**463,750**

Implementation Timeline

The host and ESCO selection is the main priority activity as soon as the project demonstration begins, at the first quarter of 2019. The audit, contract, and the agreements will be signed right after it, in the fourth quarter of the first year, followed by the implementation of the project at the same term.

Thus, the performance contracting period can begin right at the beginning of the second year and could last until the end of the project period or even earlier, depends on the feasibility study and contract agreement later. After six months operating, the saving from the LED light is expected to show an indication of a successful business model. Thus, the replication preparation by disseminating the result of the project demonstration can already be started in the fourth quarter of 2019.

Annex L: GHG Emission Reduction Estimates

Business-as-usual (BAU) estimations for energy saving and CO_2 emission have been made based on the forecast scenarios in the Electricity Power Supply Business Plan (*RUPTL*) issued by the Ministry of Energy and Mineral Resources (MEMR). In the last *RUPTL* for 2017-2026, the electricity consumption and customer growth are presented.

I. Electricity demand

The history of electricity consumption in Indonesia in the period 2008 to 2016 is shown in **Table L.1**. The history of number of customer is shown in **Table L.2**.

Table L.1. History of electricity consumption

Year	Sector *(in TWh)*						Total
	Residential	**Industry**	**Commercial**	**Social**	**Gov. Office**	**Street Lighting**	
2008	50.18	47.97	22.93	3.08	2.10	2.76	129.02
2009	54.95	46.20	24.53	3.31	2.33	2.89	134.21
2010	59.83	50.99	26.39	3.64	2.63	3.00	146.46
2011	65.11	54.73	28.31	3.99	2.79	3.07	157.99
2012	72.13	60.18	30.99	4.50	3.06	3.14	173.99
2013	77.21	64.38	34.50	4.94	3.26	3.25	187.54
2014	85.42	87.25	36.30	5.55	3.45	3.33	221.30
2015	88.68	64.08	36.98	5.94	3.72	3.45	202.85
2016	95.33	68.93	39.53	6.36	3.96	3.71	217.83

Note: 2016 data is an estimation

Table L.2. Number of Customers

Year	Sector						Total
	Residential	**Industry**	**Commercial**	**Social**	**Gov. Office**	**Street Lighting**	
2008	36,025,071	47,536	1,716,046	838,129	103,821	113,483	38,844,086
2009	37,099,830	47,900	1,879,429	873,721	108,749	120,269	40,129,897
2010	39,324,520	48,675	1,912,150	909,312	113,676	127,054	42,435,387
2011	42,577,542	50,365	2,049,361	963,766	120,246	133,865	45,895,145
2012	46,219,780	52,661	2,218,342	1,032,830	128,252	143,384	49,795,249
2013	50,116,127	55,456	2,418,431	1,110,450	137,762	157,892	53,996,118
2014	53,309,325	58,350	2,626,160	1,181,779	146,321	171,299	57,493,234
2015	56,605,260	63,314	2,894,990	1,261,516	156,782	186,118	61,167,980
2016	59,539,981	69,381	3,064,079	1,346,633	273,305	324,103	64,617,482

Note: 2016 data is an estimation

Energy share of electricity production in the year 2016 is shown in **Figure L.1**. Based on the data released by *PLN*, the emission factor for electricity production in Indonesia in year 2016 is 0.808 ton CO_{2eq}/MWh. This emission factor was used for CO_2 emission calculation.

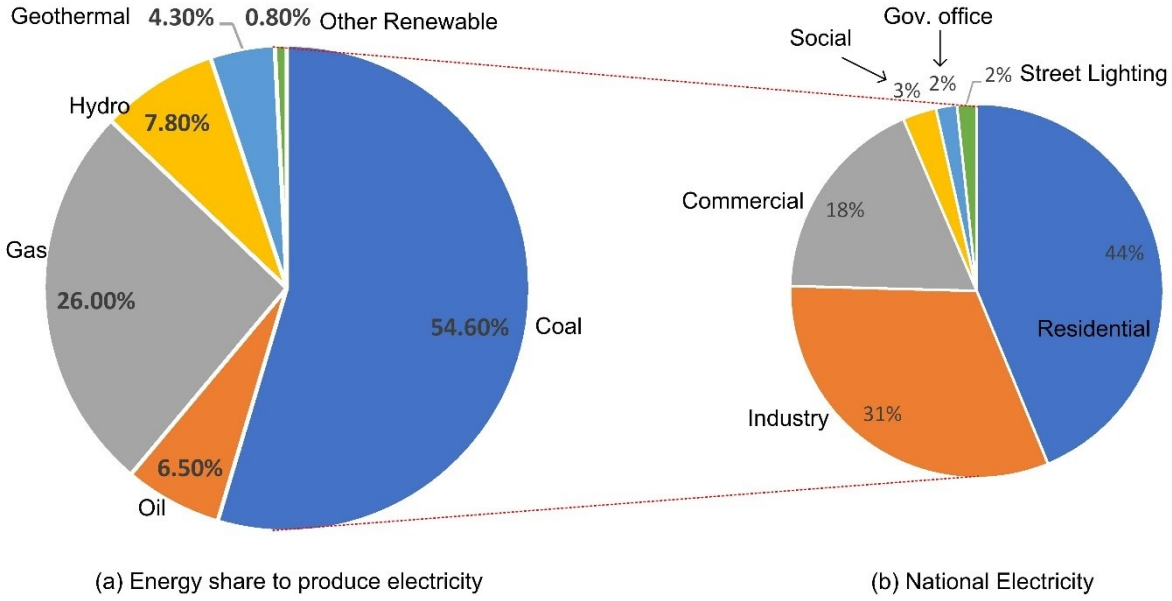


Figure L.1: Energy Share to Produce Electricity and Sectoral Electricity Usage Share in Year 2016

Based on the historical data and GDP, population growth, the projection of electricity consumption and customers has been developed in *RUPTL*. In the *RUPTL*, the projection is made until 2026. The average growth rate from year 2023 to 2026 is used to estimate the projection in year 2026 to 2030. The projection of electricity consumption and number of customer in each sector are shown in **Table L.3** and **Table L.4**, respectively.

Table L.3: Projection of Electricity Consumption

Year	Sector *(in TWh)*						Total
	Residential	Industry	Commercial	Social	Gov. Office	Street Lighting	
2017	102.16	74.41	43.17	6.81	4.24	3.97	234.77
2018	109.64	80.97	47.70	7.30	4.55	4.25	254.40
2019	117.86	88.15	52.22	7.84	4.88	4.57	275.52
2020	126.30	99.20	57.59	8.45	5.26	4.92	301.72
2021	136.24	109.95	63.31	9.11	5.67	5.30	329.58
2022	145.80	120.27	69.10	9.82	6.11	5.72	356.81
2023	155.81	131.21	75.41	10.58	6.59	6.16	385.75
2024	166.37	143.24	82.26	11.40	7.10	6.64	417.00
2025	177.50	155.43	89.82	12.29	7.65	7.15	449.84
2026	189.23	166.61	97.94	13.23	8.24	7.71	482.97

2027	201.58	178.72	106.76	14.26	8.88	8.30	518.50
2028	214.56	190.51	116.49	15.36	9.57	8.95	555.44
2029	228.19	200.60	126.94	16.55	10.31	9.63	592.22
2030	242.50	211.35	138.27	17.82	11.10	10.38	631.42

Table L.4: Projection of Number of Customers

Year	Sector						Total
	Residential	Industry	Commercial	Social	Gov. Office	Street Lighting	
2017	62,251,000	72,000	3,252,000	1,419,456	175,923	208,621	67,379,000
2018	64,782,000	74,000	3,461,000	1,496,566	185,480	219,954	70,219,000
2019	66,703,000	77,000	3,699,000	1,580,758	195,914	232,328	72,488,000
2020	68,572,000	79,000	3,967,000	1,672,818	207,324	245,858	74,744,000
2021	69,911,000	82,000	4,257,000	1,770,386	219,416	260,198	76,500,000
2022	70,960,000	85,000	4,571,000	1,874,248	232,289	275,463	77,998,000
2023	71,893,000	88,000	4,911,000	1,984,405	245,941	291,653	79,414,000
2024	72,685,000	91,000	5,279,000	2,100,071	260,276	308,653	80,724,000
2025	73,419,000	94,000	5,676,000	2,223,604	275,587	326,809	82,015,000
2026	74,118,000	98,000	6,105,000	2,353,432	291,677	345,890	83,312,000
2027	74,665,644	102,052	6,569,793	2,489,695	308,565	365,917	84,501,667
2028	75,148,792	106,157	7,071,746	2,635,182	326,597	387,300	85,675,773
2029	75,591,654	111,445	7,614,720	2,788,019	345,539	409,762	86,861,140
2030	75,875,974	116,863	8,203,587	2,948,363	365,411	433,329	87,943,526

II. Electricity Demand for Lighting

To estimate electricity demand for lighting, several assumptions were used. The assumptions are presented in **Table L.5**.

Table L.5: Assumptions for Estimating Energy for Lighting

Parameter	Sector					
	Residential	Industry	Commercial	Social	Gov. Office	Street Lighting
No of Lamps/costumer (En.lighten survey)	8.35	600	60	20	100	-
Average Wattage	13.8	17.2	17.2	13.8	13.8	-
Yearly operation hours	2,920	2,880	4,015	3,168	3,456	4,380

In the survey of the Indonesian lighting market, the typical wattages of lamps sold in Indonesia are shown in **Figure L.2**.

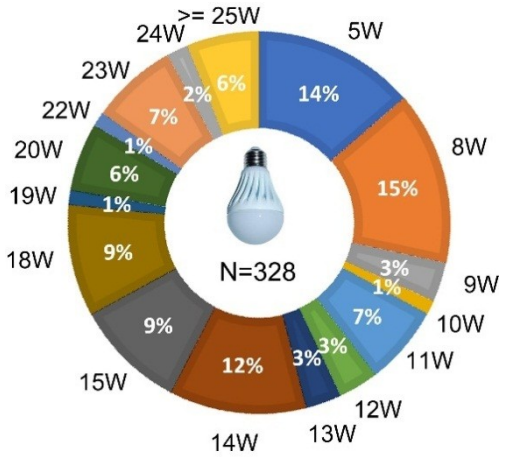


Figure L.2: Wattage lighting survey in Indonesia

The energy for lighting in each sector is calculated using the following equation:

$$E_i = N_i \overline{P}_i t_i \quad (1)$$

Where, N_i is number of costumer, L number of lamps in costumer i,

$\overline{P}$ [Watt] is the average capacity of lamp and

t [hours] is the average operating hours of the lamps in a year.

Based on those assumptions, the energy share for all lighting in Indonesia is estimated to be 18.81% in year 2016.

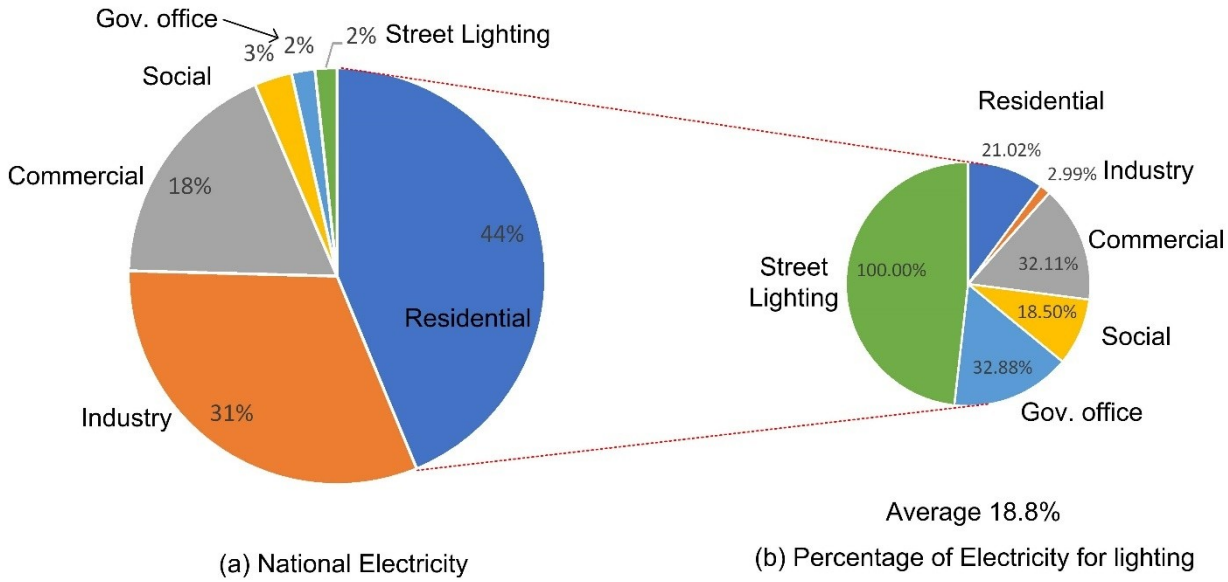


Figure L.3: Electricity for Lighting in each Sector in Year 2016

It is assumed that typical Indonesian lamps market is similar until 2030. The average number of lamps and wattage in every sector are constant until 2030. The projections of electricity for lighting until 2030 are estimated and shown in Table L.6.

Table L.6: Projection of Electricity for Lighting in every Sector

Year	Sector *(in TWh)*						Total
	Residential	Industry	Commercial	Social	Gov. Office	Street Lighting	
2017	20.95	2.14	13.47	1.24	0.84	3.97	42.609
2018	21.80	2.20	14.34	1.31	0.88	4.25	44.780
2019	22.44	2.29	15.33	1.38	0.93	4.57	46.942
2020	23.07	2.35	16.44	1.46	0.99	4.92	49.231
2021	23.52	2.44	17.64	1.55	1.05	5.30	51.498
2022	23.88	2.53	18.94	1.64	1.11	5.72	53.805
2023	24.19	2.62	20.35	1.74	1.17	6.16	56.222
2024	24.46	2.70	21.87	1.84	1.24	6.64	58.750
2025	24.70	2.79	23.52	1.94	1.31	7.15	61.428
2026	24.94	2.91	25.30	2.06	1.39	7.71	64.303
2027	25.12	3.03	27.22	2.18	1.47	8.30	67.330
2028	25.29	3.16	29.30	2.30	1.56	8.95	70.550
2029	25.43	3.31	31.55	2.44	1.65	9.63	74.019
2030	25.53	3.47	33.99	2.58	1.74	10.38	77.693

III. ADLIGHT Pilot Demonstration Projects

There are five (5) pilot demonstrations that will be implemented in the ADLIGHT project in developing and promoting ESCOs in the application of EELs in various potential groups of consumers: two (2) in buildings, two (2) in city street lighting and one (1) for the residential sector which will be participated in by MEMR in cooperation with government housing projects. These pilot demonstrations and saving potentials are shown in **Table L.7**.

Table L.7 ADLIGHT Pilot Demonstrations on ESCO to Boost LED Market Penetration

	Pilot 1	Pilot 2	Pilot 3	Pilot 4	Pilot 5 (residential)
Name	LKPP	BRI	City 1	City 2	MEMR 250,000 lamps
Type	Gov. office	Commercial	Street Lighting	Street Lighting	Change 7W CFL to 3W LED
Energy Baseline (MWh/Year)	470.06	139.52	19,710	19,710	7,665

	Pilot 1	Pilot 2	Pilot 3	Pilot 4	Pilot 5 (residential)
Percent Saving	48%	48%	64%	64%	57%
Source of Percent saving	Field survey		Field Survey		$= \frac{7W - 3W}{7W} = 57\%$
Energy Saving (MWh/Year)	225.12	65.76	12,614.4	12,614.4	4,369.05
Emission Reduction (ton CO_{2eq}/year)	181.9	53.13	10,192.43	10,192.43	3,530.19
Year when Implemented	2018	2018	2018	2019	2018
Number of Replication	1	7	1	1	-
Year when replications were Implemented	2019	2019	2019	2019	

Results of Impact Calculations

The goal of ADLIGHT project is the reduction of GHG emissions due to reduced electricity consumption in lighting system in Indonesia. Assuming the project will start in 2018, the ADLIGHT Project will result in a cumulative electricity savings of 77.45 GWh (278,820,000 MJ) by 2020 or at the end of the project in Year 3 against a baseline electricity consumption of 40.98 TWh/year in 2016. The equivalent reduction of CO_2 emissions is estimated at 62.57 kilotons (kt) CO_2 by the year 2020.

Table L.8: Energy Saving and Emission Reduction of the Project

ADLIGHT Project	**Value**	**Notes**
I - Direct		
Direct CO_2 emission reductions (t CO_2)	548,766	(1)
III– Indirect		
Indirect CO_2 emissions - top-down (tCO_2)	1,097,533	(2)
GEF finance (US $)	3,895,872	
Total emission reductions (t CO_2)	1,646,299	(3)
Unit Abatement Cost (UAC) ($/t$CO_2$) - (Direct savings) as per GEF CO_2 emission reduction calculation methodology ($/t$CO_2$)	7.0	
Unit Abatement Cost (UAC) ($/t$CO_2$) - (Direct + Indirect) as per GEF CO_2 emission reduction calculation methodology ($/t$CO_2$)	2.36	

(1) Must be considered for 10 years of direct CO_2 emission reductions from all pilot projects and replications. Since this project focuses entire sector, cumulative emission reductions during economic lifetime (considered 10 years including project duration of 3 years)

(2) Based on GEF CO_2 emission reduction calculation methodology, as it focuses entire sector. In this case it is the cumulative emission reduction for 10 year from best case of all sectors multiple by factor 0.5.

(3) Considered for 10 years of economic lifetime as per GEF guidelines (Direct + Indirect). This is calculated as (3) = (1)+(2).

IV. Energy saving and emission reduction of post projects

It is assumed that during the project, if all of the activities are implemented as planned, the lighting market in Indonesia will be significantly transformed into using high efficient lighting systems. In the case of ADLIGHT, there are two driving forces to attain energy savings. The first is the increasing efficiency of the lamps in the market due to technology development. The second saving comes from the rate of progress of replacing the old lamps into high efficient lamps, referred to as penetration rate. In the first kind of saving, the average of saving in each sector is different. It depends on the current baseline condition and the technology improvement in the future. The present saving and yearly increase in efficiency are presented in **Table L.9**. In the second kind of saving due to the increasing penetration rate, it will be divided into two scenario assumptions, i.e. conservative and optimistic penetration rates.

Table L.9: Percentages in Baseline Saving, Yearly Increase in Saving, and Projected Saving in 2040

No	Sector	Baseline Saving	Yearly additional saving [2020-2039]	Final Saving in 2040
1	Residential	57%	1.20%	79.8%
2	Industry	48%	1.50%	76.5%
3	Commercial	48%	1.50%	76.5%
4	Social	48%	1.50%	76.5%
5	Government Office	48%	1.50%	76.5%
6	Street Lighting	64%	1.00%	83.0%

- Conservative Penetration Rate

For the conservative penetration scenario, the penetration rate increases 2% yearly and it is assumed to start in 2021. In this assumption, at the present and during the ADLIGHT project, the installed high efficient lighting is very small in comparison to number of lamps in Indonesia. ADLIGHT project will affect the market and the number will increase linearly. The calculation of the conservative penetration has been made during 2020 to 2040. The baseline and projection during 2016 – 2030 are shown in **Table L.10**.

- Optimistic Penetration Rate

In this case, the percent share of implemented high efficient lighting (penetration rate) varies as a function of time (year) and calculated using the following logistic function.

$$P(t) = \frac{1}{1 + a \times e^{-bt}} \quad (2)$$

For Indonesia case, Batih and Sorapipatana (2016) suggested the constant *a* and *b* which are validated in Thailand case. The constants are: $a = 46.51$ and $b = 0.31$ (3)

The calculation results for this case are presented in **Table D.11**.

Table L.10: Energy Saving and Emission Reduction during 2016 to 2030 for Conservative Penetration Rate Scenario

Parameter		Base		Year 1	Year 2	Year 3	ADLIGHT Post-Project								
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
BAU Electricity for Lighting [TWh/year]	40.98	42.61	44.78	46.94	49.23	51.50	53.80	56.22	58.75	61.43	64.30	67.33	70.55	74.02	77.69
BAU Emission [$MtCO_{2eq}$/year]	33.11	34.43	36.18	37.93	39.78	41.61	43.47	45.43	47.47	49.63	51.96	54.40	57.00	59.81	62.78
Saving and Emission reduction from Demo															
Electricity saving [GWh/year]	-	-	-	69.10	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69
Cumulative saving [GWh]	-	-	-	69.10	309.79	550.47	791.16	1,031.85	1,272.54	1,513.23	1,753.91	1,994.60	2,235.29	2,475.98	2,716.67
Emission reduction [kilo tCO_{2eq}/year]	-	-	-	55.83	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48
Cumulative emission reduction [kilo tCO_{2eq}]	-	-	-	55.83	250.31	444.78	639.26	833.73	1,028.21	1,222.69	1,417.16	1,611.64	1,806.11	2,000.59	2,195.07
Saving and Emission reduction affected by ADLIGHT for all sectors															
Penetration rate for all sectors [%]		-	-	-	-	2%	4%	6%	8%	10%	12%	14%	16%	18%	20%
Saving due to increasing lighting efficacy [%]															
Residential			57%	58.2%	59.4%	60.6%	61.8%	63.0%	64.2%	65.4%	66.6%	67.8%	57%	58.2%	59.4%
Industry			48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Commercial			48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Social			48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Gov. Office			48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Street Lighting			64%	65.0%	66.0%	67.0%	68.0%	69.0%	70.0%	71.0%	72.0%	73.0%	64%	65.0%	66.0%
Total saving affected by the project in all sectors [GWh/year]						554	1,184	1,898	2,703	3,610	4,632	5,777	7,062	8,504	10,117
Total emission reduction affected by the project in all sectors [$ktCO_{2eq}$/year]						447	957	1,533	2,184	2,917	3,743	4,668	5,706	6,871	8,174
Total Energy Saving and Emission Reduction from DEMO and affected by the ADLIGHT Project															

Total energy saving [GWh/year]	-	-	69	241	794	1,425	2,138	2,944	3,851	4,873	6,018	7,302	8,745	10,357
Cumulative energy saving [GWh]	-	-	69	310	1,104	2,529	4,667	7,611	11,461	16,334	22,352	29,654	38,399	48,757
Total Emission reduction [$ktCO_{2eq}$/year]	-	-	56	194	642	1,151	1,728	2,378	3,111	3,937	4,863	5,900	7,066	8,369
Cumulative Emission Reduction[$ktCO_{2eq}$]	-	-	56	250	892	2,043	3,771	6,150	9,261	13,198	18,060	23,961	31,027	39,395

Table L.11: Energy Saving and Emission Reduction during 2016 to 2030 for Optimistic Penetration Rate Scenario

Parameter		Base		Year 1	Year 2	Year 3	ADLIGHT Post Project								
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
BAU Electricity for Lighting [TWh/year]	40.98	42.61	44.78	46.94	49.23	51.50	53.80	56.22	58.75	61.43	64.30	67.33	70.55	74.02	77.69
BAU Emission [$MtCO_{2eq}$/year]	33.11	34.43	36.18	37.93	39.78	41.61	43.47	45.43	47.47	49.63	51.96	54.40	57.00	59.81	62.78
Saving and Emission reduction from Demo															
Electricity saving [GWh/year]	-	-	-	69.10	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69	240.69
Cumulative saving [GWh]	-	-	-	69.10	309.79	550.47	791.16	1,031.85	1,272.54	1,513.23	1,753.91	1,994.60	2,235.29	2,475.98	2,716.67
Emission reduction [kilo tCO_{2eq}/year]	-	-	-	55.83	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48	194.48
Cumulative emission reduction [kilo tCO_{2eq}]	-	-	-	55.83	250.31	444.78	639.26	833.73	1,028.21	1,222.69	1,417.16	1,611.64	1,806.11	2,000.59	2,195.07
Saving and Emission reduction affected by ADLIGHT for all sectors															
Penetration rate for all sectors [%]	-	-	-	-	-	2.85%	3.84%	5.17%	6.92%	9.20%	12.14%	15.85%	20.43%	25.93%	32.31%
Saving due to increasing lighting efficacy [%]															
Residential			57%	58.2%	59.4%	60.6%	61.8%	63.0%	64.2%	65.4%	66.6%	67.8%	57%	58.2%	59.4%

Industry		48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Commercial		48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Social		48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Gov. Office		48%	49.5%	51.0%	52.5%	54.0%	55.5%	57.0%	58.5%	60.0%	61.5%	48%	49.5%	51.0%
Street Lighting		64%	65.0%	66.0%	67.0%	68.0%	69.0%	70.0%	71.0%	72.0%	73.0%	64%	65.0%	66.0%
Total saving affected by the project in all sectors [GWh/year]					788	1,137	1,635	2,337	3,321	4,684	6,539	9,017	12,250	16,342
Total emission reduction affected by the project in all sectors [$ktCO_{2eq}$/year]					637	919	1,321	1,888	2,683	3,785	5,284	7,285	9,898	13,204
Total Energy Saving and Emission Reduction from DEMO and affected by the ADLIGHT Project														
Total energy saving [GWh/year]	-	-	69	241	1,029	1,378	1,875	2,577	3,561	4,925	6,780	9,257	12,491	16,583
Cumulative energy saving [GWh]	-	-	69	310	1,339	2,717	4,592	7,170	10,731	15,656	22,436	31,693	44,184	60,767
Total Emission reduction [$ktCO_{2eq}$/year]	-	-	56	194	832	1,114	1,515	2,083	2,877	3,979	5,478	7,480	10,093	13,399
Cumulative Emission Reduction[$ktCO_{2eq}$]	-	-	56	250	1,082	2,195	3,711	5,793	8,671	12,650	18,128	25,608	35,701	49,100

Annex M. Annual Targets

Indicator	Baseline	Year 1	Year 2	Year 3
Project Goal: To reduce the growth rate of annual GHG emissions from lighting energy use by means of reducing electricity consumption in the energy end-use sectors				
Project Objective: To increase the penetration of high quality energy efficient lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions.				
• Cumulative electricity savings by EOP, GWh	0	10.0 (36,000,000 MJ)	17.27 (62,172,000 MJ)	77.45 (278,820,000 MJ)
• Direct project GHG emissions mitigated by EOP, kt CO_2 (GEF Core Indicator 6.2)	0	8.08 kt $CO_{2\ (direct\ by\ EOP)}$ 73.36 kt $CO_{2\ (Lifetime\ direct\ by\ EOP)}$	13.96 kt $CO_{2\ (direct\ by\ EOP)}$ 121.76 kt $CO_{2\ (Lifetime\ direct\ by\ EOP)}$	62.58 kt $CO_{2\ (direct\ by\ EOP)}$ 548.77 kt $CO_{2\ (Lifetime\ direct\ by\ EOP)}$
• Number of women and men participating in capacity building trainings throughout the project (GEF Core Indicator 11)	0	15/35	30/70	60/140
Outcome 1: Improved quality, energy efficient and affordable locally-produced EEL products and systems				
• Efficiency of locally produced lighting systems increased, (lm/W) and Production cost decreased from baseline level (%) o *Indoor Type*	• Eff: **70** lm/W; • Supply chain cost: 100 % (Baseline supply chain cost to be determined at project inception)		• Eff: **120** lm/W; • Supply chain cost: **80%** of Baseline	• Eff: **150** lm/W; • Supply chain cost: **70%** of Baseline
o *Outdoor Type*	• Eff: **100** lm/W; • Supply chain cost:100% (at project inception)		• Eff: **130** lm/W; • Supply chain Cost: **80%** of Baseline	• **170** lm/W; • Supply chain cost: **70%** of Baseline
Cumulative number of lighting manufacturers who received technical	0		3 manufacturers (1 for each associations)	6

assistance to upgrade production facilities				
• Investment grade proposal for business transformation plans ○ *Submitted by manufacturers* ○ *Approved by banks.*	• 0 • 0		• 3 • 3	• 6 • 6
• Ratio of women and men who believe they have the capacity to ○ *Submit (by local manufacturers)* ○ *Approve (by banks)* investment grade proposal for business transformation plans	• 0 • 0	• 30/70 • 30/70	• 30/70 • 30/70	• 30/70 • 30/70
Outcome 3: Increased market penetration of high quality and efficient lighting				
• Cumulative no. of innovative financial support schemes developed to accelerate penetration of EE lighting systems	• 0	• 1	• 2	• 4[30]
• Cumulative no. of pilot demonstration completed and replication plans developed and approved for implementation	• 0		• 3 (1 cities and 2 buildings)	• 5 (2 cities, 2 buildings and a residential area) With possible replication in 2 cities and 7 buildings
• Number of stakeholders engaged ○ *Municipalities* ○ *clients (project developers/*	• 0 • 0	• 1 • 1	• 2 • 3	• 4 • 6

[30] 1. Sustainable development pathways; 2. Inclusive and effective democratic governance; 3. Resilience building

building owners) ○ *technology providers* ○ *financial institutions*	 • 0 • 0	 • 1 • 1	 • 3 • 1	 • 5 • 2
• Women's and men's level of satisfaction with EEL systems provided (reliability, affordability, convenience, efficiency)	• 0	• At least 50 % of women and men beneficiaries are highly satisfied with the EEL systems provided (minimum score 7/10)	• At least 60 % of women and men beneficiaries are highly satisfied with the EEL systems provided (minimum score 7/10)	• At least 70 % of women and men beneficiaries are highly satisfied with the EEL systems provided (minimum score 7/10)

Gender Analysis and Gender Action Plan for the ADLIGHT Project

Gender analysis

Introduction

This assessment aims to provide an overview of the state of gender equality in Indonesia and identify gender gaps, challenges and opportunities for UNDP and Government partners through the ADLIGHT project. This assessment illustrates how gender equality can be advanced through the transformation of Indonesia's lighting market. The assessment was based on available data from global and Indonesian studies, donor agencies and interviews.

ADLIGHT aims to promote the increased deployment of high efficiency lighting technologies in Indonesia through the transformation of the national market, thereby reducing electricity demand and the related greenhouse gas (GHG) emissions.

This will be achieved through implementation of three Components, namely:
(1) Support to local lighting industry to prepare the market for high quality, high efficient lighting systems,
(2) Regulatory mechanisms; and market monitoring, verification, and enforcement, and
(3) New business models and awareness raising for high efficiency lighting technology penetration.

Gender equality in Indonesia: contextual analysis

Status of gender equality in Indonesia

Vast inequalities between men and women persist in Indonesia, at every level of society. In 2016, Indonesia ranked 88th out of 144 countries in the World Economic Forum's Global Gender Gap Report, scoring .682. This reveals a gap for female legislators, senior officials and managers. Indonesia has a Gender Inequality Index value of 0.467, ranking it 105 out of 159 countries in the 2015 index. The GII reflects gender-based inequalities in three areas: reproductive health, empowerment, and economic activity.[31]

According to the National Labour Data in Indonesia's national labour survey (SAKERNAS) in 2015, women's labor force participation rates 48.9% as opposed to 82.7% for men[32]. When compared to previous years,

[31] sustainable production technologies, access to modern energy services and energy efficiency, natural resources management, extractive industries, urbanization, citizen security, social protection, and risk management for resilience

[32] This question will have a "Not Applicable" checkbox which can be activated for global and regional projects, or country projects with clearance from the RBx desk officer.

the data reveals that over the last ten years, there is an increase of tertiary educated young women gaining meaningful wage work in the formal sector, mostly in urban areas. However women in poorer families are becoming more likely to resort to informal work to make ends meet. This inequality gap is reflected in the increasing GINI coefficient of .397 in Indonesia. Over the past 20 years inequality has risen faster than any other country in south-east Asia. [33] Workers in the informal sector, do not have access to minimum wage rights, or benefits of wage work like insurance, parental leave, or sick leave, and are most vulnerable to exploitation, injury and financial instability. The informal sector is mostly made up of low-skilled and low-educated women, either working for family members or running their own businesses.

In Indonesia, small to medium enterprises (SMEs) make up 57% of the GDP. About one-third of medium-sized enterprises are owned by women and this number is increasing by 8%[34] annually while the number of SMEs owned by men is dropping. Women-led SMEs are an important untapped opportunity that can support economic growth and innovation. However, barriers are in the way to unlocking this potential. The results of preliminary research into the barriers faced by women-led SMEs conducted by UNDP Indonesia found that women-led enterprise often lack access to finance, including credit due to a range of challenges: lack of education around managing financing, lending agencies and banks exercise discrimination, combined with women's tendency to put forward her husband or brother's name when applying for credit. This results in women having far less agency and decision-making power when operating her enterprise, and spending family income.

Women are underrepresented in positions of leadership in every sector in Indonesia. Particularly at the political level, indicators surrounding progress in female political participation has shown both progress and need for change. There has been an increase in women representation in national parliament from 17.32 to 18.04 per cent in 2016. At the sub-national level women hold less than 15 percent of seats (14.6 percent in provincial and 25.8 percent in regional). In the executive branch, almost half of civil servants are women, yet less than 20% hold decision-making positions. President Joko Widodo has been congratulated for appointing eight women in his cabinet of 34 ministers.

Legal and Administrative Framework Protecting Women and Promoting Gender Equality

At the international level, Indonesia has shown commitment to advancing gender equality. In 1984, it ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) which committed nation states to harmonizing domestic laws with international law.

The Government of Indonesia has committed to achieving 17 Sustainable Development Goals (SDGs), all of which have specific indicators related to gender in addition to Goal 5 for Gender Equality. Gender mainstreaming is an approach public and private sectors are more commonly utilizing to achieve the SDGs, and in particular the Government of Indonesia's agenda through special regulations on Gender Mainstreaming and Gender Responsive Planning and Budgeting. Specifically, Gender mainstreaming has become national policy since the year of 2000. The commitment to integrate gender equality into national development was confirmed by Presidential Decree No. 9/2000 on "Gender Mainstreaming in National Development" regulating that all government sectors, at national and sub-national level, including the Commander of the Armed Forces, the Chief Police, and the Supreme Attorney, shall integrate a gender mainstreaming policy and strategy into its respective tasks and functions. All sectors

[33] This question will have a "Not Applicable" checkbox which can be activated for global and regional projects, or country projects with clearance from the RBx desk officer.

[34] This question will have a "Not Applicable" checkbox which can be activated for global and regional projects

shall report the implementation of the policy to the Ministry of Women's Empowerment and Child Protection annually which reports the progress of gender mainstreaming to the President.

While there is strong commitment to the Gender Mainstreaming Regulation in Indonesia, there are still challenges in its implementation such as lack of mindset and political will for gender-responsive budgeting as well as the lack of investment in sex-disaggregated data. Further to this, improving institutional knowledge, and strengthening monitoring and evaluation, and gender-related accountability mechanisms are necessary for gender mainstreaming at policy and implementation levels.[35]

Gender and energy access

The correlations between poverty, access to affordable and effective energy resources and gender equality has been widely explored. Access to affordable, reliable, sustainable and modern energy is a human development enabler – it provides men and women with more time and money in and outside of the household, at work, while curbing GHGs and improving the health and safety of communities. For example, improved access to time-saving electric appliances has been significantly more transformative for women than men due to women's primary role relative to household chores.[36] Due to persistent gender roles in Indonesia and around the world, lighting technology directly impacts men and women differently, while also impacting structured systems that hinder gender equality.

Ministry of Energy and Mineral Resources states that Indonesia has reached 95.35% of electrification ratio at national level and 97.10% at rural areas. Approximately 2,300 villages are still living without access to electricity,[37] either using firewood for cooking or kerosene lamps for visibility. Kerosene is harmful, particularly for women who are primarily responsible for cooking, cleaning and using electrical sources for household duties. The World Bank estimates that breathing kerosene fumes is the equivalent of smoking two packets of cigarettes a day and two-thirds of adult females with lung cancer in developing nations are non-smokers.[38] Kerosene lamps are widely used for lighting in rural areas of Asia and Africa where electrical distribution is either not available, or too costly for widespread use.[39] Yet using kerosene for lighting is extremely inefficient, dangerous and expensive, and it has extensive health and environmental drawbacks.

At some other rural areas, the hardship of collecting and using firewood for cooking in the outdoor causes great problems for women. Women cook on open fires by burning dung, wood or other biomass. The collection and transport of fuel represents a labor intensive and time-consuming task, undertaken mainly by women and children. The health impacts on women from using firewood can be enormous — smoke produced from open fires, lung diseases caused by smoke and indoor air pollution, neck and back injuries

[35] This question will have a "Not Applicable" checkbox which can be activated for global and regional projects, or country projects with clearance from the RBx desk officer.

[36] This question will have a "Not Applicable" checkbox which can be activated for global and regional projects, or country projects with clearance from the RBx desk officer.

[37] Proposed innovative financial support schemes to accelerate penetration of advanced lighting systems, may include: (a) PPP mechanism for smart city(ies) LED street lighting projects; (b) ESCO model Feasibility Study (FS) Template and Financial Modeling Software; (c) Credit Guarantee Facility for Energy Efficiency LED lighting investment; and (d) Bundling LED lighting projects as accepted collateral for banking sector

[38] http://www.id.undp.org/content/dam/indonesia/2017/doc/INS Indonesia_Country%20Explanatory%20Note_HDR2016.pdf

[39] https://www.adb.org/sites/default/files/publication/182935/ino-paper-16-2016.pdf

caused by the transportation of heavy wood loads. These ailments combined with the difficult access to health services at the remote rural areas doubles the burden placed on women.[40]

Dr. Nuzainah Gintin of Indonesia says, "Energy is very important in women's daily lives, especially for serving food for families, lighting for teaching their children at night, and energy for their small food industries… cooking and small business are the major challenges facing women around access to energy… for the women in my country, a change that would allow them to be financially independent would have the most impact in their lives."

Gender and energy efficient lighting

Street lighting

Indonesian cities typically have sub-optimal street lighting systems, with illegal connections, limited metering coverage and poor service standards.[41] Increasing the energy efficiency of street lighting is not only cost effective, it provides gender-sensitive results by improving nigh-time safety in cities. According to UN-Habitat, women's vulnerability to sexual harassment and assault makes them in higher need for security.[42] In Indonesia, for example, one in three women have experienced violence against them in their lifetimes. There are a range of initiatives undertaken in Indonesia to address gender-based violence, using public lighting safety as an entry point. For example, UNWomen in Indonesia is working with the Government to promote 'Safe Cities' that addresses lack of street lighting to address the high rates of violence and harassment against women in urban spaces.[43]

The GIZ project Smart Street Lighting Initiative (SSLI) which commenced in 2014 in Indonesia effectively ensured participation of women's representatives the perceptions of women and men with relation to safe street lighting. As a result of focus group discussions, results showed that there is a need to raise awareness of energy efficiency among both women and men; public perception of street lighting is in general positive, although there were comments on lamps that did not turn on, frequently malfunction, and are not immediately replaced; women and men have different perceptions on what constitutes adequate illumination, as well as the sense of safety it provides on a particular street, with men having the tendency to perceive that the illumination had been adequate while the women tended to claim that it was not bright enough; and in the area where the SSLI was implemented, more men than women accessed the street at night due to both cultural reasons around women staying at the home, and also because it was not lit well.[44]

Residential areas

[40] https://www.oxfam.org/sites/www.oxfam.org/files/bp-towards-more-equal-indonesia-230217-en_0.pdf

[41] http://www.id.undp.org/content/indonesia/en/home/presscenter/articles/2017/09/15/no-sdg-untouched-by-women-owned-social-enterprises.html

[42] https://sustainabledevelopment.un.org/memberstates/indonesia

[43] https://www.energytoday.net/economics-policy/policies/women-energy-economic-empowerment-applying-gender-lens-amplify-impact-energy-access/

[44] Ministry of Energy and Mineral Resources – Strategi Pemerintah Capai Rasio Elektrifikasi 99% di Tahun 2019, https://www.esdm.go.id/id/berita-unit/direktorat-jenderal-ketenagalistrikan/strategi-pemerintah-capai-rasio-elektrifikasi-99-di-tahun-2019

A pilot study from 2013 on residential energy consumption in the city of Bandung, Indonesia, demonstrated the importance of women's role in household energy consumption. The results showed that in households where the woman (wife) was the dominant decision maker on decisions about energy expenditure and control of energy consumption in the household, the energy consumption tended to be the lowest. The mean energy consumption of these households was 23-25% lower than in the households where energy decisions were made by the husband or by joint decision.[45]

Representation of women in the energy industry

Beyond household energy consumption, there are many examples of women as producers, technicians and entrepreneurs in sustainable energy. However, the traditional energy sector is still one of the least gender-inclusive sectors to date globally. According to one estimate, women represent only 6 percent, 4 percent and less than 1 percent of the technical, decision-making and top management positions, respectively, in the energy sector. For women entrepreneurs who would like to thrive within the energy industry, existing structural inequality, which manifests itself in the form of discrimination in law and practice, also poses barriers.

UNEP has reported on the underrepresentation of women in the energy sector at a global level, and the impact this has on gender equality. Gender aspects in the planning and policy cycles and sectors often has little visibility in formal and centralized policy frameworks. Men over-represent at the global policy level, for example At COP21, where 140 countries presented their plans for emission reductions (Intended Nationally Determined Contributions or INDCs), only 50 countries referred to gender as important in combatting climate change.[46] Energy policies typically focus on issues of investment, tariffs, privies, access, availability, infrastructure development, participation and environment – all of which are linked to gender roles. However energy policy is often considered gender-neutral.

Consequently, the forums in which energy issues are identified and potential solutions proposed are likely to have an inadvertent male bias (UNIDO and UN Women 2013). In addition to the leadership gap, the share of women in the workforce in the energy sector is generally quite low however greater sex-disaggregated data on this is needed. To address this disparity, governments should consider reserving dedicated spots for women as community representatives and within electricity entities, whether a rural electricity cooperative board of directors or a distribution billing and collections team. Intentionally seeking women's views and insights will lead to innovations and enhance a program's effectiveness.

Lack of sex-disaggregated data presents challenges for promoting gender equality in the energy sector. As for Indonesia, there is little data available about the representation of women in the lighting industry, at various levels of the market. However, in a focus group discussion held by Institute for Essential Services Reform with Ministry of Women's Empowerment and Child Protection on gender mainstreaming in Indonesia's renewable energy sector, several gender gaps were identified[47] that can be applied to the wider energy sector in general. Access to information related to energy is limited to women, given the

[45] *Ibid*

[46] James Melik, *"Solutions sought to end use of kerosene lamps"*, http://www.bbc.com/news/business-18262217

[47] Energia - Gender Training Manual, http://www.energia.org/cm2/wp-content/uploads/2015/04/04.-1319143629-Gendertrainingmanual.pdf

male-dominated industry, which in turn limits women's participation and contribution to shaping policy. This emphasizes that women are underrepresented at the decision making level in public and private spaces resulting in poor understanding of gender implications of policy.

Performance in Indonesia's labour market has been weak, according to a report by Asia Development Bank[48] Slow job growth and high levels of labor underutilization affect different groups of workers, particularly young people, women and those in rural areas. Investment in education in recent decades has not resulted in the expansion of labor productivity. For men and women, up-skilling and re-skilling opportunities are limited and access to career mobility is limited. Women in particular are restricted in the workforce due to gender roles ensuring their efforts are spent in the home.

In both developed and developing countries women's best chance of becoming involved in sustainable energy provision is at the community level (UNEP). Business models for small-scale energy production range from consignment arrangements, to linking of entrepreneurs to microfinancing institutions (possibly through the use of loan guarantee funds, which lowers the risk for financing institutions), to women individually or in groups taking on the manufacture or assembly of devices (sometimes as part of family businesses), to women's networks raising awareness of, for example, policy, options, pricing and safety.

Gender gaps related to access to energy, finances, training, employment and entrepreneurship need to be addressed in Indonesia. Creating policies that include both men and women, will result in more sustainable outcomes. Infrastructure and market transformation projects such as ADLIGHT have the opportunity to promote new skills, training, increased employment and entrepreneurship opportunities for women, as well as more equitable benefit-sharing at the community level.

Conclusion

ADLIGHT presents unique opportunities to strengthen women's participation in the energy efficient electricity sector along the entire value chain, from manufacturing, installation, maintenance, distribution, and billings.[49] However deeper gendered assessments need to be conducted in order to illustrate how women and men interact with the local economy differently. Energy sectors are primarily male-dominated in Indonesia and globally, however women are an untapped resource, as key beneficiaries of lighting technologies, and consultants on how to most effectively raise awareness about new lighting technologies. Finally, addressing gender gaps at leadership, policy and industrial levels in the electricity sector is an opportunity to advance gender equality, and the economy of Indonesia at large.

[48] https://www.transparency-partnership.net/sites/default/files/ssli_nama_mrv.pdf

[49] https://unhabitat.org/wp-content/uploads/2014/07/Gender-Responsive-Urban-Planning-and-Design.pdf

Proposed Gender Action Plan

Objective	Action	Indicator	Responsible institution
Overall objective			
Active participation of women and men in capacity building trainings throughout the project	Require and encourage active participation of women and men in capacity building trainings throughout the project	Cumulative number of women and men participating in capacity building trainings throughout the project Ratio of women and men participating in capacity building trainings throughout the project	PMU MEMR/DGNREEC
Outcome 1. Improved quality, energy efficient and affordable locally-produced EE Lighting (EEL) products and systems			
Enhanced gender balanced employment in the energy sector	Encourage local manufacturers to increase the number of female interns and request lighting manufacturer associations also to encourage the increase of female interns in the local manufactories Contracting women in consultancy, financial and customer services, as appropriate	Ratio of women and men interns in local EEL manufacturing companies Ratio of women and men employed through jobs created from the project (40/60)	PMU MEMR/DGNREEC PMU UNDP UNEP
Active participation of women and men in the periodic market surveys	In the design of periodic market surveys on production of EEL, incorporate gender sensitive questions In the design of periodic market surveys on application of EEL, develop surveys that specifically address women's needs, preferences and habits in acquirement, consumption and disposal of EEL products	Ratio of women and men participating in the periodic market surveys	PMU MEMR/DGNREEC

Objective	Action	Indicator	Responsible institution
	Ensure that the periodic market surveys conducted amongst the beneficiaries of the EEL systems are targeting women and men separately (i.e. both men and women in the household rather than one household as a whole)		
Women and men have the capacity to submit (from manufacturers) and approve (by banks) investment grade proposals for business transformation plans	Require and encourage active participation of women and men in trainings, workshops and focus group dialogues	Cumulative number of women and men who believe they have the capacity to submit and approve of investment grade proposals for business transformation plans Ratio of women and men participating in trainings, workshops and focus group dialogues	PMU MEMR/DGNREEC
Outcome 3: Increased penetration of high quality and efficient lighting			
Enhanced gender balanced employment in the energy sector	Contracting women in consultancy, financial and customer services, as appropriate Require and encourage contractors to employ women, as appropriate	Ratio of women and men employed through jobs created from the project	PMU MEMR/DGNREEC UNDP UNEP
Equal participation of women and men in stakeholder consultation meetings on installation of EEL systems in pilot areas	Require and encourage active participation of women and men in stakeholder consultation meetings for the general public on installation of EEL systems in pilot areas Ensure that stakeholder	Ratio of women and men participating in stakeholder consultation meetings in pilot areas	PMU MEMR/DGNREEC

Objective	Action	Indicator	Responsible institution
	consultation meetings are held at times and locations convenient for women		
Women and men in pilot areas are benefiting from EEL technologies equally	Collect information on number of beneficiaries and their level of satisfaction through periodic market surveys, monitoring visits etc.	Cumulative number of women and men benefiting from EEL technologies as influenced by the project Ratio of women and men benefiting from EEL technologies as influenced by the project Women's and men's level of satisfaction with EEL systems provided by the project (reliability, affordability, convenience, efficiency)	PMU MEMR/DGNREEC
Women and men have equal access to knowledge about energy efficient lighting technologies and their benefits	Require and encourage active participation of women in consultation meetings on their information needs on EEL systems and preferred channels for information sharing Ensure that the material developed under the awareness and promotion program on the benefits of LED lighting is designed and communicated in a way that is easily understood by women and men with disabilities as well as different levels of education Conduct online survey on the use of the information system (on the website) to capture	Cumulative number of women and men participating in consultation meetings Ratio of women and men participating in consultation meetings Cumulative number of women and men with access to knowledge about EEL technologies and their benefits developed Ratio of women and men with access to knowledge about EEL technologies and their benefits developed Cumulative number of women and men users of the information system Ratio of women and men	PMU MEMR/DGNREEC

Objective	Action	Indicator	Responsible institution
	users' gender	users of the information system	
		Number of women's groups involved	
	Encourage participation of women's groups to promote consumer energy efficiency awareness		
	Ensure that the awareness and promotion program and information system on benefits of LED lighting includes gender responsive information material such as the potential benefits of EEL for women and men	Cumulative number of women and men beneficiaries with increased awareness and access to knowledge about energy efficient lighting technologies	

General recommendations for the ADLIGHT project

- Monitor benefits, participation, and feedback among women and men and incorporate remedial action that promotes gender equality as appropriate
- Ensure that evaluation forms are distributed during each training/workshop collecting information on participant's gender, previous engagement in similar trainings/workshops and perceived relevance of the workshop/training
- Collect sex-disaggregated data and project's gender issues/impact through surveys, stakeholder interviews and other appropriate stakeholder engagement methods, as identified in the Stakeholder Engagement Plan
- Consider inclusion of sex-disaggregated data collected during the project in midterm evaluation, project closure reports, annual reports and other project and UNDP information material
- Recruit a gender specialist/expert or ensure at least one member of project management team is gender sensitized
- Create ToRs for project staff that ensure equal opportunity for women and men, and where applicable, require skills/expertise in gender
- Ensure that the implementation of the gender action plan is included as one requirement in the TORs for project staff recruited throughout the project
- Invite universities to focus group dialogues and encourage them to provide more internship positions for women in the energy efficiency sector
- Review, monitor and report regularly on the gender action plan

Proposed implementation arrangements

The project manager should have the overall responsibility for implementing, monitoring and reporting of the gender action plan. In the beginning of the project, a full-time or part time gender focal point could be recruited to provide additional technical assistance for the project management unit on gender action plan annual planning, implementation, monitoring and reporting.

In order to successfully mainstream gender in the project, a portion of the project budget should be allocated for the implementation of the gender action plan. Resources for the implementation of the gender action plan could be drawn from the budgets for project activities such as market surveys, capacity building activities, installation of EEL systems, awareness raising program on the benefits of LED lighting and from the consultancy package.

Annex O: Knowledge Management Strategy for the ADLIGHT Project

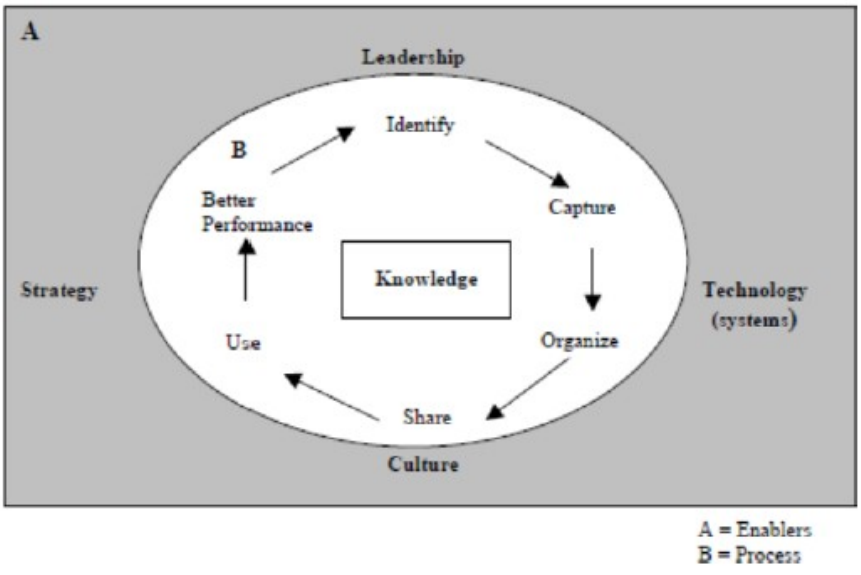
1.0 Introduction

The main driver for implementing development projects, such as ADLIGHT, is to serve as a demonstration of how a certain concept can work under given circumstances in order to solve or alleviate a particular challenge. Therefore, it is imperative that, to the extent possible, all information acquired during project design and execution is well documented for purposes of replication in the future. However, more importantly, the aforementioned information is required during project execution in order to determine the extent to which the project is meeting its original objectives and outcomes.

Collectively, the process of handling the aforementioned information is referred to as *Knowledge Management*. According to the United Nations Joint Inspection Unit (2007), knowledge management is:

> "*...the systematic processes, or range of practices, used by organizations to identify, capture, store, create, update, represent, and distribute knowledge for use, awareness and learning across the organization.*"

Based on the above definition, this document aims at guiding the ADLIGHT project on how to conduct knowledge management during project implementation. In the case of this project, knowledge management will be key due to the fact that there are several parties involved at different stages of the project. Therefore, there is need to harmonize the information gathered as well as capture the knowledge acquired as well as the knowledge brought to the project by the various experts and stakeholders participating in the project. It is anticipated that the knowledge management process for the ADLIGHT project will adopt the approach developed by International Labour Organization (ILO) as summarized in the following diagram:



Source: ILO, 2004

As depicted in the previous diagram, knowledge management is a somewhat cyclical process but requires an enabling environment to thrive. In the case of the ADLIGHT Project, in order to successfully undertake knowledge management, there will be need for a high level of commitment by the Project Manager. In

addition, some form of information technology will be required in order to support knowledge acquisition, storage and sharing. In addition, stemming from good project management, there will be the need for fostering a culture of knowledge management based on good knowledge management practices and strategic direction.

Consequently, the overall objective of this Knowledge Management Strategy is to ensure the effective use of the knowledge acquired and availed to the ADLIGHT Project. The specific objectives will be:

1. To collate and retain knowledge in order to enhance the project's institutional memory;
2. To ensure timely and open access to knowledge so as to foster high staff performance and effective work flow processes; and
3. To nurture knowledge sharing for more effective project implementation as well as to inform decision-making processes.

2.0 Guiding Principles

In order to ensure the successful implementation of this Knowledge Management Strategy, a number of guiding principles will have to be adhered to. These are discussed in this section:

2.1 Collective Responsibility

While the overall responsibility of knowledge management is that of the Project Manager, it is, however, everyone's obligation to contribute to the knowledge management process. This is important because different project staff, stakeholders and experts will be involved at different stages of the project. Therefore, having an effective system for collating and sharing information will be crucial.

2.2 Range of Knowledge

It is anticipated that a wide range of knowledge will be acquired and shared during the project lifetime. This includes explicit knowledge acquired from stakeholders during surveys, standard procedures, progress reports as well as implicit knowledge such as the one held by individuals such as experts which is unwritten. It should be noted that some of the progress report formats to be adopted in the project will attempt to elicit the knowledge acquired and, therefore, will require particular attention.

2.3 Simplicity

Due to limitation of financial resources and to ensure ease of utilization, the knowledge management process should be simple yet effective. For instance, there will be no need to acquire complex knowledge management software. Rather, the project could adopt the use of existing tools such as the information sharing systems used by UNDP such as Yammer, OneDrive, Outlook calendar, Web server, etc. It is worth noting that knowledge management is NOT technology-focussed but people-focussed.

2.4 Use of Various Modalities

It is incumbent on the Project Team to use a wide range of knowledge acquisition, storage and sharing modalities based on ease of doing so. For example, a bilateral meeting can provide a forum for the generation and exchange of a lot of knowledge and information. Therefore, in order to capture this knowledge for the benefit of all relevant stakeholders, brief "Meeting Notes" can be drafted and shared after every meeting. This is also important in the case of Workshops and Seminars. Alternatively, voice

recording can be utilized to quickly and easily capture the entire meeting proceedings, albeit in a non-distilled format.

2.5 Multifunctional Approach

There is need to recognize that the knowledge acquired can have diverse uses. For example, Workshop Proceedings Report can be evidence that a particular milestone has been attained. In addition, the report can be a source of information for a particular group of stakeholders. Therefore, in the process of knowledge management, care should be taken to ensure that the "packaging" of the knowledge is done in a way to ensure that the products are suitable for the foreseeable different uses.

2.6 Constant Review

For any system to be effective, its performance has to be constantly reviewed and, where need be, adjusted. Therefore, the Project Manager could delegate monitoring and maintenance of knowledge management to the individual allocated the Monitoring and Evaluation responsibility. This will ensure that the opportunities of knowledge acquisition and sharing are not lost.

2.7 Publicity and Public Engagement

By its nature, the ADLIGHT Project has an inherent interest among various stakeholders, the public included. Therefore, an effective knowledge management process should include means of publicizing the knowledge garnered as well as using a simple and effective way of engaging with the public e.g. using mass media (TV, radio, newspapers, website, etc) and social media, such as Twitter and Facebook.

2.8 Alignment to Overall Project Plan

In a number of instances, outputs of the project will contribute as inputs to the ADLIGHT's Project knowledge management. In other cases, products of knowledge management will contribute to achieving certain outcomes of the project. Therefore, it will be important that the process of knowledge management is aligned to the overall Project Plan. Specifically, the project roll-out plan should include an activity for the development of a Knowledge Management Action Plan.

3.0 Establishment of the EEL Knowledge Management Center

The project in its Component 1, Activity 1.1.1, will support the establishment of an Efficient Lighting Knowledge Center for collecting market information on lighting products, quality of products, consumer perception, including sex-disaggregated consumer data where available and other related information regarding efficient lighting technology and technology providers. Existing agency with relevant mandate and organization, such as BSN, will be assisted in preparing a strategy and design to meet the requirements and purpose of the Knowledge Center and in establishing it in the selected agency that will continue to be responsible in carrying out a sustainable program ushered by the project in providing not only knowledge but also a categorized quality product listing (QPL) and database in cooperation with a network of qualified laboratories.

Annex P: Stakeholder engagement plan

The stakeholder engagement plan provides a description of stakeholders identified during the project design stage, their interest in the project and the most relevant engagement techniques that will be used throughout the project. The stakeholders are not strictly limited to the list as additional stakeholders and stakeholder engagement needs are likely to be identified during the project implementation.

Stakeholder/ Stakeholder group	Why included (interest)	Participation method		Timing
		Method (most common)	Responsibility	
GEF	Donor of funds for the ADLIGHT project	Formal meetings (i.e. supervision missions) Project reports	UNDP UNEP	Annually
Ministry of Energy and Mineral Resources (MEMR)	Provides overall direction and oversight as the Executing Agency; acts as Chairman of the National Steering Committee; provides the lead role in coordinating with other ministries and agencies in pushing forward the goals and objectives of the efficient lighting industry program and resolving inter-ministry barriers	Project board meetings Project reports Workshops, trainings focus group dialogues One-on-one meetings, Phone/email	PMU DGNREEC UNDP UNEP	Annually, quarterly or more frequently when necessary To be confirmed in the annual work plan As frequently as necessary
Directorate General of New-Renewable Energy and Energy Conservation (DGNREEC)	Acts as the Responsible Party for the project and leads the Project Management Unit for the day-to-day operations and resolution of issues to ensure smooth implementation of the project; provides technical support for the technology programs; coordinates with other government agencies as well as UNDP and UNEP	Project board meetings Project reports Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	PMU UNDP UNEP	Quarterly or more frequently when necessary To be confirmed in the annual plan As frequently as necessary
MEMR P3TK EBTKE (Research and Development of DGNREEC)	Provides technical support and laboratory services for the needs of the LED industry; leads the capacity building and accreditation of laboratory facilities in meeting the MEPS/*SNI* standard requirement, benchmarking and round-robin testing	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequent as necessary
Min of	Acts as the integrator of the	Project board	MEMR/	Quarterly or more

Stakeholder/ stakeholder group	Why included (interest)	Participation method		Timing
Development Planning (*Badan Perencanaan Pembangunan Nasional or BAPPENAS)*	related government programs within the overarching objectives of the EEL program which crosses many sectoral boundaries and policy/organizational areas considering overall national interest	meetings Project reports Focus group dialogues One-on-one meetings, Phone/email	DGNREEC PMU	frequently when necessary Annually To be confirmed in the annual plan As frequently as necessary
Public Procurement Agency – *Lembaga Kebijakan Pengadaan Barang/Jasa Pemerintah* (LKPP)	Acts as main resource agency for the development and adoption of government procurement policies in order to hasten the market penetration of locally produced high efficiency and affordable lighting products	Project board meetings Project reports Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	Quarterly or more frequently when necessary Annually To be confirmed in the annual plan As frequently as necessary
Financial Service Authority (*Otoritas Jasa Keuangan* or *OJK*)	Coordinates among financial institutions and banks and provides overall policy direction and guidance in implementation of financing mechanisms that will be adopted by the project	Project board meetings Project reports Workshops, trainings, focus group dialogues, business forums One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	Quarterly or more frequently when necessary Annually To be confirmed in the annual plan As frequently as necessary
PT Sarana Multi Infrastruktur - Persero (PT SMI)	Acts as resource group for green banking and green infrastructure as it applies to buildings and street lighting finance	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
City Government – Smart Cities	Act as pilot demonstration hosts in EEL application in street lighting and the evaluation of the impacts of the transformation toward	Workshops, trainings, focus group dialogues Surveys	MEMR/ DGNREEC PMU	To be confirmed in the annual plan First year, before mid-term review

Stakeholder/ Stakeholder	Why included (interest)	Participation method		Timing
	EELs			and before project closure
		One-on-one meetings, Phone/email		As frequently as necessary
Energy Service Company (ESCO)	Provides the main link between the EEL manufacturers and users and facilities project development, financing and EPC requirements	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
State Owned Banks & Financial Institution	Acts as catalysts in the development and implementation of the business transformation plans to transform the industry to produce and market EELs locally and potentially abroad	Workshops, trainings, focus group dialogues and business forums One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
Perusahaan Listrik Negara (*PLN*) State Electricity Company)	Acts as resource organization on energy supply, pricing, and policy in support of the increased market for EELs and transition from the conventional lamps as well as piloting a LED distribution scheme and billing system through PLN infrastructure for the residential sector via the ESCO model.	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
Ministry of Trade	Act as resource ministry in the harmonization of policies and regulations regarding trading of locally produced and imported EELs; provides active support in the MVE schemes and market control to encourage efficient and reasonably priced lighting products that meet MEPS/*SNI*	Project board meetings Project reports Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	Quarterly or more frequently when necessary Annually To be confirmed in the annual plan As frequently as necessary
Ministry of Industry	Act as resource ministry for the policies and regulations governing the lighting industry sector and the implementation of business transformation plans towards producing affordable, high quality lighting products	Project board meetings Project reports Workshops, trainings, focus group dialogues	MEMR/ DGNREEC PMU	Quarterly or more frequently when necessary Annually To be confirmed in the annual plan

Stakeholder/ Stakeholder	Why included (interest)	Participation method		Timing
		One-on-one meetings, Phone/email		As frequently as necessary
Ministry of Public Works and Housing	Acts as resource ministry in the application of EELs in the street lighting and building consumer groups	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
Ministry of Finance/Bureau of Customs	Provides lead role in the application and regulation of policies regarding custom duties and importation of lighting product components as well exportation of standard products whenever viable; participated in the development and implementation of MVE schemes vis-à-vis MEPS/*SNI* adopted and the corresponding inspection software to discourage entry or usage of inefficient lighting products	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
Badan Standardisasi Nationa (*BSN*) or National Standardization Agency	Provides technical support and policy coordination in the development and adoption of the *SNI* and MEPS within the timeframe of the project considering ASEAN and other international commitments in MEPS and labeling	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
BPPT (Research Agency for Technology Development and Implementation)	Acts as resource agency in the R&D and product development for EEL technology and their application	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
Tingkat Kandungan Dalam Negeri (*TKDN*) or Local Content certifiers	Act as resource agency in the policy review and recommendations in enhancing the local content of locally manufactured lighting products	Workshops, trainings, focus group dialogues One-on-one meetings, Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
Credit Facility Guarantee/Credit Insurance Companies	Act as resource groups for determining appropriate financing support schemes in support of the full scale ESCO model regarding credit	Workshops, trainings, focus group dialogues One-on-one meetings,	MEMR/ DGNREEC PMU	To be confirmed in the annual work plan As frequently as

		Participation method		
	guarantee facility	Phone/email		necessary
Lighting Industry Associations (*GAMATRINDO, ALINDO and APERLINDO*)	Provides technical support and coordination among local manufacturers; leads the industry associations and its members in policy discussions to achieve timely progress in project implementation and performance targets; lead the transition to higher efficiency and cost-effective EEL production and acts as coordinator and focal point in the monitoring, verification and enforcement representing the EEL industry for measuring impact and corrective actions	Workshops, trainings, focus group dialogues Surveys One-on-one meetings Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual work plan First year, before mid-term review and before project closure As frequently as necessary
Development Partners (MTRE3, ICED2, PFAN, and other bilateral projects)	Provide technical support and coordination regarding the implementation of relevant activities towards achievement of outputs that have common project objectives	Workshops, trainings, focus group dialogues One-on-one meetings Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual work plan As frequently as necessary
Beneficiaries of LED distribution in selected cities such as residents in government buildings, owners/visitors of commercial buildings and residents benefitting from street lighting	Stakeholders who will directly benefit from the installation of EEL lighting	Public meetings Surveys Reading material Email, phone, fax or mail Radio/TV Social media, UNDP's website, other project related websites	MEMR/ DGNREEC PMU	To be confirmed in the annual work plan First year, before mid-term review and before project closure Throughout project implementation
Energy efficient lighting manufacturers (local and imported lighting)	Will be involved in trainings on how to increase the quality and value of their LED products	Workshops, trainings, focus group dialogues Forums, conferences and exhibitions Surveys	MEMR/ DGNREEC PMU	To be confirmed in the annual work plan First year, before mid-term review

Stakeholder/ Stakeholder group **Why included (interest)** **Timing**

Stakeholder/ Stakeholder group	Why included (interest)	Participation method		Timing
				and before project closure
		Other reading material Email, phone, fax or mail Radio/TV Social media, UNDP's website, other project related websites		Throughout project implementation
Masyarakat Konservasi dan Efisiensi Energi Indonesia MASKEEI or IECES (*Indonesia Energy Conservation* and Efficiency *Society*)	Will be involved in the development of an awareness program on the benefits of energy efficient lighting technologies	Workshops, focus group dialogues One-on-one meetings Phone/email	MEMR/ DGNREEC PMU	To be confirmed in the annual plan As frequently as necessary
General Public, NGOs and other relevant organisations	The general public who does not participate directly in the project activities but has an interest in being informed on project results and lessons learned	Public meetings Annual reports Forums, conferences and exhibitions Other reading material Email, phone, fax or mail Radio/TV Social media, UNDP's website, other project related websites	MEMR/ DGNREEC PMU UNDP	Before every pilot project start Annually To be confirmed in the annual plan Throughout project implementation
Coal power plant workers, conventional lighting manufacturers and other stakeholders	Stakeholders who might have a concern about the project's potential impacts on them	Public consultation meetings Other reading material Email, phone, fax or mail Radio/TV Social media, UNDP's	MEMR/ DGNREEC PMU	Before every pilot project start Throughout project implementation

Stakeholder/ Stakeholder group	Why included (interest)	Participation method		Timing
		website, other project related websites		

Grievance mechanism

The project management unit (PMU) is suggested to take the role of Grievance Redress Mechanism (GRM) with the mandate to:

(i) Receive and address any concerns, complaints, notices of emerging conflicts, or grievances (collectively“ Grievance”) alleging actual or potential harm to affected person(s) (the “Claimant(s)”) arising from Project;
(ii) Assist in resolution of Grievances between and among Project Stakeholders; as well as the various government ministries, agencies and commissions, CSOs and NGOs, and others in the context of the Project; and
(iii) Conduct itself at all times in a flexible, collaborative, and transparent manner aimed at problem solving and consensus building.

The GRM will receive and consider all comments and complaints associated with the project. Any individual or organisation may send comments and/or complaints in person, by phone, fax, email or mail, or in any other manner that the PMU may deem fit. All Grievances and reports of conflict will be received, assigned a tracking number, acknowledged to Claimant, recorded electronically, and subject to periodic updates to the Claimant as well as the office file.

Normally all grievances will be acknowledged within 3-5 working days in the form of a standard letter or email providing a brief description of the process that will be followed. A response will be communicated to the Claimant within 14-21 working days of receiving the grievance.

Files for each Grievance will be available for review by the Claimant and other Stakeholders involved in the Grievance, or their designated representative(s). Appropriate steps will be taken to maintain the confidentiality of the Claimant, if previously requested. The GRM will provide periodic updates to the Claimant regarding the status and current actions to resolve the Grievance. Not including the acknowledgment of receipt of the Grievance, such updates will occur within reasonable intervals (not greater than every 30 days).

Bi-annually, the GRM will make available to the public, a report describing the work of the GRM, listing the number and nature of the Grievances received and processed in the past six months, a date and description of the Grievances received, resolutions, referrals and ongoing efforts at resolution, and status of implementation of ongoing resolutions. The level of detail provided with regard to any individual Grievance will depend on the sensitivity of the issues and Stakeholder concerns about confidentiality, while providing appropriate transparency about the activities of the GRM.

The aforementioned report will also highlight key trends in emerging conflicts, Grievances, and dispute resolution, and make recommendations regarding: (i) Measures that can be taken by the Government to avoid future harms and Grievances; and (ii) Improvements to the GRM that would enhance its effectiveness, accessibility, predictability, transparency, legitimacy, credibility, and capacity.

In addition to the GRM described above, other channels can be utilized by Complaints through UNDP's Accountability Mechanism (Stakeholder Response Mechanism [SRM] and Social and Environmental Compliance Unit [SECU]). These channels can be utilized for grievances purporting that UNDP is not complying with the social and environmental policies or commitments that apply to its projects (i.e. corruption, coercion or major systematic violations of rights and/or policies), or in other cases where the Complaint is not satisfied with the GRM's response. All SECU and SRM Requests can be submitted via an online webform.

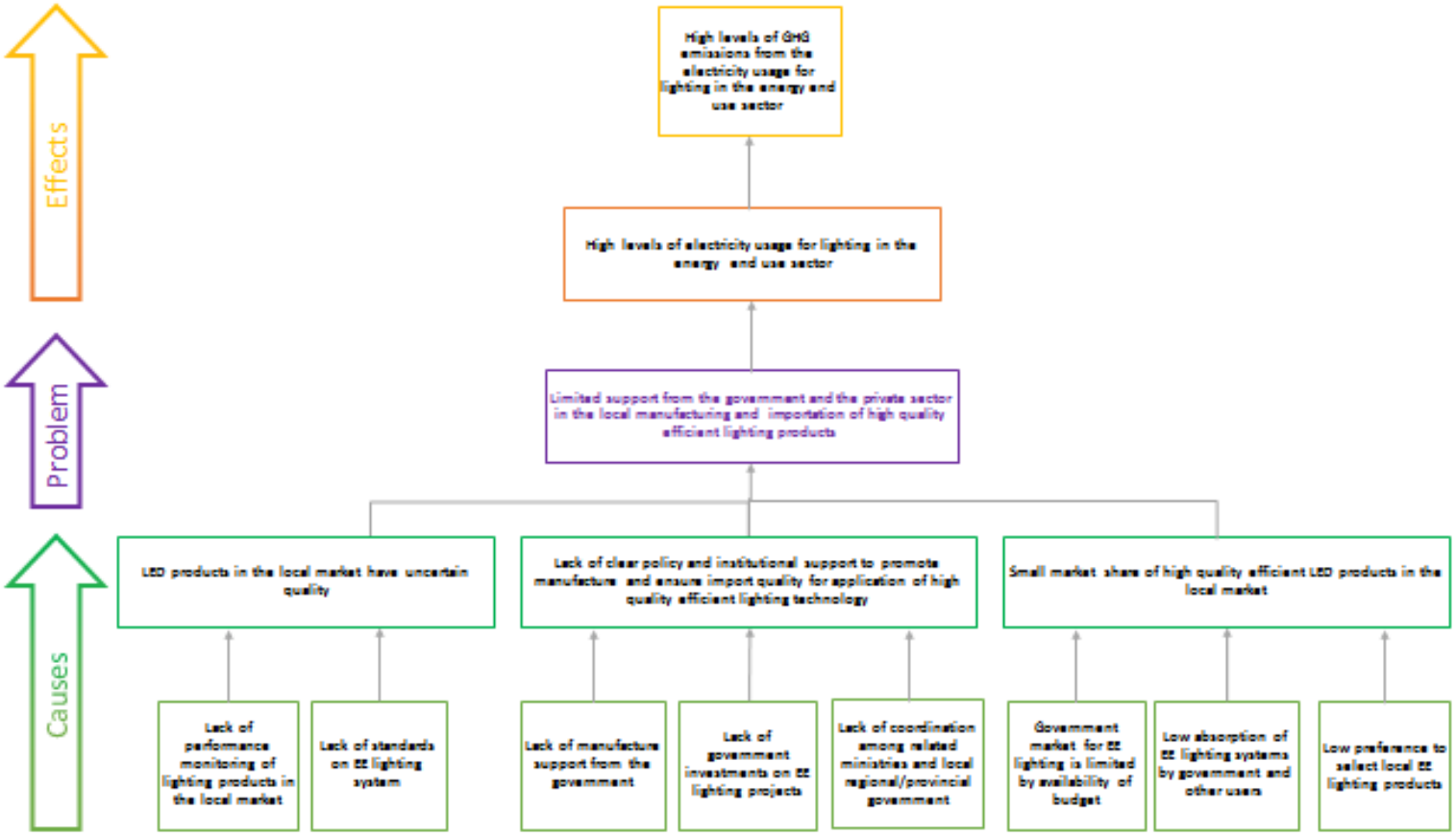
Monitoring and reporting

The stakeholders will be engaged in project's monitoring and reporting process through various activities and communication channels as outlined in the project document's Monitoring and Evaluation Plan and in Table 3. Results of stakeholder engagement activities will be reported back to the stakeholders throughout the project implementation.

The Stakeholder Engagement Plan itself is a living document and will be reviewed and updated regularly as response to changed project circumstances that might arise during the project implementation period. The overall responsibility of Monitoring and Reporting will be that of the Project Manager who will delegate the responsibility to one of the project staff assigned to undertake Monitoring and Evaluation.

Annex Q:

Problem Tree: Advancing Indonesia's Lighting Market to High Efficient Technologies (ADLIGHT)



Annex R: Supplemental Provisions to the Project Document[50]: The Legal Context

General responsibilities of the Government, UNDP and the executing agency

1. All phases and aspects of UNDP assistance to this project shall be governed by and carried out in accordance with the relevant and applicable resolutions and decisions of the competent United Nations organs and in accordance with UNDP's policies and procedures for such projects, and subject to the requirements of the UNDP Monitoring, Evaluation and Reporting System.

2. The Government shall remain responsible for this UNDP-assisted development project and the realization of its objectives as described in this Project Document.

3. Assistance under this Project Document being provided for the benefit of the Government and the people of Indonesia, the Government shall bear all risks of operations in respect of this project.

4. The Government shall provide to the project the national counterpart personnel, training facilities, land, buildings, equipment and other required services and facilities. It shall designate the Government Co-operating Agency named in the cover page of this document (hereinafter referred to as the "Co-operating Agency"), which shall be directly responsible for the implementation of the Government contribution to the project.

5. The UNDP undertakes to complement and supplement the Government participation and will provide through the Executing Agency the required expert services, training, equipment and other services within the funds available to the project.

6. Upon commencement of the project the Executing Agency shall assume primary responsibility for project execution and shall have the status of an independent contractor for this purpose. However, that primary responsibility shall be exercised in consultation with UNDP and in agreement with the Co-operating Agency. Arrangements to this effect shall be stipulated in the Project Document as well as for the transfer of this responsibility to the Government or to an entity designated by the Government during the execution of the project.

7. Part of the Government's participation may take the form of a cash contribution to UNDP. In such cases, the Executing Agency will provide the related services and facilities and will account annually to the UNDP and to the Government for the expenditure incurred.

(a) Participation of the Government

1. The Government shall provide to the project the services, equipment and facilities in the quantities and at the time specified in the Project Document. Budgetary provision, either in kind or in cash, for the Government's participation so specified shall be set forth in the Project Budgets.

2. The Co-operating Agency shall, as appropriate and in consultation with the Executing Agency, assign a director for the project on a full-time basis. He shall carry out such responsibilities in the project as are assigned to him by the Co-operating Agency.

3. The estimated cost of items included in the Government contribution, as detailed in the Project Budget, shall be based on the best information available at the time of drafting the project proposal. It is understood that price fluctuations during the period of execution of the project may necessitate an adjustment of said contribution in monetary terms; the latter shall at all times be determined by the value of the services, equipment and facilities required for the proper execution of the project.

4. Within the given number of man-months of personnel services described in the Project Document, minor adjustments of individual assignments of project personnel provided by the Government may be made by the Government in consultation with the Executing Agency, if this is found to be in the best interest of the project. UNDP shall be so informed in all instances where such minor adjustments involve financial

[50] http://asiapacific.unwomen.org/ru/countries/indonesia

implications.

5. The Government shall continue to pay the local salaries and appropriate allowances of national counterpart personnel during the period of their absence from the project while on UNDP fellowships.

6. The Government shall defray any customs duties and other charges related to the clearance of project equipment, its transportation, handling, storage and related expenses within the country. It shall be responsible for its installation and maintenance, insurance, and replacement, if necessary, after delivery to the project site.

The Government shall make available to the project - subject to existing security provisions - any published and unpublished reports, maps, records and other data which are considered necessary to the implementation of the project.

7. Patent rights, copyright rights and other similar rights to any discoveries or work resulting from UNDP assistance in respect of this project shall belong to the UNDP. Unless otherwise agreed by the Parties in each case, however, the Government shall have the right to use any such discoveries or work within the country free of royalty and any charge of similar nature.

8. The Government shall assist all project personnel in finding suitable housing accommodation at reasonable rents.

9. The services and facilities specified in the Project Document which are to be provided to the project by the Government by means of a contribution in cash shall be set forth in the Project Budget. Payment of this amount shall be made to the UNDP in accordance with the Schedule of Payments by the Government.

10. Payment of the above-mentioned contribution to the UNDP on or before the dates specified in the Schedule of Payments by the Government is a prerequisite to commencement or continuation of project operations.

(b) Participation of the UNDP and the executing agency

1. The UNDP shall provide to the project through the Executing Agency the services, equipment and facilities described in the Project Document. Budgetary provision for the UNDP contribution as specified shall be set forth in the Project Budget.

2. The Executing Agency shall consult with the Government and UNDP on the candidature of the Project Manager[51] who, under the direction of the Executing Agency, will be responsible in the country for the Executing Agency's participation in the project. The Project Manager shall supervise the experts and other agency personnel assigned to the project, and the on-the-job training of national counterpart personnel. He shall be responsible for the management and efficient utilization of all UNDP-financed inputs, including equipment provided to the project.

3. The Executing Agency, in consultation with the Government and UNDP, shall assign international staff and other personnel to the project as specified in the Project Document, select candidates for fellowships and determine standards for the training of national counterpart personnel.

4. Fellowships shall be administered in accordance with the fellowships regulations of the Executing Agency.

5. The Executing Agency may, in agreement with the Government and UNDP, execute part or all of the project by subcontract. The selection of subcontractors shall be made, after consultation with the Government and UNDP, in accordance with the Executing Agency's procedures.

6. All material, equipment and supplies which are purchased from UNDP resources will be used exclusively for the execution of the project, and will remain the property of the UNDP in whose name it will be held by the Executing Agency. Equipment supplied by the UNDP shall be marked with the insignia of

[51] http://www.nama-database.org/index.php/Smart_Street_Lighting_Initiative_(SSLI)

the UNDP and of the Executing Agency.

7. Arrangements may be made, if necessary, for a temporary transfer of custody of equipment to local authorities during the life of the project, without prejudice to the final transfer.

8. Prior to completion of UNDP assistance to the project, the Government, the UNDP and the Executing Agency shall consult as to the disposition of all project equipment provided by the UNDP. Title to such equipment shall normally be transferred to the Government, or to an entity nominated by the Government, when it is required for continued operation of the project or for activities following directly therefrom. The UNDP may, however, at its discretion, retain title to part or all of such equipment.

9. At an agreed time after the completion of UNDP assistance to the project, the Government and the UNDP, and if necessary the Executing Agency, shall review the activities continuing from or consequent upon the project with a view to evaluating its results.

10. UNDP may release information relating to any investment oriented project to potential investors, unless and until the Government has requested the UNDP in writing to restrict the release of information relating to such project.

Rights, Facilities, Privileges and Immunities

1. In accordance with the Agreement concluded by the United Nations (UNDP) and the Government concerning the provision of assistance by UNDP, the personnel of UNDP and other United Nations organizations associated with the project shall be accorded rights, facilities, privileges and immunities specified in said Agreement.

2. The Government shall grant UN volunteers, if such services are requested by the Government, the same rights, facilities, privileges and immunities as are granted to the personnel of UNDP.

3. The Executing Agency's contractors and their personnel (except nationals of the host country employed locally) shall:

(a) Be immune from legal process in respect of all acts performed by them in their official capacity in the execution of the project;

(b) Be immune from national service obligations;

(c) Be immune together with their spouses and relatives dependent on them from immigration restrictions;

(d) Be accorded the privileges of bringing into the country reasonable amounts of foreign currency for the purposes of the project or for personal use of such personnel, and of withdrawing any such amounts brought into the country, or in accordance with the relevant foreign exchange regulations, such amounts as may be earned therein by such personnel in the execution of the project;

(e) Be accorded together with their spouses and relatives dependent on them the same repatriation facilities in the event of international crisis as diplomatic envoys.

4. All personnel of the Executing Agency's contractors shall enjoy inviolability for all papers and documents relating to the project.

5. The Government shall either exempt from or bear the cost of any taxes, duties, fees or levies which may impose on any firm or organization which may be retained by the Executing Agency and on the personnel of any such firm or organization, except for nationals of the host country employed locally, in respect of:

(a) The salaries or wages earned by such personnel in the execution of the project;

(b) Any equipment, materials and supplies brought into the country for the purposes of the project or which, after having been brought into the country, may be subsequently withdrawn

therefrom;

(c) Any substantial quantities of equipment, materials and supplies obtained locally for the execution of the project, such as, for example, petrol and spare parts for the operation and maintenance of equipment mentioned under (b), above, with the provision that the types and approximate quantities to be exempted and relevant procedures to be followed shall be agreed upon with the Government and, as appropriate, recorded in the Project Document; and

(d) As in the case of concessions currently granted to UNDP and Executing Agency's personnel, any property brought, including one privately owned automobile per employee, by the firm or organization or its personnel for their personal use or consumption or which after having been brought into the country, may subsequently be withdrawn therefrom upon departure of such personnel.

6. The Government shall ensure:

(a) prompt clearance of experts and other persons performing services in respect of this project; and

(b) the prompt release from customs of:

(i) equipment, materials and supplies required in connection with this project; and

(ii) property belonging to and intended for the personal use or consumption of the personnel of the UNDP, its Executing Agencies, or other persons performing services on their behalf in respect of this project, except for locally recruited personnel.

7. The privileges and immunities referred to in the paragraphs above, to which such firm or organization and its personnel may be entitled, may be waived by the Executing Agency where, in its opinion or in the opinion of the UNDP, the immunity would impede the course of justice and can be waived without prejudice to the successful completion of the project or to the interest of the UNDP or the Executing Agency.

8. The Executing Agency shall provide the Government through the resident representative with the list of personnel to whom the privileges and immunities enumerated above shall apply.

9. Nothing in this Project Document or Annex shall be construed to limit the rights, facilities, privileges or immunities conferred in any other instrument upon any person, natural or juridical, referred to hereunder.

Suspension or termination of assistance

1. The UNDP may by written notice to the Government and to the Executing Agency concerned suspend its assistance to any project if in the judgement of the UNDP any circumstance arises which interferes with or threatens to interfere with the successful completion of the project or the accomplishment of its purposes. The UNDP may, in the same or a subsequent written notice, indicate the conditions under which it is prepared to resume its assistance to the project. Any such suspension shall continue until such time as such conditions are accepted by the Government and as the UNDP shall give written notice to the Government and the Executing Agency that it is prepared to resume its assistance.

2. If any situation referred to in paragraph 1, above, shall continue for a period of fourteen days after notice thereof and of suspension shall have been given by the UNDP to the Government and the Executing Agency, then at any time thereafter during the continuance thereof, the UNDP may by written notice to the Government and the Executing Agency terminate the project.

3. The provisions of this paragraph shall be without prejudice to any other rights or remedies the UNDP may have in the circumstances, whether under general principles of law or otherwise.